

Azure Data Factory (ADF) Notes

ADF Tutorial Video1 Link:

<https://youtu.be/8zIVOdKyoDA?si=JJl7iSIbnG1M2nDM>

Data Source Link : <https://github.com/anshlambagit/Azure-Data-Factory/tree/main/Raw%20Data>

Timestamps:

0:00 Introduction

05:56 What is Azure Data Factory

13:02 Free Azure Account

15:41 Azure Data Fundamentals

19:10 Azure Data Lake

27:58 Create Azure Data Factory Workspace

30:18 Dataset and Linked Service in Azure Data Factory

39:07 Data Ingestion in Data Lake

40:05 Azure Data Factory Tutorial

46:02 Copy Activity in Azure Data Factory

1:01:11 Copy Data Using REST API in ADF

1:16:43 Get Metadata Activity in Azure Data Factory

1:24:12 IF Condition in ADF

1:26:42 For Each Activity in Azure Data Factory

1:37:34 Expression Builder in ADF

1:46:31 Parameterized Pipeline in Azure Data Factory

2:04:06 Data Flow in Azure Data Factory

2:13:03 Data Transformation in Azure Data factory using Data Flow

2:26:30 Schedule Trigger in Azure Data Factory

2:35:59 Set Variable Activity ADF

2:41:46 Storage Event Trigger in ADF

2:43:32 Real Time Scenarios in Azure Data Factory

2:51:40 Delete Activity in ADF

2:53:14 Debugging Pipeline in Azure Data Factory

3:01:22 Execute Pipeline Activity in Azure Data Factory

3:10:14 End-To-End Data Pipeline using ADF

Lets Start

Prerequisite

- Laptop or pc wit stable internet connection
- Azure Account (Don't worry will create one)
- Excitement to learn Azure Date Factory

WHAT IS AZURE DATA FACTORY?

ADF is the ETL Tool

ADF most commonly refers to [Azure Data Factory](#), Microsoft's cloud-based, serverless data integration service for creating data-driven workflows (pipelines) to move and transform data at scale, acting as an ETL/ELT tool. However, ADF can also mean [Automatic Direction Finding](#), a radio navigation system in aviation, or an [Automatic Document Feeder](#), a printer/scanner feature. The specific meaning depends on the context, but in tech, it's usually Azure Data Factory.

1. Azure Data Factory (ADF) is a cloud-based ETL AND data integration service provided by Microsoft Azure.
2. It enables you to create, schedule, and manage data pipelines that move and transform data from various sources to destinations

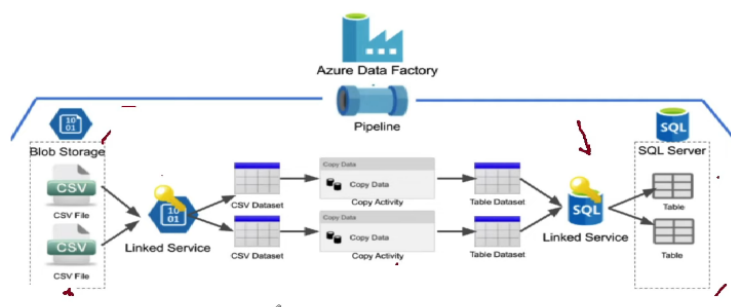
Azure Data Factory use cases:

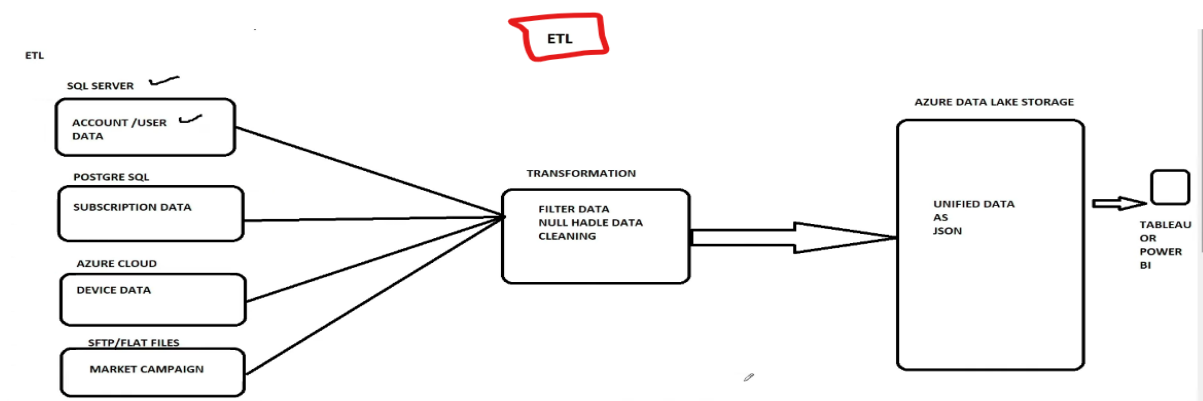
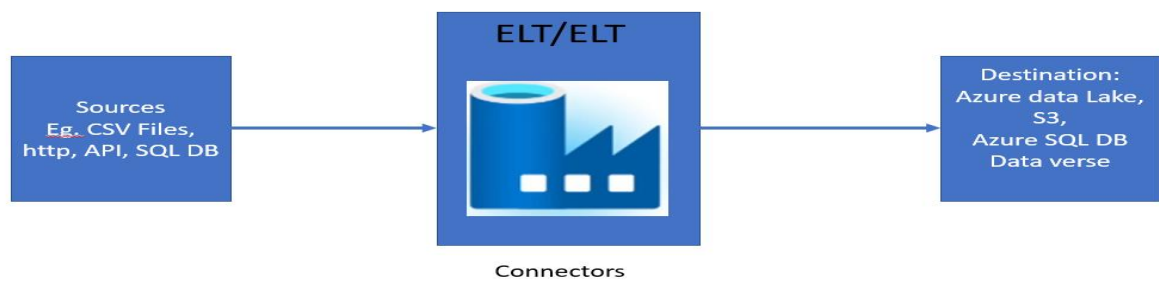
ADF can be used for:

- Supporting data migrations
- Getting data from a client's server or online data to an Azure Data Lake
- Carrying out various data integration processes

ADF COMPONENTS & TERMINOLOGY

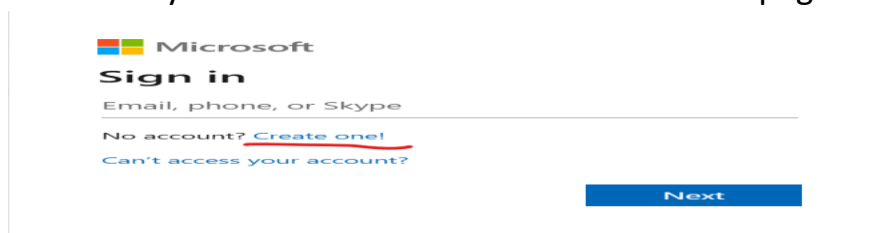
1. activity -
2. dataset -
3. linked service-
4. source & sink
5. trigger
6. integration run time



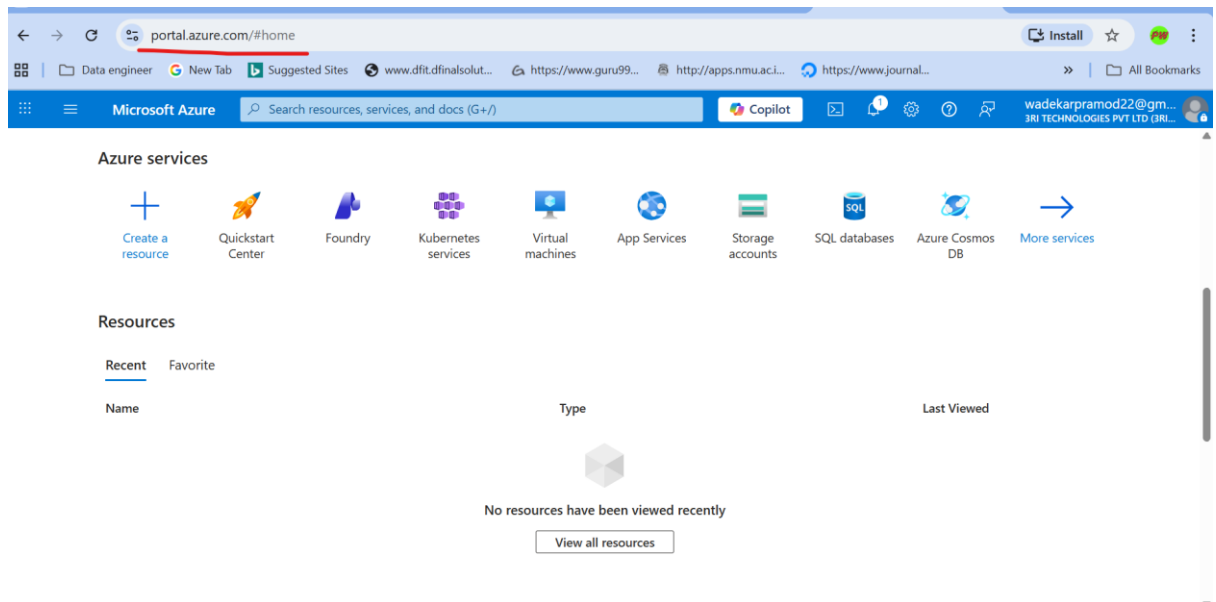


Create Free Azure Account:

1. Go to <https://azure.microsoft.com/en-us/pricing/purchase-options/azure-account>
2. Click on Try Azure for free after that click on next page create one link



3. After creating account go to chrome and search portal.azure.com



Resource Group:

✓ What is a Resource Group in Azure?

A **Resource Group (RG)** is a **logical container** in Azure that **holds related resources together** so they can be **managed, monitored, and controlled as a single unit**.

👉 Think of it as a **folder** for Azure resources.

◆ What kind of resources go inside a Resource Group?

- Virtual Machines (VM)
- Azure Data Factory
- Storage Accounts
- Azure SQL Database
- App Services
- Virtual Networks
- Key Vaults

First stage operation. If we can create any resource we need a resource group, you need a folder, resource group similar to a folder in the PC and we store the same files. Similar, resource group means a store a resource

Resource is service you can create and used in Azure. Eg, ADF, Azure Storage Account, Azure snapshot. These services are stored in the resource group.

Diff between Azure Data Lake and Blob Storage:

- If we can use **Blob storage** then we can only use container
- If we can use Azure Data Lake hierarchy name spaces, we can use folder within folders, we can store data in folders

Blob Storage Vs Data Lake in Azure

- Pre-requisite: [Azure Storage](#)

Azure Storage is a cloud-based storage solution provided by Microsoft Azure, which allows users to store and access data objects in the cloud. It offers a variety of storage options for different data types and scenarios, such as blobs, files, tables, and queues. It provides highly scalable, durable, and available storage services, which can be easily integrated with other Azure services and applications. Users can access Azure Storage through various methods, including the Azure Portal, Azure Storage Explorer, Azure PowerShell, Azure CLI, and the Azure Storage REST API.

Blob Storage

Blob Storage is a type of object-based cloud storage designed for unstructured or semi-structured data. Blobs are organized into containers, which are similar to folders in a file system, and can be accessed via REST APIs, client libraries, or Azure PowerShell and CLI. Blob storage offers several tiers of service to meet different performance and cost requirements, including Hot, Cool, and Archive tiers, which offer different levels of availability and access times. Additionally, Blob storage offers features such as versioning, lifecycle management, and Azure Data Lake Storage Gen2 integration.

Azure Blob Storage is a popular choice for many cloud-based applications and services that require scalable and reliable storage for unstructured data. It is optimized for storing and retrieving large files, such as images, videos, and backups, and provides access to the stored data through HTTP or HTTPS.

Features of Blob Storage

- It is an object storage service that allows you to store unstructured data as blobs. You can store different types of data such as text and binary data, images, videos, and other files.
- Data is replicated and stored in multiple locations to ensure high availability and data redundancy.

- It also provides a scalability feature, which means that you can store and access an unlimited amount of data without any constraints.
- It also contains security features such as encryption, role-based access control, and shared access signatures to ensure that your data is secure.
- It offers different access tiers, including hot, cool, and archive tiers, to help you manage the cost of storing data based on the frequency of access.

Characteristics of Blob Storage

- It is highly scalable and can store massive amounts of data.
- It is accessible via a REST API that can be accessed from anywhere.
- It provides high durability, availability, and reliability.
- It is cost-effective and provides different pricing tiers based on the frequency of access to the data.
- It provides different access tiers to help manage the cost of storing data based on the frequency of access.

Use Cases of Blob Storage

- It can be used to store and manage media files such as images, videos, and audio files.
- Used to store backup data for disaster recovery purposes.
- For storing application data such as logs, user data, and other files required by the application.
- It can be used as a data lake to store and process large amounts of unstructured data.

Data Lake in Azure

Data Lake is a hierarchical file system and a cloud-based data repository that provides scalable and secure storage for big data analytics workloads. It can store structured and unstructured data and it also uses a distributed file system to provide parallel access to data, allowing for faster processing of large datasets. It also integrates with a variety of big data processing frameworks and tools, such as Hadoop, Spark, and Azure Data Factory, enabling you to perform advanced analytics and machine learning on your data.

Azure Data Lake Storage is a powerful and flexible data repository that can help organizations extract insights from their big data with ease and efficiency.

Features of Data Lake

- It is basically designed to store and process large volumes of data in various formats.
- It uses a distributed file system to provide parallel access to data.
- It integrates with a variety of big data processing frameworks and tools enabling you to perform advanced analytics on your data.
- It can handle multiple access patterns, making it suitable for both batch and real-time processing workloads.
- It also provides strong security and compliance features, including role-based access control and encryption at rest.

Characteristics of Data Lake

- It is highly scalable, allowing organizations to store and process large volumes of data.
- It supports a wide range of data types and integrates with many big data processing frameworks and tools.
- It provides strong security and compliance features, making it suitable for organizations with strict data privacy and security requirements.
- It offers a cost-effective solution for storing and processing large volumes of data in the cloud.

Use Cases of Data Lake

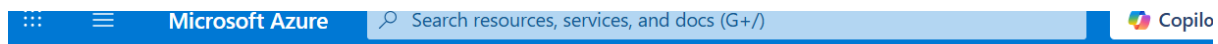
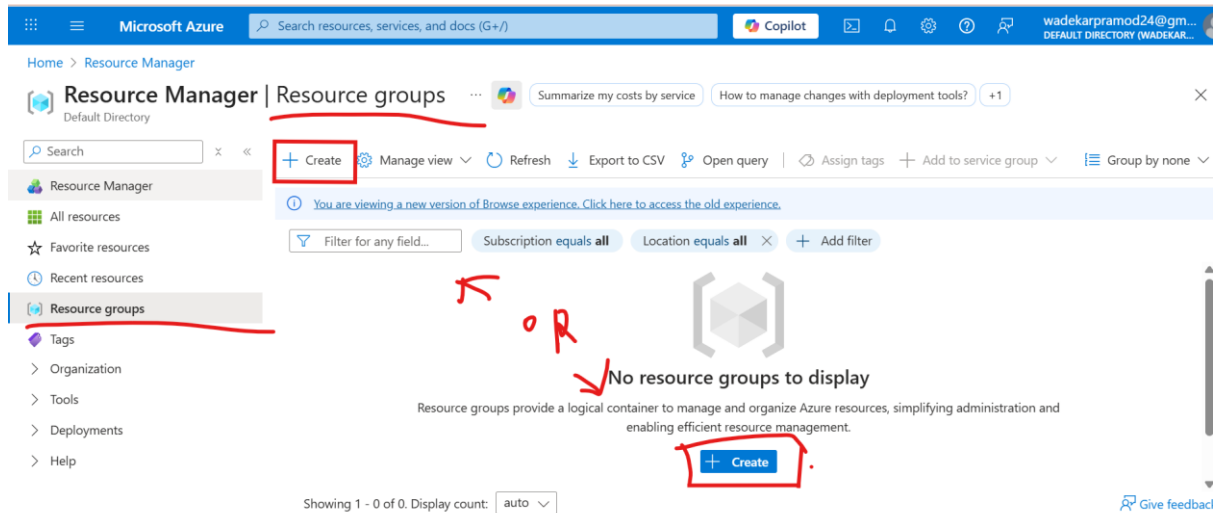
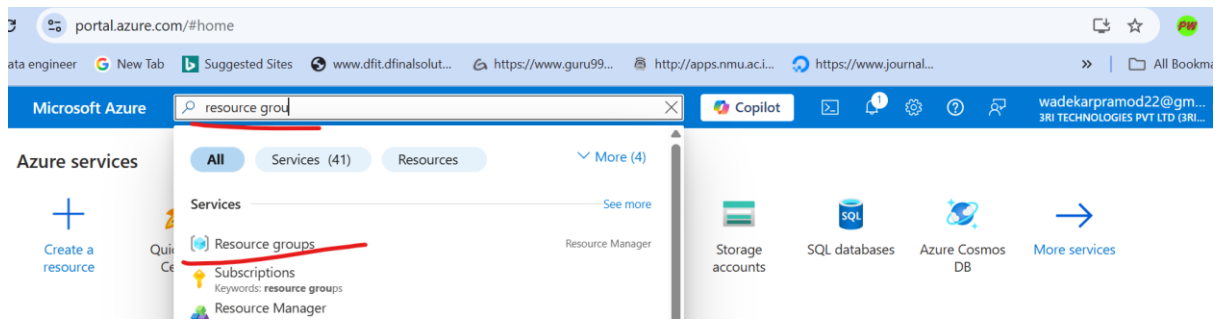
- It is ideal for storing and processing large volumes of data, making it a natural fit for big data analytics use cases.
- It can be used to store and process data for machine learning models, providing a scalable and secure repository for training and deployment.
- It can be used to store and process data generated by IoT devices, allowing organizations to analyze and gain insights from their IoT data.

Difference between Blob Storage and Data Lake in Azure

Factors	Blob Storage	Data Lake
Purpose	Blob Storage is designed for unstructured data storage	Data Lake is designed for big data analytics

Factors	Blob Storage	Data Lake
Data type	Blob Storage stores unstructured or semi-structured data	Data Lake can store both structured and unstructured data.
File size	Blob storage supports small to large file sizes, up to several terabytes per blob	Data Lake supports small to extremely large file sizes, up to several petabytes per file
Cost	Blob storage cost is lower.	Data Lake cost is higher due to advanced features and capabilities
Integrity	Blob Storage can be easily integrated with other Azure services	Data Lake is integrated with Azure services for big data analytics and machine learning
Security	Blob Storage offers security features such as encryption at rest and in transit	Data Lake also provides additional security features for big data processing and analysis
Accessibility	Blob Storage is accessible through HTTP or HTTPS	Data Lake can be accessed through various big data processing tools and technologies
Usecase	Blob Storage is used for storing and retrieving large files, such as images, videos, and backups	Data Lake is used for IoT, big data analytics, and machine learning purposes

Now Start Practicle step by step:



Home > Resource Manager | Resource groups >

Create a resource group

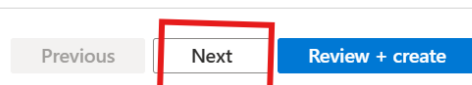
Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ Azure subscription 1

Resource group name * ⓘ RG-ADFCourse Add

Region * ⓘ (US) East US



Basics Tags Review + create

Apply tags to your Azure resources to logically organize them by categories. A tag consists of a key (name) and a value. Tag names are case-insensitive and tag values are case-sensitive. [Learn more](#)

Name	Value	Resource
	:	Resource group

out of scope now

Create a resource group ...

Basics Tags Review + create

[Automation Link](#)

Basics

Subscription	Azure subscription 1
Resource group name	RG-ADFCourse
Region	East US

Tags

None

After create refresh or click on Home -> Menu -> resource groups

Resource Manager | Resource groups ... Summarize my costs by service Export resource groups using Bicep or Terraform +1

Default Directory

Search

+ Create Manage view Refresh Export to CSV Open query Assign tags Add to service group

You are viewing a new version of Browse experience. Click here to access the old experience.

Filter for any field... Subscription equals all Location equals all Add filter

Name ↑	Subscription	Location
RG-ADFCourse	Azure subscrip...	East US

click

Create Storage account

RG-ADFCourse Resource group Help me generate Terraform for this resource group configuration. How to manage changes with deployment tools? +1

Search

+ Create Manage view Delete resource group Refresh Export to CSV Open query Group

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Essentials JS

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Home > Resource Manager | Resource groups > RG-ADFCourse > Marketplace

Get Started

Service Providers

Search with AI

Management

Private Marketplace

Private Offer Management

My Marketplace

Favorites

My solutions

Recently viewed

storage account search

Azure benefit eligible only Azure services only

Pricing: All Operating System: All Publisher Type: All Product Type: All Publisher name: All

New! Get AI-generated suggestions for 'storage account' View suggestions

Showing 1 to 20 of 177 results for 'storage account'. Clear search

Tile view

Storage account

Microsoft Azure Service

Storage Account Using ARM

DIGISTORM LTD. Azure Application

Storage Account Using ARM

VIRTUCLLOUD LTD. Azure Application

Storage task - Azure Storage Actions

Microsoft Azure Service

click

Home > Resource Manager | Resource groups > RG-ADFCourse > Marketplace > Storage account

Storage account Microsoft

Storage account Add to Favorites

Microsoft | Azure Service

★ 4.3 (1884 ratings)

Azure benefit eligible

Subscription Azure subscription 1 Plan Storage account Create

Fill The Following info carefully:

Microsoft Azure

Home > Resource Manager | Resource groups > RG-ADFCourse > Marketplace > Storage account >

Create a storage account

Basics | Advanced | Networking | Data protection | Encryption | Tags | Review + create

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below. [Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *

Resource group * [Create new](#)

Instance details

Storage account name * **unique**

Region * [Deploy to an Azure Extended Zone](#)

Preferred storage type ✓

Primary workload ✓

Performance * ☒ Standard: Recommended for most scenarios (general-purpose v2 account) ✓
☐ Premium: Recommended for scenarios that require low latency.

Redundancy * ✓

Need to search following concept:

Redendency:

[Deploy to an Azure Extended Zone](#)

Locally-redundant storage (LRS):
Lowest-cost option with basic protection against server rack and drive failures. Recommended for non-critical scenarios.

Geo-redundant storage (GRS):
Intermediate option with failover capabilities in a secondary region. Recommended for backup scenarios.

Zone-redundant storage (ZRS):
Intermediate option with protection against datacenter-level failures. Recommended for high availability scenarios.

Geo-zone-redundant storage (GZRS):
Optimal data protection solution that includes the offerings of both GRS and ZRS. Recommended for critical data scenarios.

Locally-redundant storage (LRS)

After that Click On next

Create a storage account ...

Basics Advanced Networking Data protection Encryption Tags Review + create

Security

Configure security settings that impact your storage account.

Require secure transfer for REST API operations ⓘ ☒

Allow enabling anonymous access on individual containers ⓘ ☐

Enable storage account key access ⓘ ☒

Default to Microsoft Entra authorization in the Azure portal ⓘ ☐

Minimum TLS version ⓘ

Permitted scope for copy operations (preview) ⓘ

Hierarchical Namespace

Hierarchical namespace, complemented by Data Lake Storage Gen2 endpoint, enables file and directory semantics, accelerates big data analytics workloads, and enables access control lists (ACLs) [Learn more](#) ↗

Enable hierarchical namespace ⓘ ☒

Access protocols

Blob and Data Lake Gen2 endpoints are provisioned by default [Learn more](#) ↗

Enable SFTP ⓘ ☐

Previous

Next

Review + create

Advanced Networking Data protection Encryption Tags Review + create

Click Next next ———

Click next button networking to review + create

Create a storage account ...

Basics Advanced Networking Data protection Encryption Tags Review + create

[View automation template](#)

Basics

Subscription	Azure subscription 1
Resource group	RG-ADFCourse
Location	East US
Storage account name	pwdatastorageaccount
Preferred storage type	Azure Blob Storage or Azure Data Lake Storage Gen 2
Primary workload	
Performance	Standard
Replication	Locally-redundant storage (LRS)

Advanced

Enable hierarchical namespace	Enabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Enable large file shares	Enabled

Security

Secure transfer	Enabled
-----------------	---------

[Previous](#) [Next](#) **Create**

Now storage account is created -> click on home or go to resource button

The screenshot shows the Azure Resource Manager console. On the left, the 'Resource groups' list is visible, with 'RG-ADFCourse' selected. A red arrow points from this selection to the main content area. In the main content area, the 'RG-ADFCourse' resource group is displayed. Under the 'Resources' tab, a table lists the resources within the group. The resource 'pwdatastorageaccount' is highlighted with a red box. A red arrow points from the 'RG-ADFCourse' resource group header to this resource.

Name	Type	Location
pwdatastorageaccount	Storage account	East US

Now in storage account -> create Azure Data factory

Search Create Manage view Delete resource group Refresh Export to CSV Open query

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Cost Management
- Monitoring

Essentials

Resources Recommendations

Filter for any field...	Type equals all	Location equals all	+ Add filter
☐	Name ↑		Type
☐	pwdatastorageaccount		Storage account

search azure data factory

Pricing: All Operating System: All Publisher Type: All Product Type: All Publisher name: All

☐ Azure benefit eligible only ☐ Azure services only

New! Get AI-generated suggestions for 'azure data factory' View suggestions

Showing 1 to 20 of 95 results for 'azure data factory' Clear search

Data Factory Monitoring Dashboard
In2Intel
Azure Application
A lightweight web-based dashboard for realtime monitoring and analytics of your Azure Data Factories

Data Factory
Microsoft
Azure Service
Hybrid data integration service that simplifies ETL pipeline

Factory Operations Agent in Azure AI (preview)
Microsoft
Azure Service
Bring your manufacturing data into the era of AI with the Factory Operations Agent in Azure AI in

FulkrumCloud: Automated Azure Data Factory & Azure
Bitwise Inc - US
SaaS
FulkrumCloud automates the migration of ADF & Synapse data pipelines & dataflows to Microsoft

Modern Data Mart
Ceteris AG
Azure Application
Small Data-Warehouse Architecture integrating a single source using Azure Data Factory

Sight Machine Factory Namespace MCP
Sight Machine, Inc.
SaaS
Provides MCP Server support for the Factory Namespace Agent, built on our Factory Namespace SI M

Data Factory Add to Favorites
Microsoft | Azure Service

★ 3.7 (612 ratings)

Azure benefit eligible

Subscription

Azure subscription 1

Plan

Data Factory

Create

Home > Resource Manager | Resource groups > RG-ADFCourse > Marketplace > Data Factory >

Create Data Factory ...

Basics Git configuration Networking Advanced Tags Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ ▼

Resource group * ⓘ ▼

[Create new](#)

Instance details

Name * ⓘ ✓ *unique*

Region * ⓘ ▼

Version * ⓘ ▼

i We recommend you **upgrade to Fabric Data Factory** for a modern, unified data integration experience. [Try Fabric instead!](#)

Basics Git configuration Networking Advanced Tags Review + create

All next and click on create

Home > Resource Manager | Resource groups > RG-ADFCourse > Marketplace > Data Factory >

Create Data Factory ...

Basics Git configuration Networking Advanced Tags Review + create

[View automation template](#)

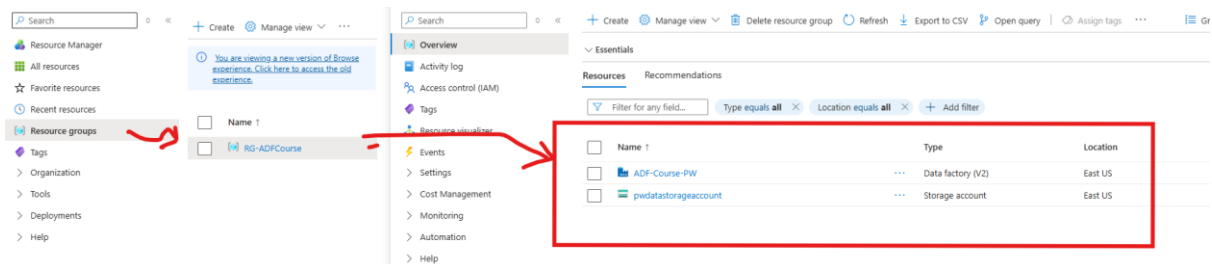
Basics

Subscription Azure subscription 1
Resource group RG-ADFCourse
Name ADF-Course-PW
Region East US
Version V2

Networking

Connect via Public endpoint

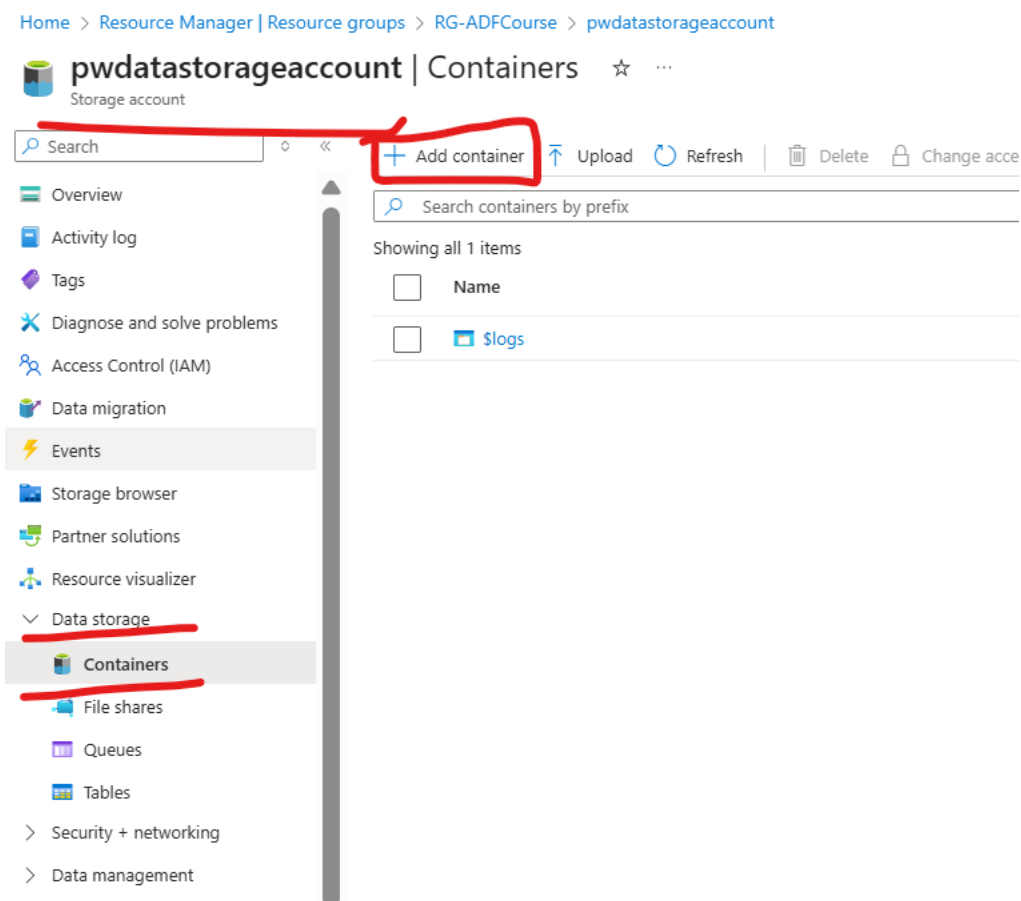
Go to home -> resource group ->

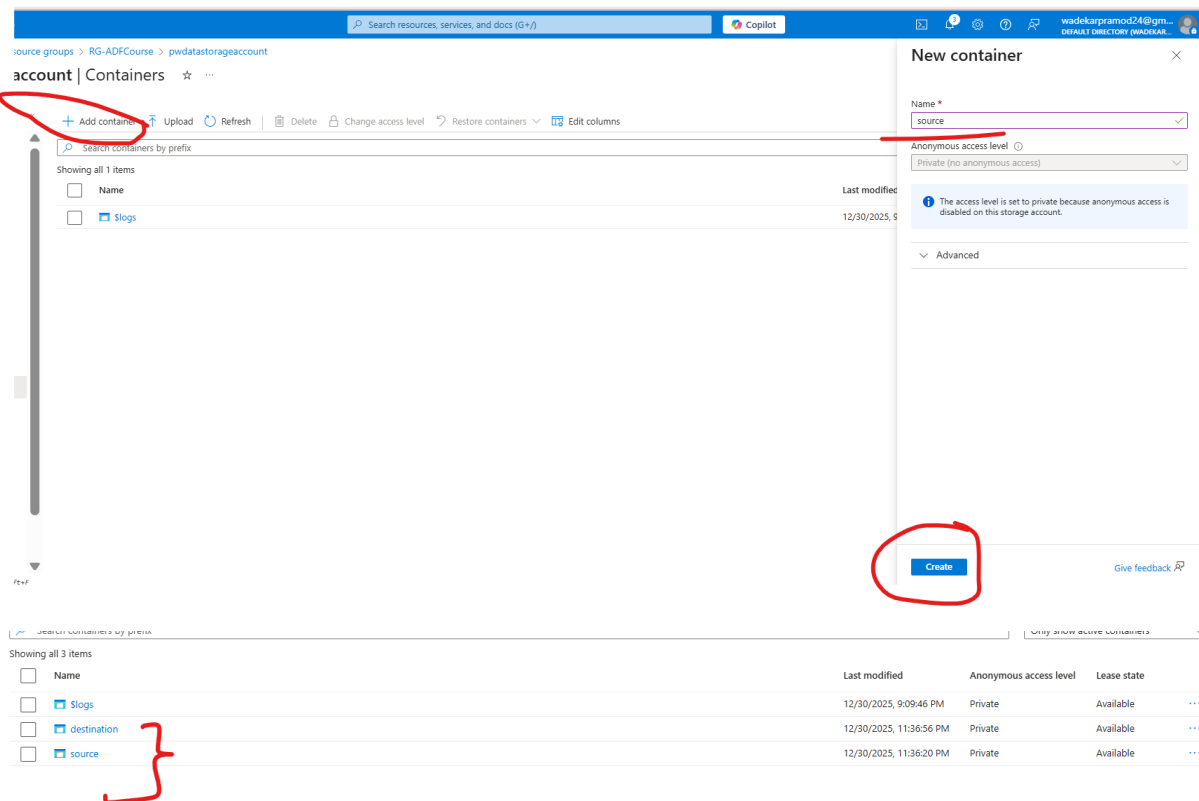


Linked services and dataset: (IMP) **Add notes def and discription:**

Now go to storage account and Data Storage -> container in left menu

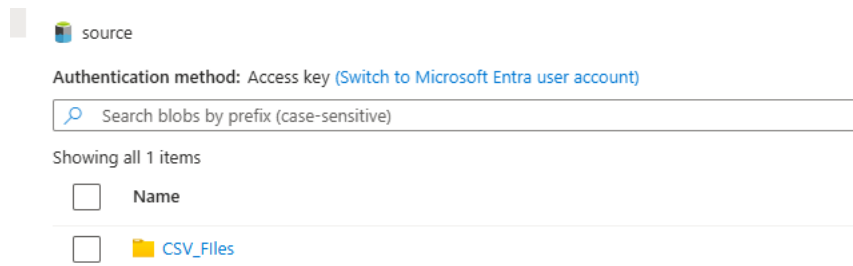
Now create contener click on create:



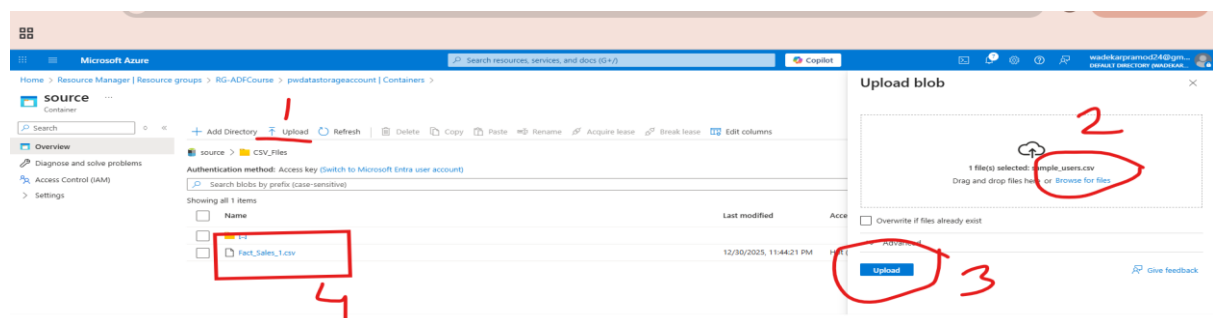


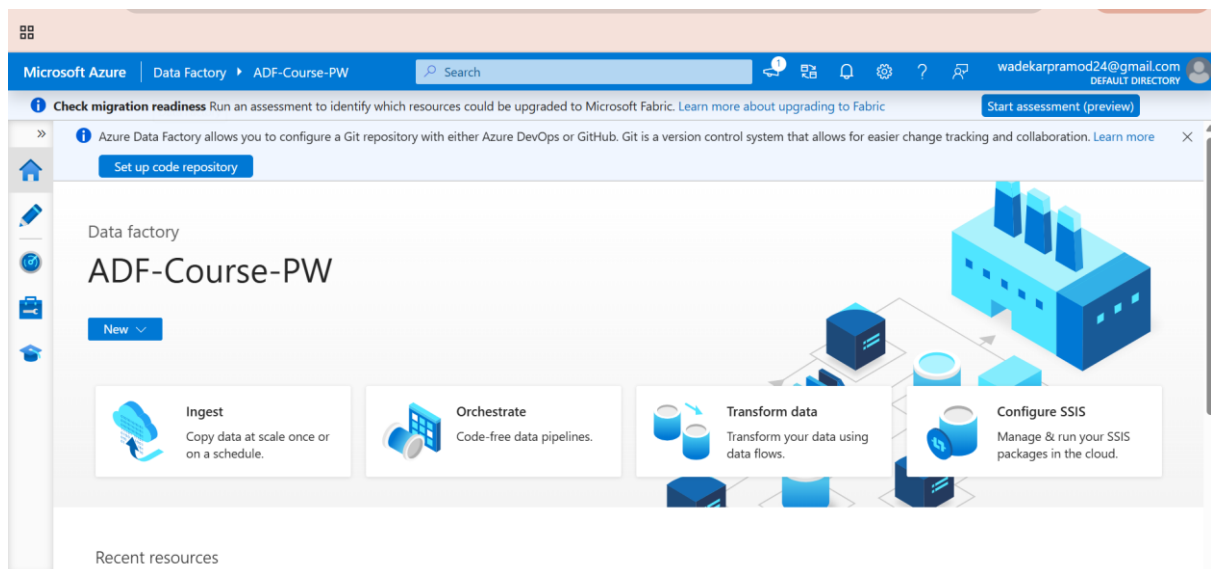
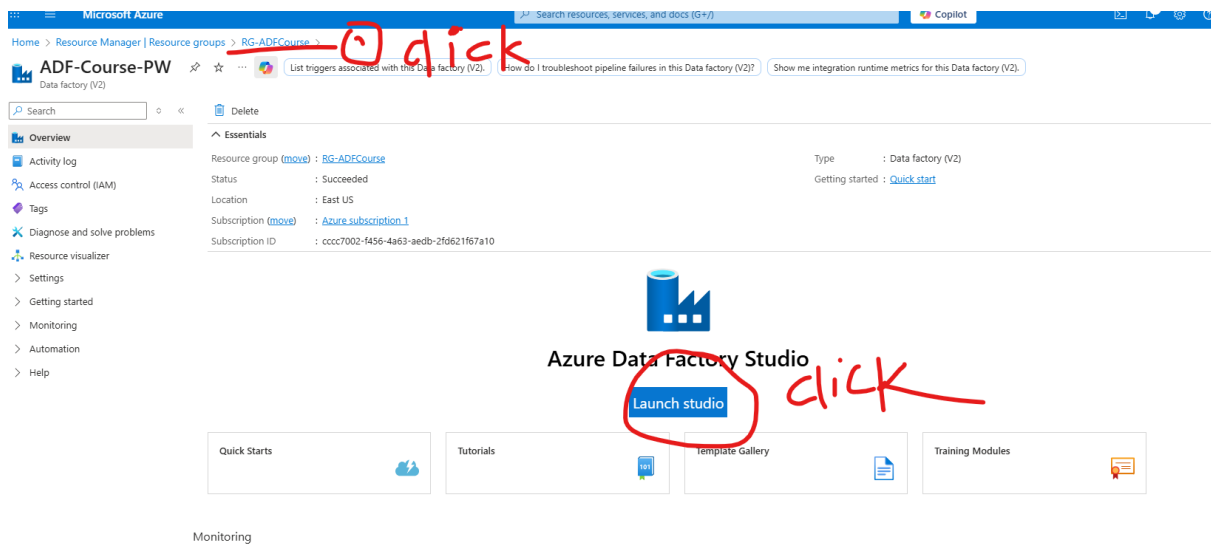
Now we learn add data / migrate data :

Now click on source -> add one direactory name is CSV_Files under this directory add data

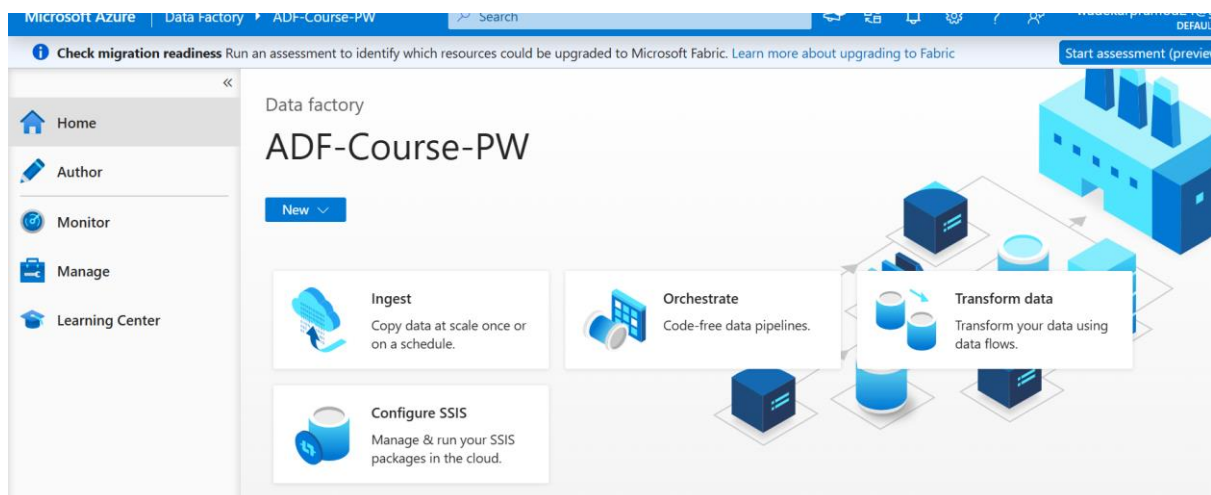


Now open this directory click on upload and add csv file in our browser done. This is simple method





This is the Azure data factory (ADF) Dashboard



Author Tab: this is your go to tab main used creating pipelines here and perform multiple activities in ADF.

Monitor tab: After you creating pipeline and perform activity and monitor them, scheduling task.

Manage Tab: create linked services and datasets manager create all this. If manager devops, github, git configuration all this services manage in this tab.

Create Linked services:

✓ What is Linked Service in Azure Data Factory (ADF)?

A **Linked Service** in ADF is a **connection configuration** that defines **how ADF connects to external resources** like databases, storage, or compute services.

👉 Simply:

Linked Service = Connection String in ADF

◆ Why are Linked Services used?

1 To Connect to Data Sources & Destinations

ADF uses Linked Services to connect to:

- Databases (Azure SQL, SQL Server, MySQL, Oracle)
- Storage (ADLS, Blob, S3)
- Compute (Databricks, Synapse)
- APIs (REST, HTTP)

Without Linked Service → **ADF cannot access data**

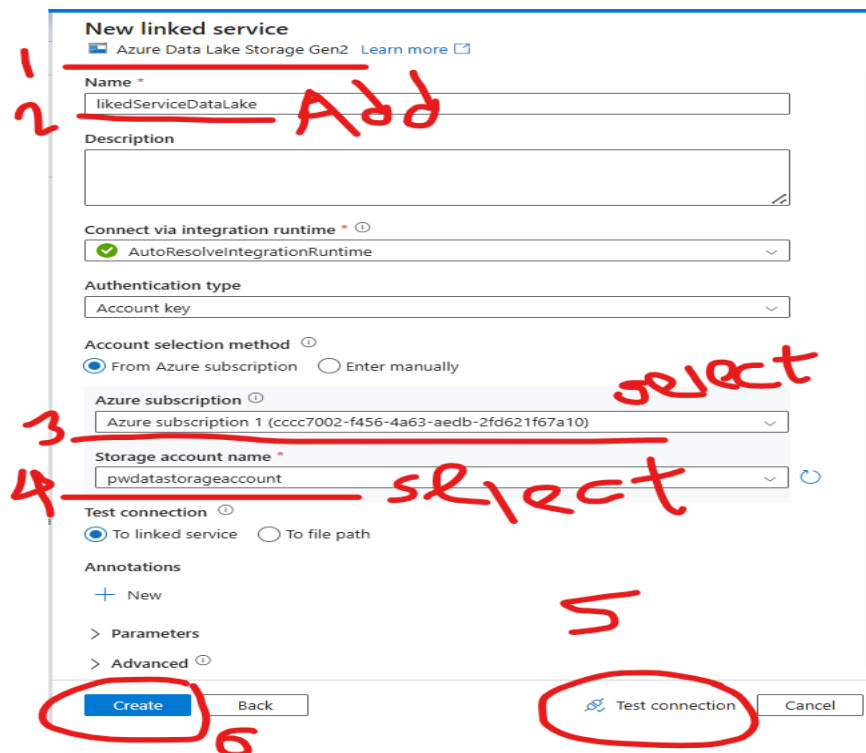
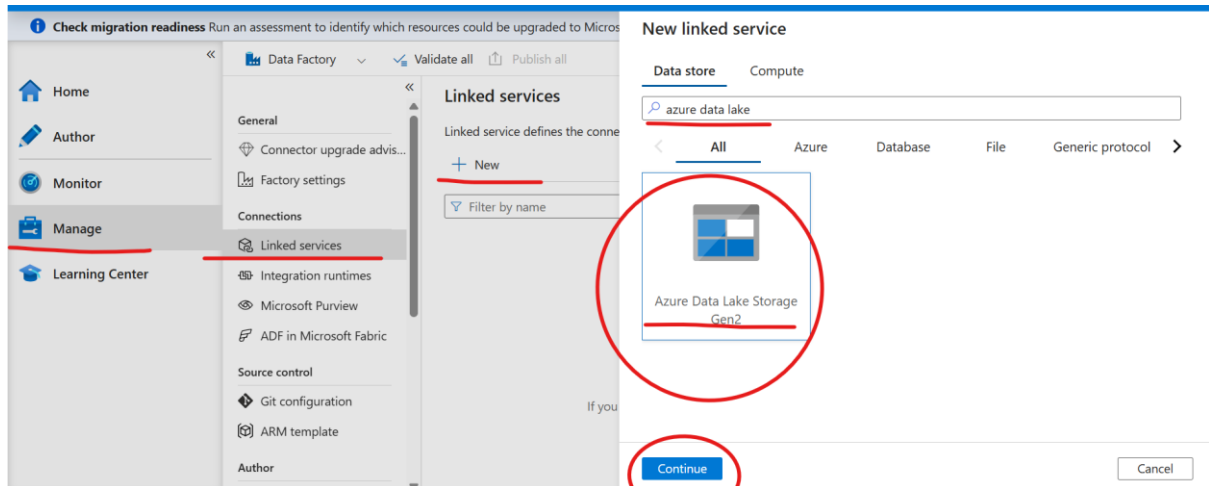
◆ Example (Simple)

👉 Scenario: Copy data from **Azure SQL DB** to **ADLS**

You need:

1. Linked Service – Azure SQL (source)
2. Linked Service – ADLS (sink)
3. Dataset – SQL table
4. Dataset – ADLS file
5. Copy Activity

Click on manage tab -> click on new -> search azure data lake -> select





Data factory connected to storage account using this linked service :

[ADF-Course-PW](#) -> connected to -> [pwdatastorageaccount](#) using [likedServiceDataLake](#)

Now all setup is done. Now we can perform activities and perform the operation, create pipeline.

1. COPY Activity:

- **Now we are copy this data in data lake to data lake**

What is Copy Activity in Azure Data Factory (ADF)?

Copy Activity is one of the core activities in Azure Data Factory used to **move (copy) data from a source to a destination.**

Examples:

- SQL Server → Azure Data Lake
- MySQL → Azure SQL Database
- Blob Storage → Data Warehouse
- On-premises DB → Azure Cloud
- CSV / JSON / Parquet → Table

✓ Key Components of Copy Activity

Component	Description
Source	Where data comes from
Sink	Where data goes
Linked Service	Connection details
Dataset	Structure of data

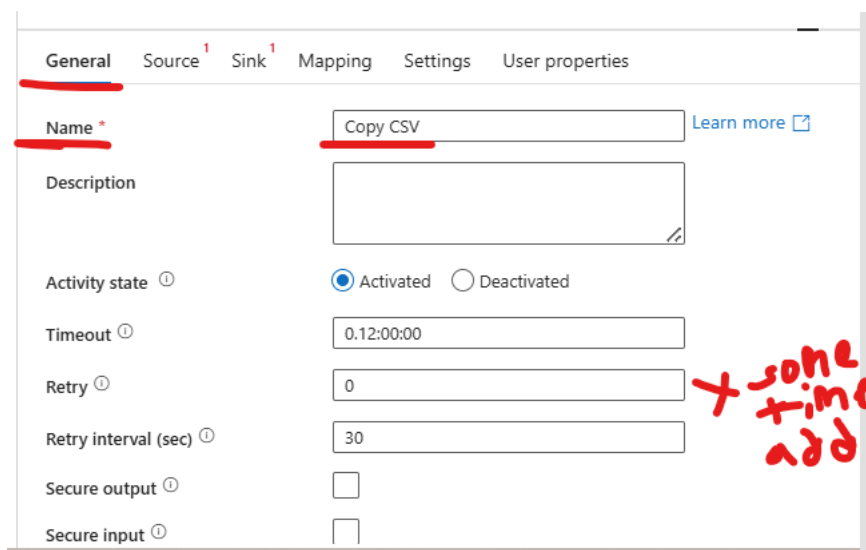
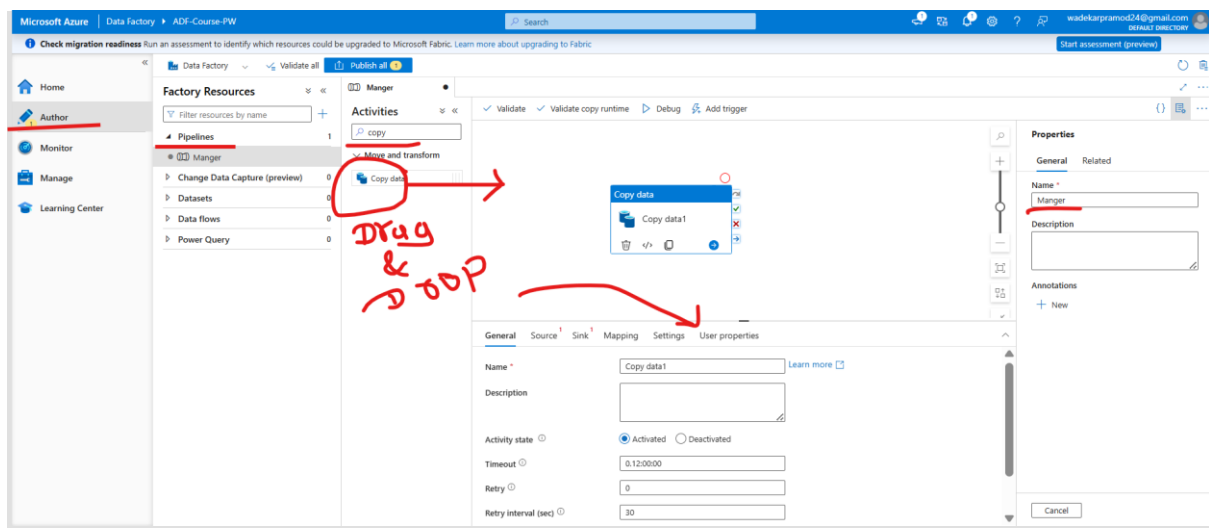
Component	Description
Mapping	Column-to-column mapping
Performance	Parallel copy, batch size

Copy data one page to another:

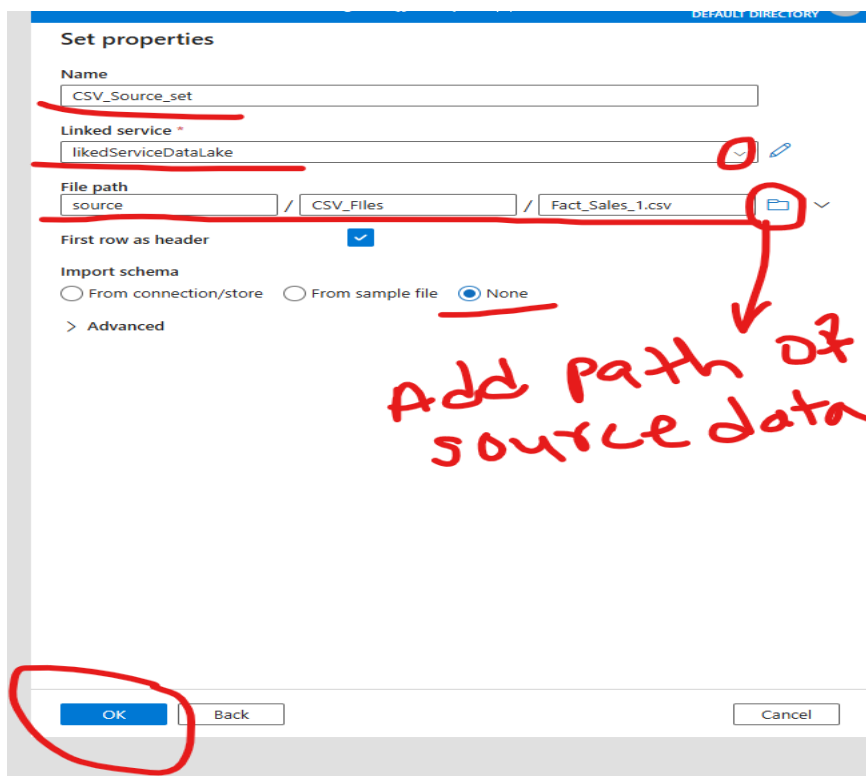
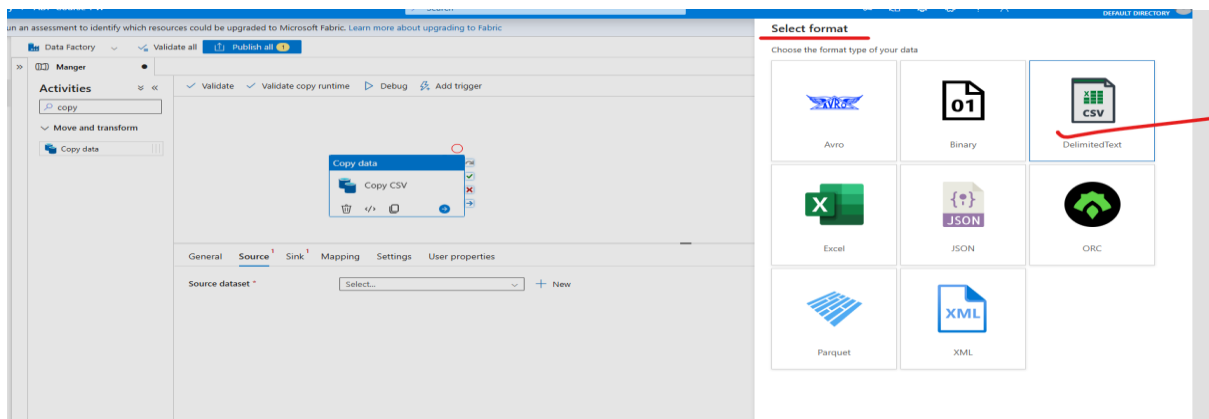
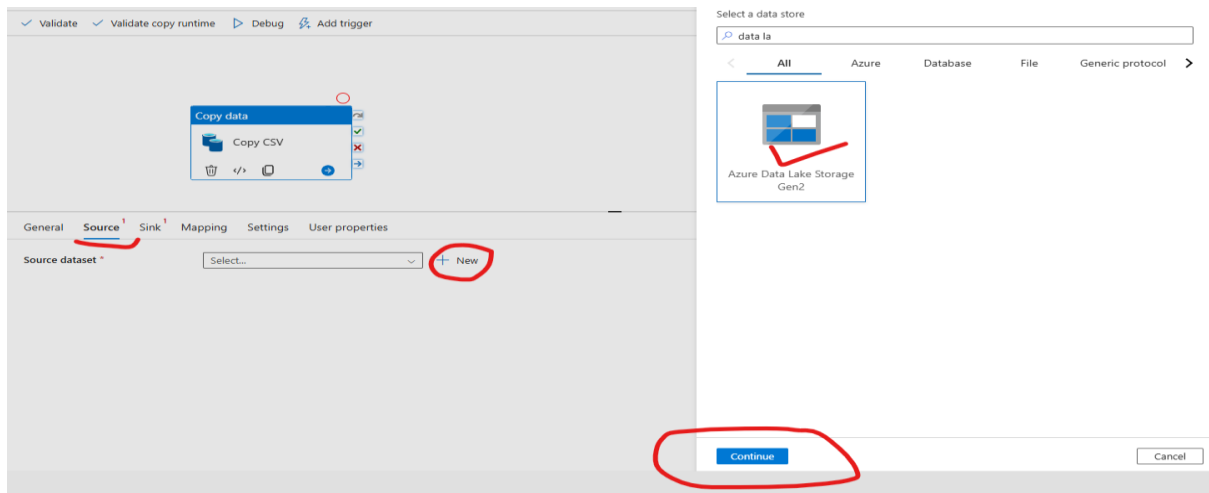
One page means in the github, browser, one directory to another

Now perform the COPY Activity:

Go to the Author tab -> click on the pipeline1 three dots ... -> click on new pipeline



General Tab is done



Source tab is done

In Sink Tab we add to destination path not a same path of source

The screenshot shows the 'Copy data' task configuration in the 'Sink' tab. The 'Sink dataset' dropdown is set to 'Select...', and the 'New' button is circled in red. On the right, the 'Select a data store' panel shows 'data lake' selected, with 'Azure Data Lake Storage Gen2' highlighted by a red checkmark. The 'Continue' button at the bottom right is also circled in red.

Copy data

Copy CSV

General Source Sink Mapping Settings User properties

Sink dataset * Select... New

Select a data store

data lake

All Azure Database File Generic protocol

Azure Data Lake Storage Gen2

Continue Cancel

The 'Select format' dialog box is shown, with the instruction 'Choose the format type of your data'. Three formats are visible: 'Avro', 'Binary', and 'DelimitedText' (which is selected and highlighted with a red checkmark). Below these, there are three partially visible options: 'JSON', 'Parquet', and 'Text'. A red arrow points to the 'Continue' button at the bottom.

Select format

Choose the format type of your data

Avro Binary DelimitedText

JSON Parquet Text

click Continue

✓ Validate Cancel options Add trigger

Copy data
Copy CSV

Parameters Variables Settings **Output**

Pipeline run ID 551bc7fd-4abc-42e5-9dbd-49f99cb0754e Pipeline status In progress

All status

Showing 1 - 1 of 1 items

Activity name	Activity stat...	Activity ...	Run start	Duration	Integration runtime	User proper...	Activity run ID	Log
Copy CSV	Queued	Copy data	12/31/2025, 1:24:10 AM	2s			c20693f6-2c84-4876-bfb6-42cae9412d28	

✓ Validate Debug Add trigger

Copy data
Copy CSV

Parameters Variables Settings **Output**

Pipeline run ID 551bc7fd-4abc-42e5-9dbd-49f99cb0754e Pipeline status Succeeded

All status

Showing 1 - 1 of 1 items

Activity name	Activity stat...	Activity ...	Run start	Duration	Integration runtime	User proper...	Activity run ID	Log
Copy CSV	Succeeded	Copy data	12/31/2025, 1:24:10 AM	16s	AutoResolveIntegrationRuntime (East US)		c20693f6-2c84-4876-bfb6-42cae9412d28	

Microsoft Azure

Home > RG-ADFCourse > pwdatastorageaccount | Containers

destination

Search

Overview

destination > CSV_FILES_NEW

Authentication method: Access key (switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

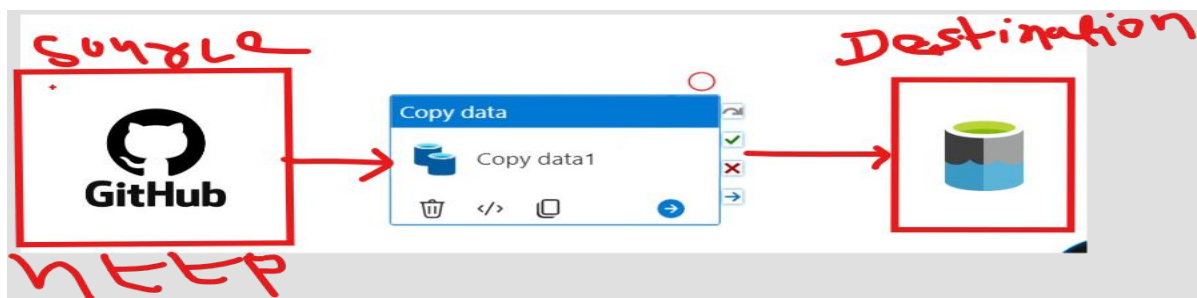
Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
Fact_Sales_1.csv	12/31/2025, 1:24:24 AM	Hot (inferred)	Block blob	319.93 KiB	Available

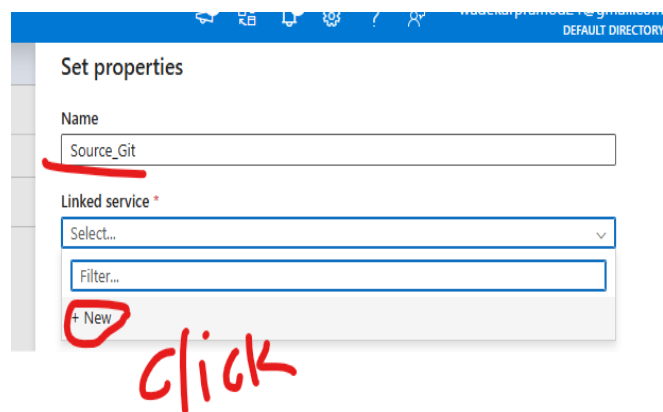
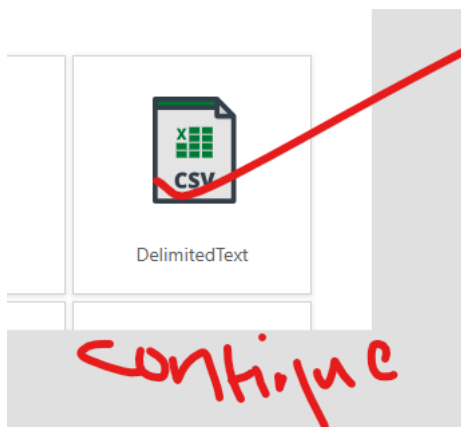
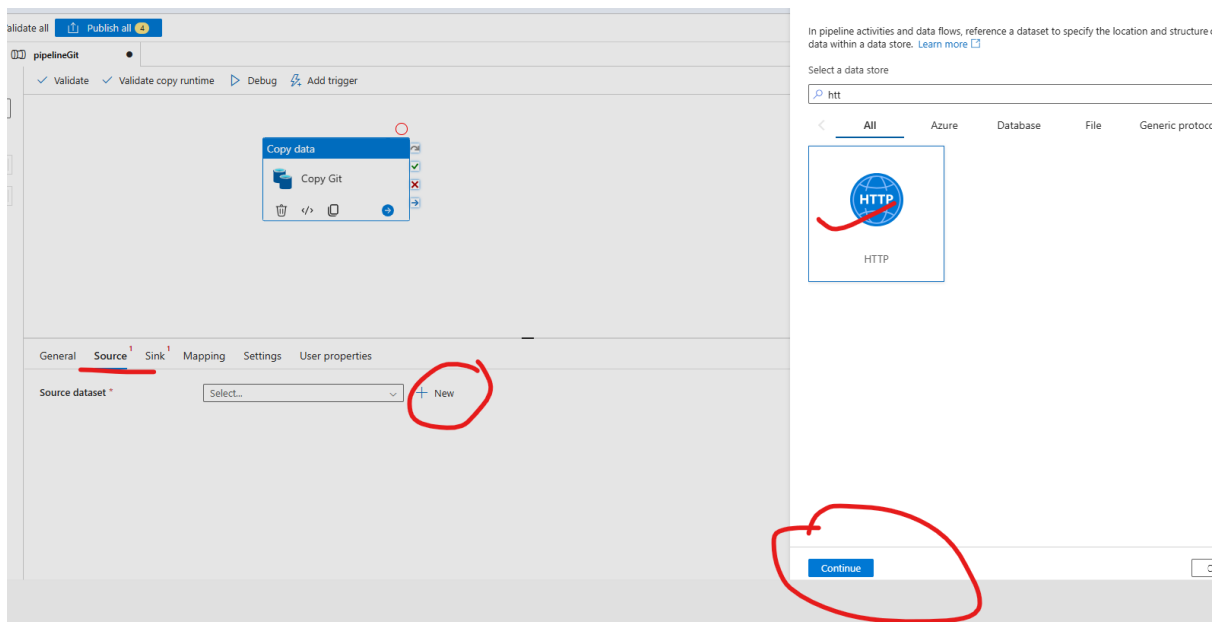
Directory and file created automatically in destination

Now we are copy this data in data lake to data lake its done

Now we can copy data from http connection to data lake



Same step to add new pipeline -> search copy -> drag and drop -> fill same general tab add name click next source tab



How can you find the github link: go to github, open csv file and upper side you have showing row button click on that and copy url **base** and **Relative URL**

```
Preview Code Blame 41 lines (41 loc) · 2.8 KB
1 transaction_id,transactional_date,product_id,customer_id,payment,credit_card,loyalty_card,cost,quantity,price,
2 4411,5/1/2022 1:02,P0305,6,,30271790522719,F,8.74,3,9.99
3 4412,5/1/2022 4:00,P0242,7,visa,4041591026711540,F,1.60,2,1.79
4 4413,5/1/2022 5:49,P0529,2,mastercard,5048373491517664,F,17.59,2,18.79
5 4414,5/1/2022 6:58,P0336,7,mastercard,5048377130633352,T,8.17,4,8.89
6 4415,5/1/2022 8:47,P0399,6,visa,4041595611872,F,9.83,2,11.29
7 4416,5/1/2022 12:06,P0097,4,visa,4041591008610,F,9.91,4,11.29
8 4417,5/1/2022 19:42,P0644,7,mastercard,5048374238650248,T,11.48,2,12.59
```

Full URL: https://raw.githubusercontent.com/anshlambagit/Azure-Data-Factory/refs/heads/main/Raw%20Data/Fact_Sales_2.csv

Base URL: <https://raw.githubusercontent.com>

Relative URL: anshlambagit/Azure-Data-Factory/refs/heads/main/Raw%20Data/Fact_Sales_2.csv

New linked service

HTTP Learn more

Name *
Linked_git

Description

Connect via integration runtime *
AutoResolveIntegrationRuntime

Base URL *
https://raw.githubusercontent.com
Information will be sent to the URL specified. Please ensure you trust the URL entered.

Server certificate validation
Enable Disable

Authentication type *
Anonymous

Auth headers
+ New

Annotations
+ New

> Parameters
> Advanced

Create Cancel Test connection

Set properties

Name
Source_Git

Linked service *
Linked_git

Relative URL
anshlambagit/Azure-Data-Factory/refs/heads/main/Raw%20Data/Fact_Sales_2.csv

First row as header
[checked]

Import schema
From connection/store From sample file None

> Advanced

OK Back Cancel

Done source is added and source tab is done

Now same as sink tab like we can perform last time:

Click on new -> sersch data lake -> add desination and directory name -> ok -> click on debug and check its copied or not

The screenshot shows the Azure Data Factory pipeline editor. At the top, there are tabs for 'Validate', 'Debug', and 'Add trigger'. The main canvas displays a 'Copy data' activity with a 'Copy Git' icon. On the right, the 'Properties' pane is open, showing the 'General' tab with the 'Name' field set to 'pipelineGit'. Below the main canvas, the 'Output' tab is selected, showing the pipeline run ID '463c415d-2873-46fb-851d-93b3c5f40b12' and the pipeline status 'Succeeded'. A table below shows the activity details:

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
Copy Git	Succeeded	Copy data	12/31/2025, 2:02:59 AM	17s	AutoResolveIntegrationRuntime (East US)		65aa59a4-9870-4ae6-a7a5-85

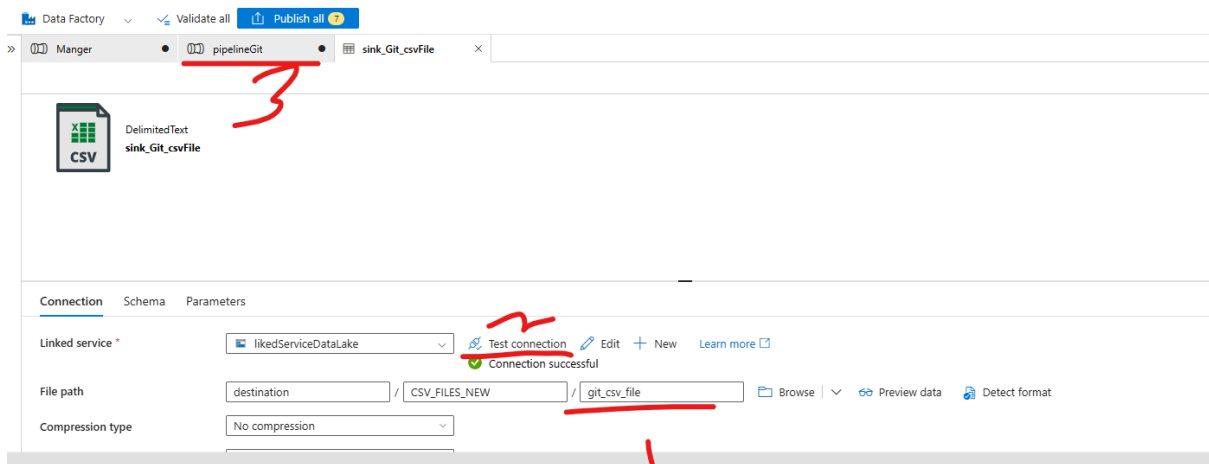
The screenshot shows the Azure Data Lake Storage Explorer. The breadcrumb navigation at the top indicates the path: 'destination > CSV_FILES_NEW > anshlambagit > Azure-Data-Factory > refs > heads > main > Raw Data'. The 'Authentication method' is set to 'Access key (Switch to Microsoft Entra user account)'. A search bar is present with the text 'Search blobs by prefix (case-sensitive)'. Below the search bar, a table shows the contents of the directory:

Name	Last modified	Access tier	Blob type	Size	Lease state
Fact_Sales_2.csv	12/31/2025, 2:03:14 AM	Hot (inferred)	Block blob	2.8 KiB	Available

Now created multiple folder if you don't want then you want add file name when we can add http relative path folder name also then not added multiple folder just try to following

Click on pipeline -> sink tab -> click on open -> add file name ->

The screenshot shows the Azure Data Factory pipeline editor with the 'Sink' tab selected. The 'Sink dataset' field is set to 'sink_Git_csvFile'. The 'Copy behavior' is set to 'Select...'. The 'Max concurrent connections' field is empty. The 'Open' button next to the 'Sink dataset' field is highlighted with a red circle.



After 3 step click on debug

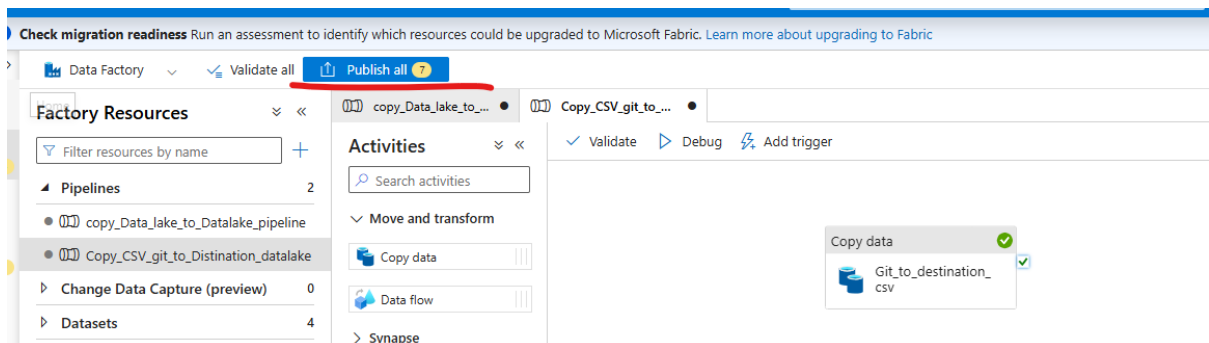
Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 3 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	[-]					
☐	anshlambagit	12/31/2025, 2:03:14 AM				
☐	Fact_Sales_1.csv	12/31/2025, 1:24:24 AM	Hot (inferred)	Block blob	319.93 KiB	Available
☐	git_csv_file	12/31/2025, 2:09:27 AM	Hot (inferred)	Block blob	2.8 KiB	Available

Now we want to save our work then click on publish all button on upper side



IMPORTANT MAIN OPERATIONS:

Now we perform some operation like if we have multiple file in same folder and some files for data science team and some files for reporting team but condition is data science team file not showing to reporting team and reporting team files not showing to data science team.

Whats the solution?

Req1: we need to create new container name is reporting for power BI or Data science team, But we creating multiple files in same folder but according to data security and data governance data science team file not showing to reporting team and reporting team files not showing to data science team. Then now we can achive this using by **GET METADATA ACTIVITY**

GET METADATA ACTIVITY:

Get Metadata activity fetches **structural and system information** about a data source such as a file, folder, table, or database object.

It tells ADF things like:

- Does this file exist?
- How many files are in a folder?
- What columns are in a table?
- What is the file size or last modified date?

Metadata You Can Fetch

Depending on the dataset type, you can retrieve:

File / Folder

- `exists`
- `size`
- `lastModified`
- `childItems` (list of files/folders inside a directory)

Database Table

- `columnCount`
- `structure` (column names & data types)

Why Is It Used?

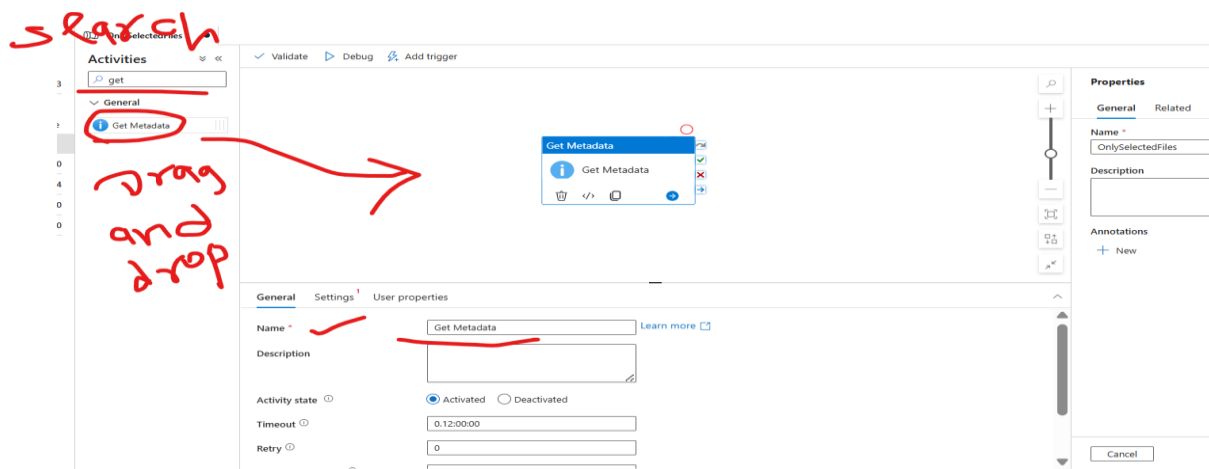
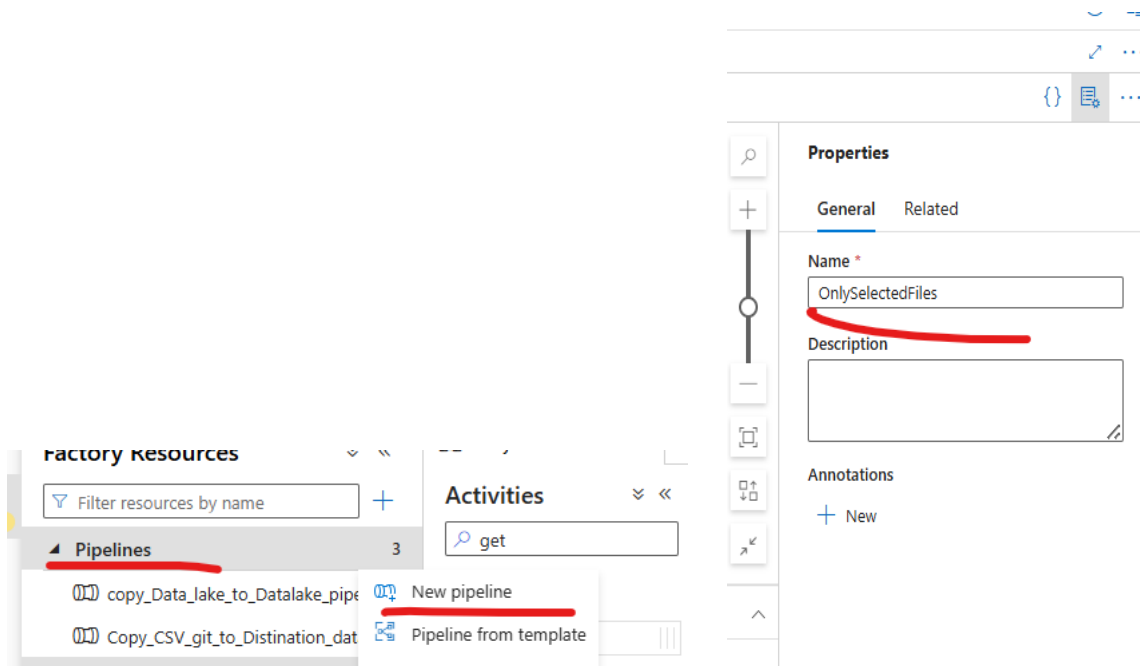
Get Metadata is mainly used for **control flow and dynamic pipelines**, not for data movement.

Key Reasons:

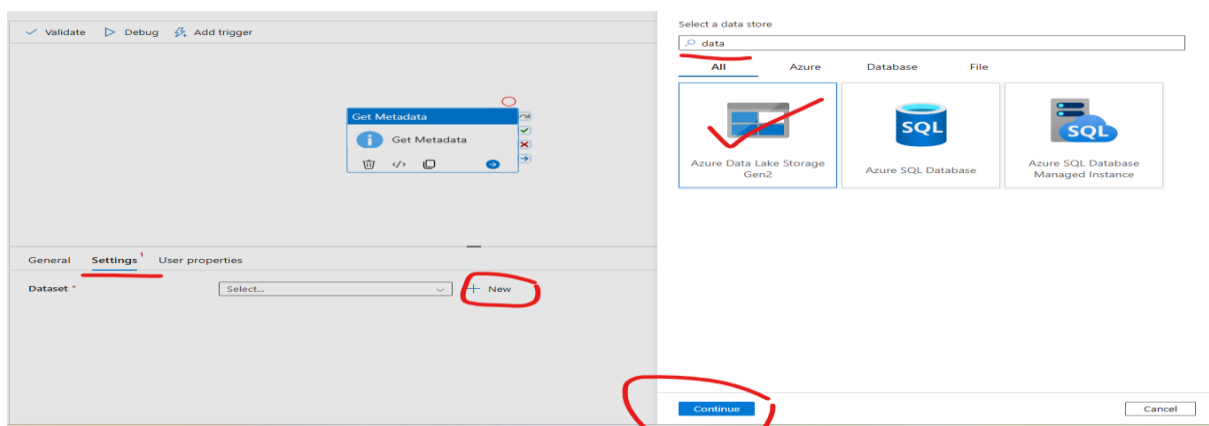
1. **Check file existence before processing**
2. **Loop through multiple files dynamically**
3. **Handle conditional logic (If Condition activity)**
4. **Validate data before copy or transformation**
5. **Avoid pipeline failures**

Create pipeline:

Click one three dots of pipeline -> click new pipeline



General is done



Select format

Choose the format type of your data

 Avro	 Binary	 DelimitedText
 Excel	 JSON	 ORC
 Parquet	 XML	

Continue Back

Cancel

Set properties

Name

CSV_Meta_Data

Linked service *

likedServiceDataLake

File path

destination

/ CSV_FILES_NEW

/ File name

First row as header



Import schema



From connection/store



From sample file



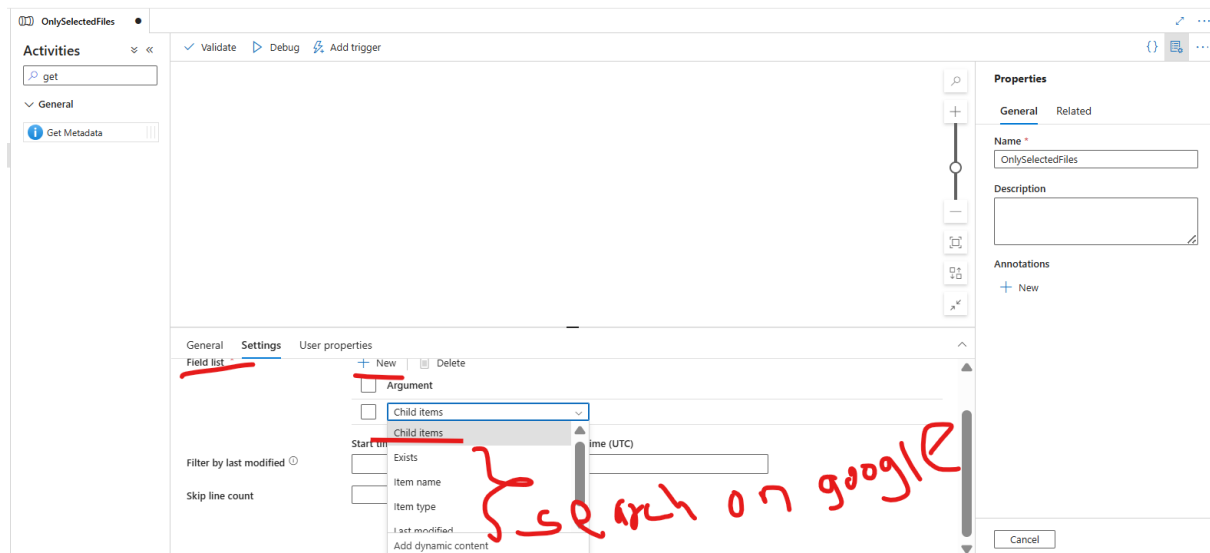
None

> Advanced

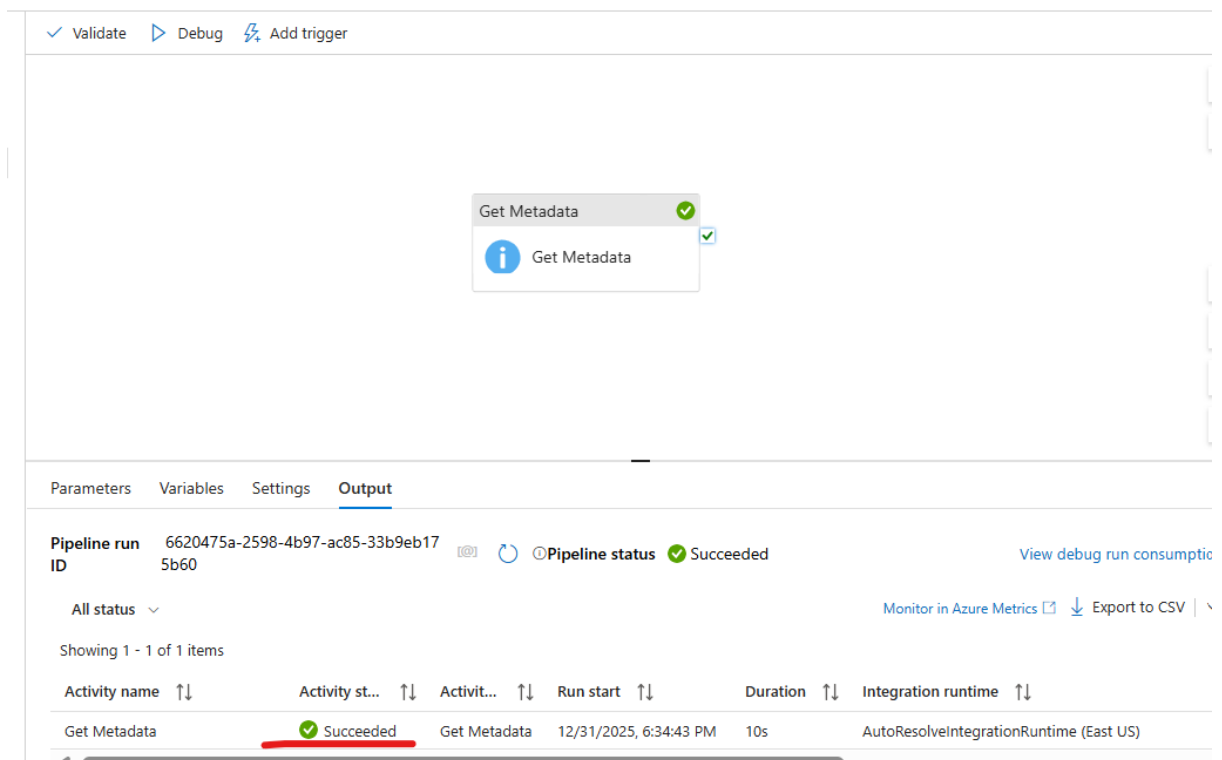
OK

Back

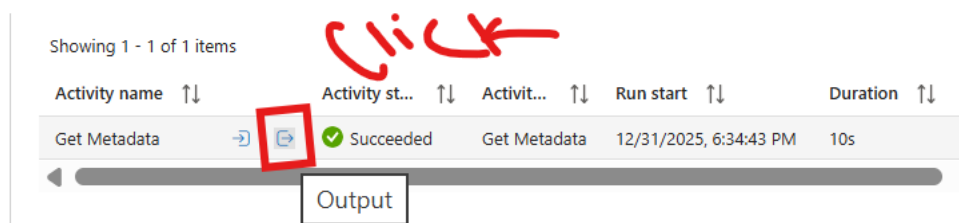
Cancel



Setting Tab is done. Click on debug



Now for output follows following conditions:



Output

Copy to clipboard

```
{
  "childItems": [
    {
      "name": "Fact_Sales_1.csv",
      "type": "File"
    },
    {
      "name": "Fact_Sales_2.csv",
      "type": "File"
    }
  ]
}
```

ⓘ Pipeline status ✔ Succeeded [View debug run cons](#)

[Monitor in Azure Metrics](#) [Export to C](#)

	Run start	Duration	Integration runtime
Get Metadata	12/31/2025, 6:34:43 PM	10s	AutoResolveIntegrationRuntime (East US)

Output

Copy to clipboard

```
{
  "childItems": [
    {
      "name": "Fact_Sales_1.csv",
      "type": "File"
    },
    {
      "name": "Fact_Sales_2.csv",
      "type": "File"
    }
  ],
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (East US)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 0
  },
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": [
      {
        "meterType": "AzureIR",
        "duration": 0.016666666666666666,
        "unit": "Hours"
      }
    ]
  }
}
```

Now we want to create in storage account new container the name is reporting

Storage | Containers

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix

Showing all 4 items

Name	Created
logs	12/30/2025, 9
destination	12/30/2025, 1
reporting	12/31/2025, 6
source	12/30/2025, 1

New container

Name

Anonymous access level

Private (no anonymous access)

Last modified

12/30/2025, 9

12/30/2025, 1

12/31/2025, 6

12/30/2025, 1

Advanced

create

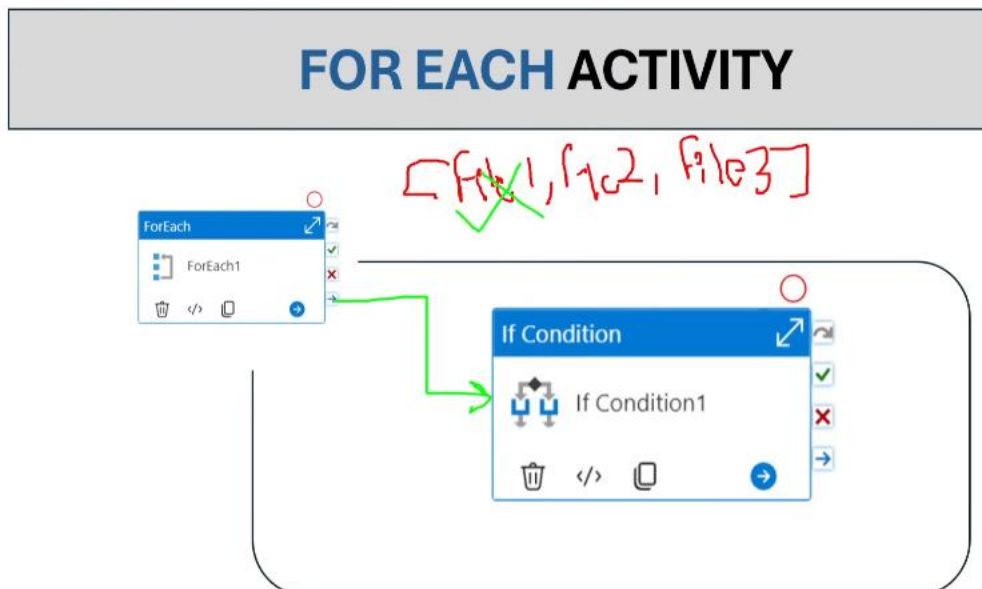
Now we can add in this folder file for reporting team and start name is fact means condition is the name is starting with fact this files are stored in reporting folder for this condition we need to add one pipeline the name is if condition activity

If Condition Activity:



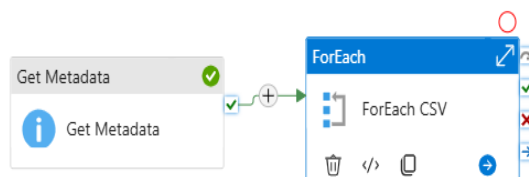
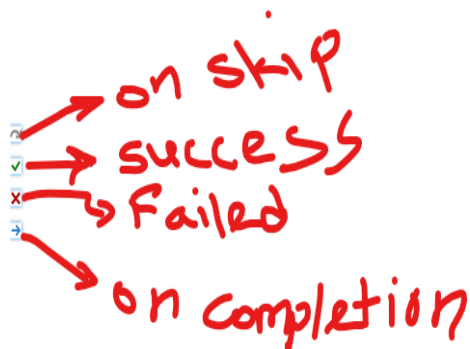
Now for multiple files or array of file in loop we need to add for each condition or activity

For Each Activity:

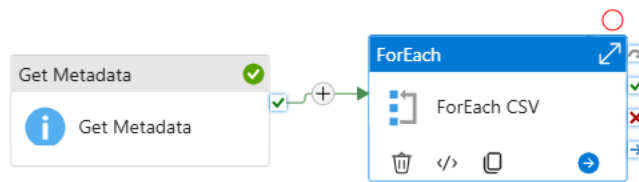


Now Create one more pipeline the name is For each Activity

If metadata is success the connect to the for each condition



✓ Validate ▶ Debug ⚡ Add trigger



General Settings¹ Activities (0) User properties

Name *

ForEach CSV

[Learn more](#)

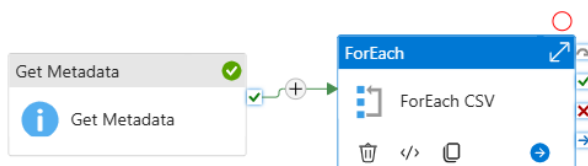
Description

Activity state ⓘ

☒ Activated ☐ Deactivated

General is done.

✓ Validate ▶ Debug ⚡ Add trigger



General Settings¹ Activities (0) User properties

Sequential



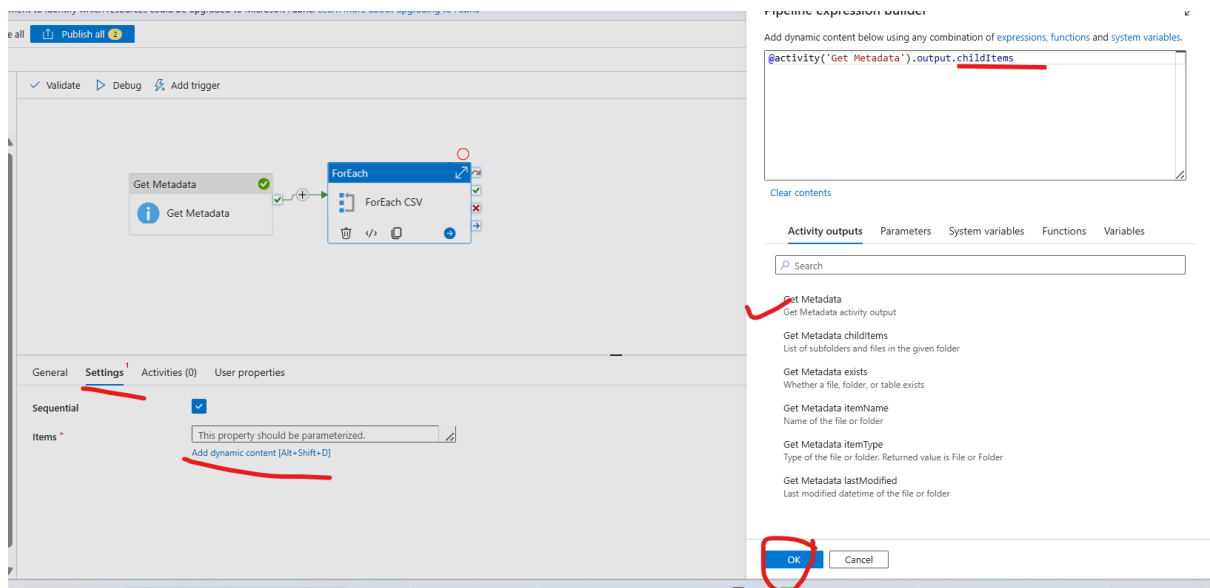
Items *

This property should be parameterized.

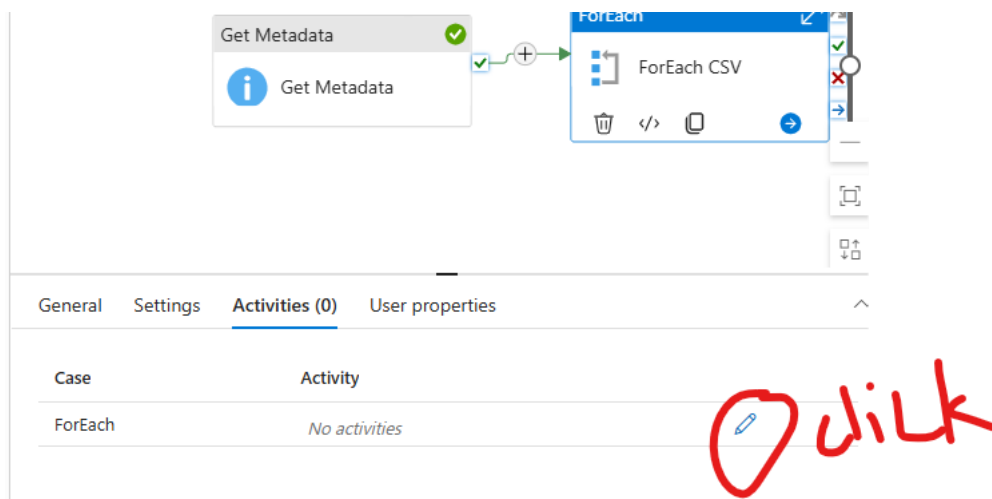
[Add dynamic content \(Alt+Shift+D\)](#)

click ↴

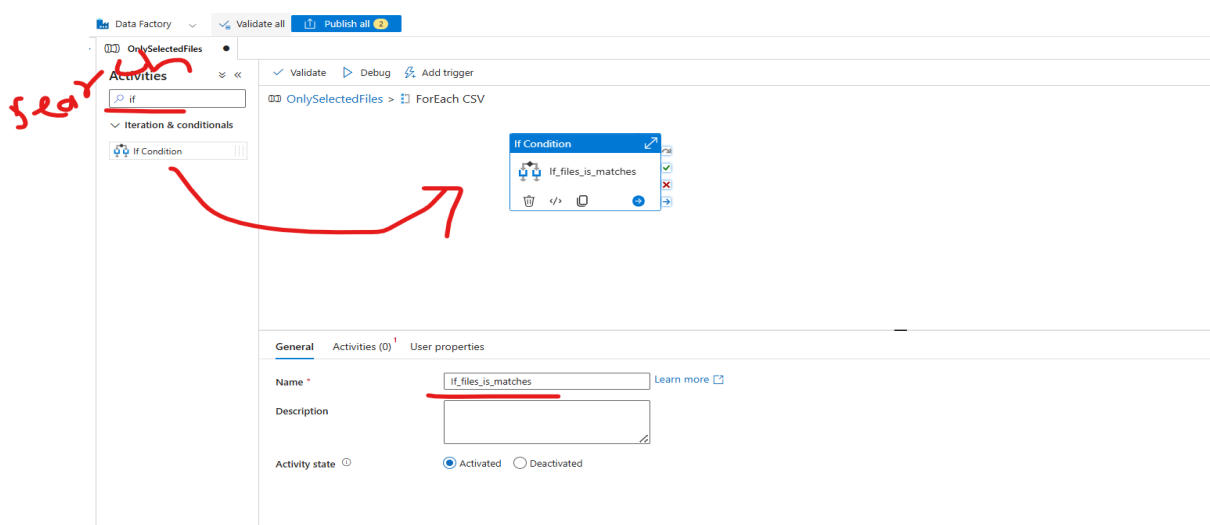
click on order box

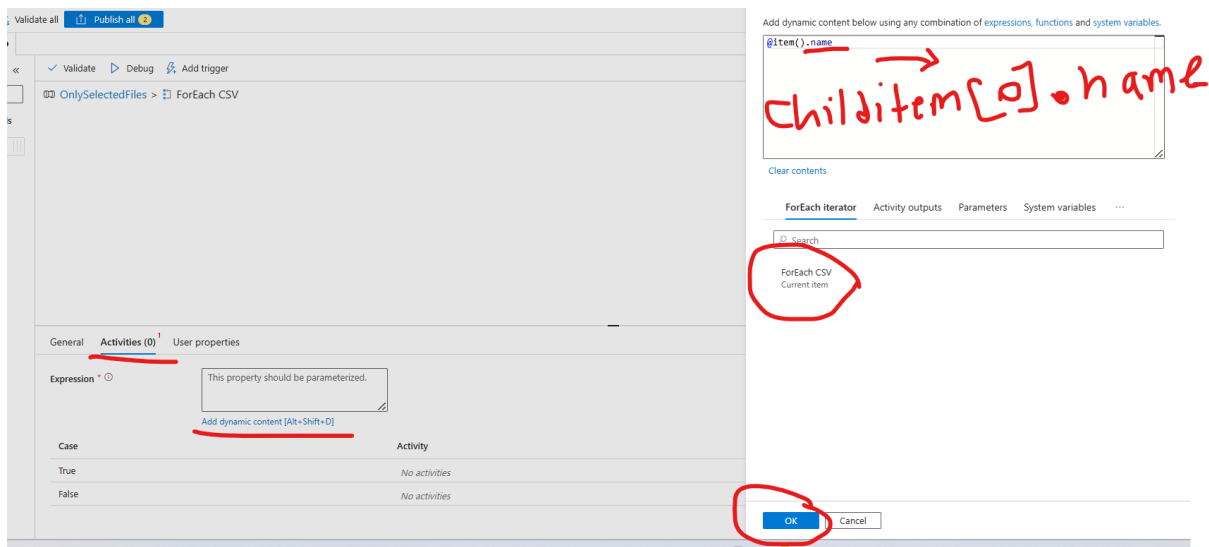
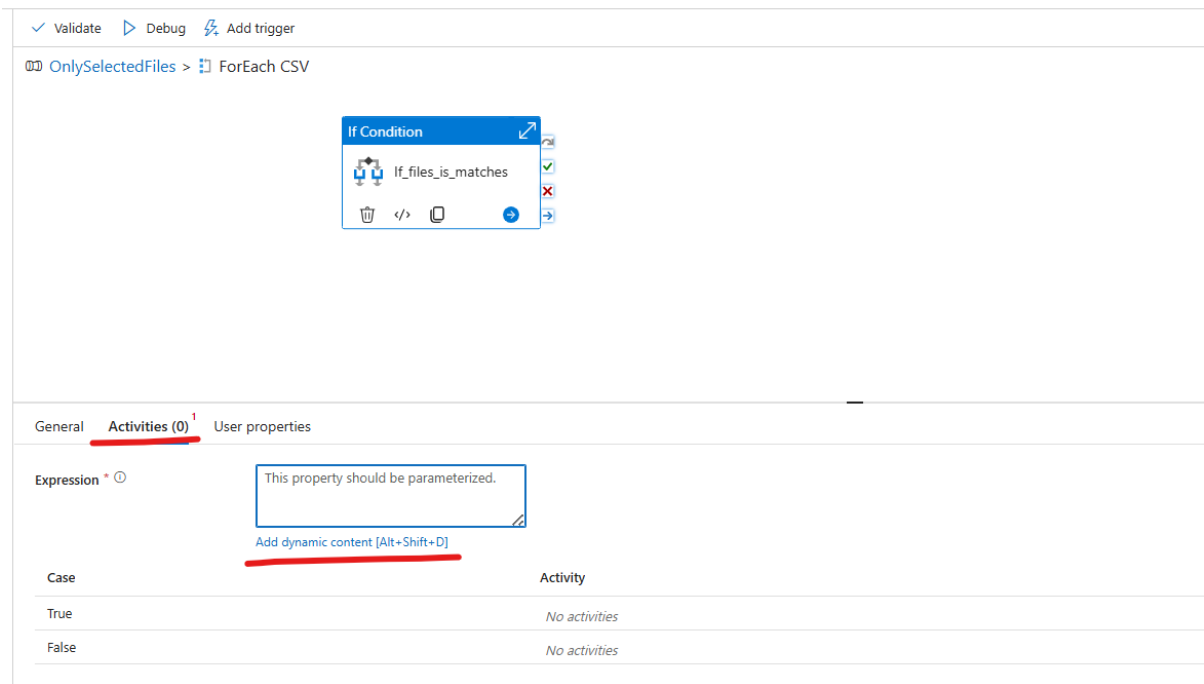


After that click on Activity

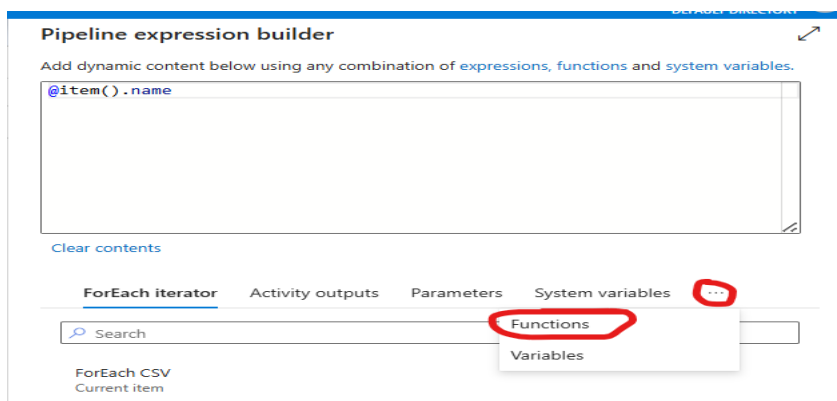


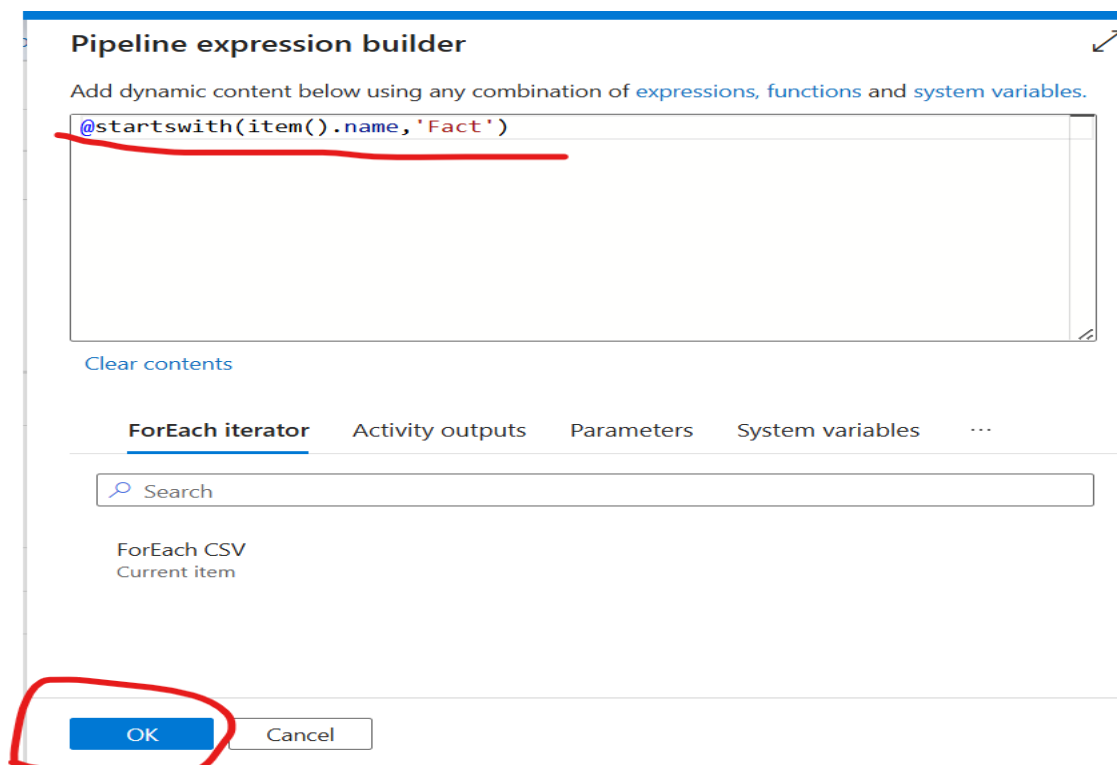
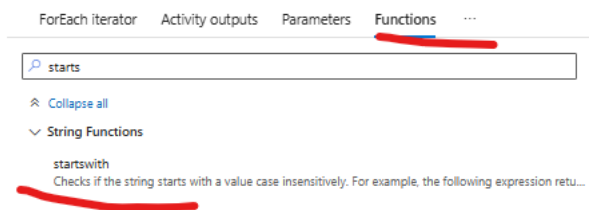
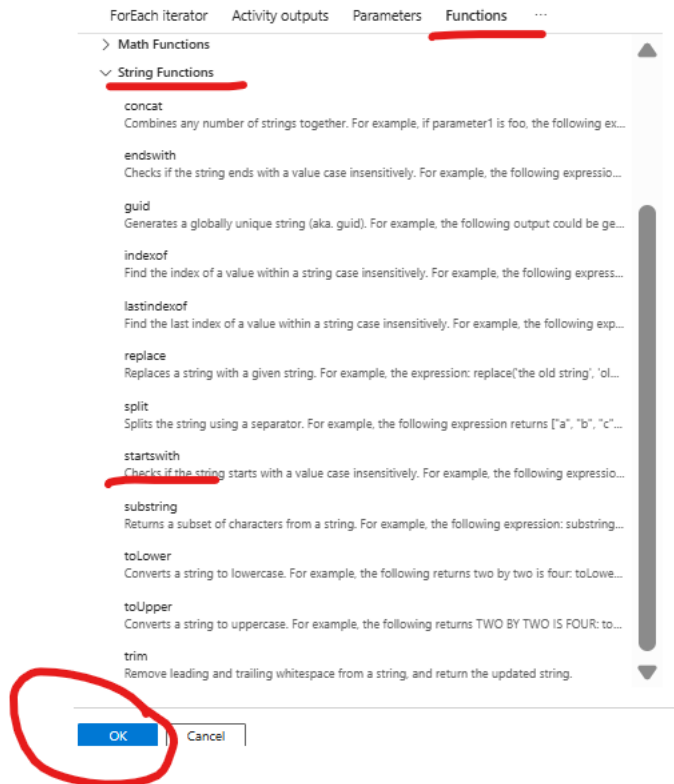
After that





Now we can can add contion is start with fact using function





Now we can perform copy activity:

Validate all [Publish all](#) 2

« [✓ Validate](#) [▶ Debug](#) [⚡ Add trigger](#)

[OnlySelectedFiles](#) > [ForEach CSV](#)

General **Activities (0)** User properties

Expression ⓘ `@startswith(item().name,'Fact')`

Case	Activity
True	No activities
False	No activities

Now if condition is true then perform copy activity:

Same follows all the steps of copy activity we already perform:

Search copy-> drag and drop -> add source name in general -> sink -> new -> search data lake -> csv format -> ok -> add destination container path -> then next step is

Set properties

Name

Linked service *

File path
 / /

First row as header ☒

Import schema
☒ From connection/store ☐ From sample file ☐ None

Advanced
[Open this dataset](#) for more advanced configuration with parameterization.

[OK](#) [Back](#) [Cancel](#)

Add Parameter

Connection

Schema

Parameters

+ New

🗑 Delete

☐	Name	Type	Default value	
☐	<u>P_File_Name</u>	<u>String</u>	Value	🗑

Connection

Schema

Parameters

Linked service *

linkedServiceDataLake

Test connection

Edit

+ New

Learn more

File path

destination

/

CSV_FILES_NEW

/

File name

Browse

Preview data

Detect format

Compression type

No compression

Column delimiter

Comma (,)

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

@dataset().P_File_Name

Clear contents

Parameters

Functions

Search

+

P_File_Name

click of select

OK

Cancel

Parameter source csv done

Data Factory | Validate all | Publish all 3

>> OnlySelectedFiles | parameter_source_csv

DelimitedText
parameter_source_csv

CSV

Connection | Schema | Parameters

Linked service * | likedServiceDataLake | Test connection | Edit | + New | Learn more

File path | destination | / | CSV_FILES_NEW | / | @dataset().P_File_Name

Compression type | No compression

Column delimiter | Comma (,)

Row delimiter | Default (\r,\n, or \r\n)

✓ Validate | ✓ Validate copy runtime | ▶ Debug | ⚡ Add trigger

OnlySelectedFiles > ForEach CSV > If_files_is_matches > True activities

Copy data
CopyFactData

General | Source | Sink | Mapping | Settings | User properties

Source dataset * | parameter_source_csv | Open | + New | Preview data | Learn m

Dataset properties

Name	Value
P_File_Name	Value Add dynamic content [Alt+Shift+D]

File path type | ☒ File path in dataset | ☐ Wildcard file path | ☐ List of files

Start time (UTC) | End time (UTC)

Filter by last modified

upgraded to Microsoft Fabric. Learn more ab

Debug | Add trigger

CSV > If_files_is_matches > True

Copy data
CopyF

Settings | User properties

parameter_source_csv

Dataset properties

Name	Value
P_File_Name	Value Add dynamic conte

File path in dataset | ☐ Wildcard file path

t time (UTC)

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

@item().name

Clear contents

Activity outputs | Parameters | System variables | Functions | Variables

Search

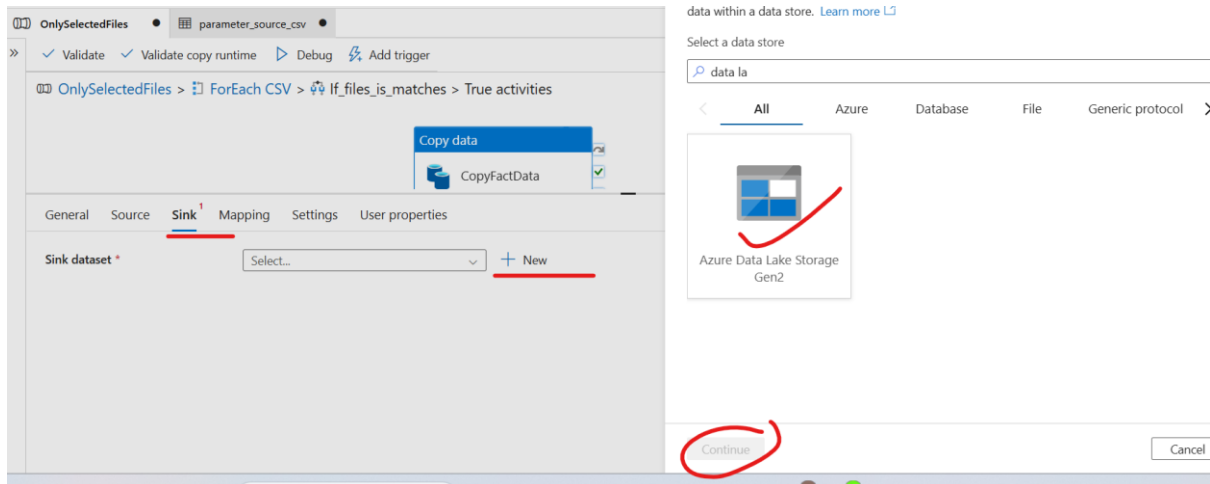
Get Metadata
Get Metadata activity output

Get Metadata childItems
List of subfolders and files in the given folder

OK | Cancel

Source tab is done.

Now click on sink tab:



Click on continue -> select csv -> and then

Set properties

Name

reporting_sink

Linked service *

likedServiceDataLake

File path

reporting

/ csv_files

/ File name

First row as header



Import schema

☐ From connection/store ☐ From sample file ☒ None

Advanced

[Open this dataset](#) for more advanced configuration with parameterization.

click

OK

Back

Cancel

Connection Schema Parameters

+ New Delete

☐	Name	Type	Default value	
☐	<u>P_File_Name</u>	<u>String</u>	Value	

Connection Schema Parameters

Linked service * linkedServiceDataLake Test connection Edit + New Learn more

File path reporting / csv_files / File name Browse Prev

Compression type No compression

Column delimiter Comma (,)

Row delimiter Default (\r\n, or \n)

Add dynamic content [Alt+Shift+D]

Click

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables

@dataset().P_File_Name

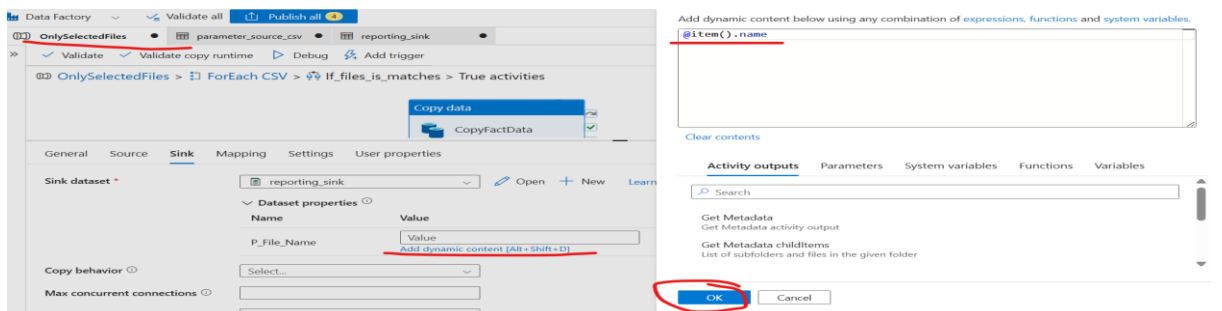
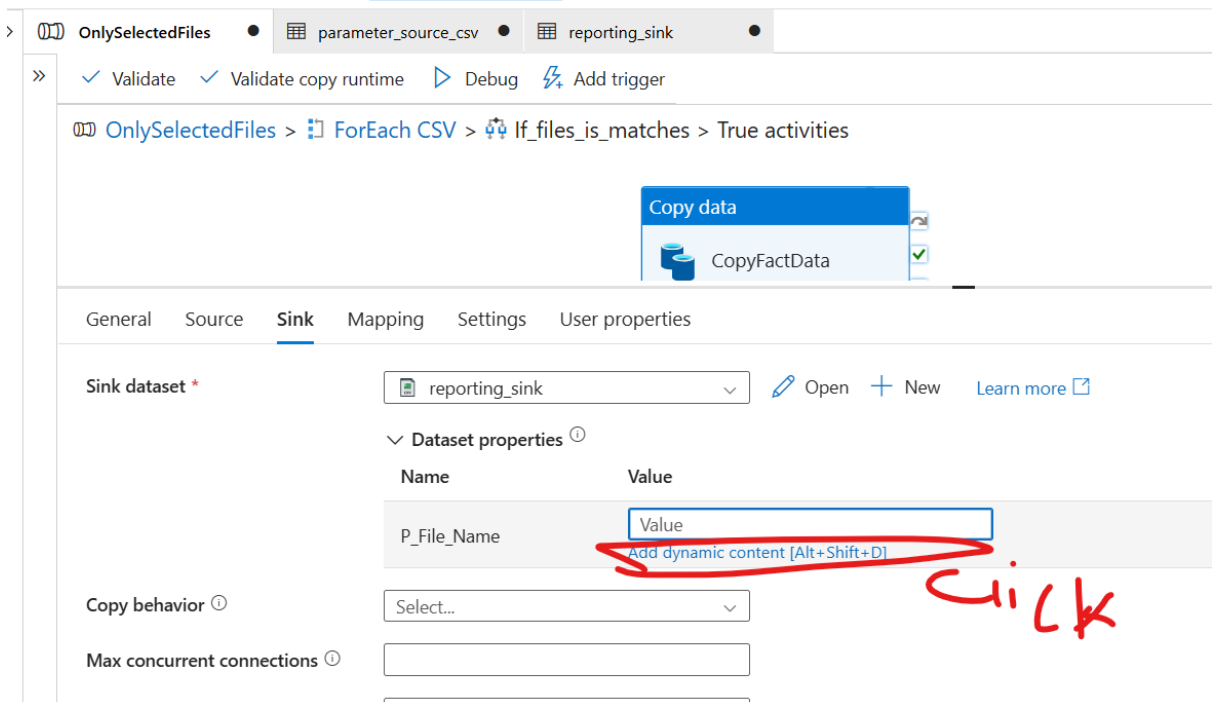
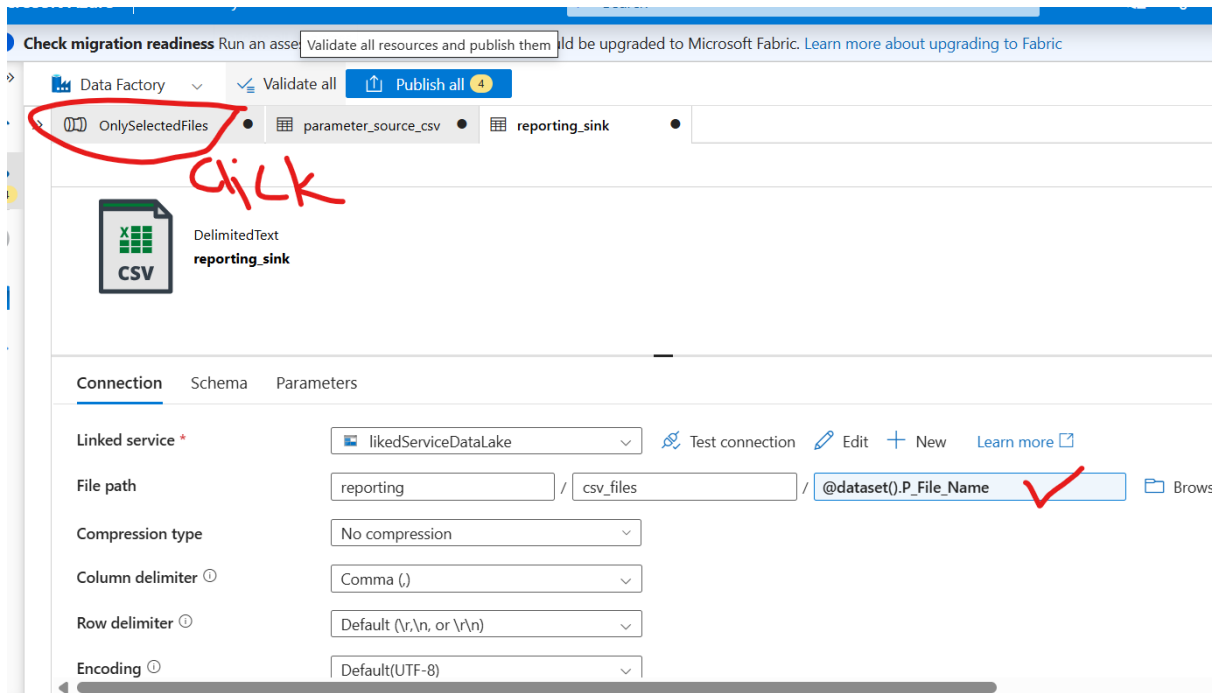
Clear contents

Parameters Functions

Search

P_File_Name select

OK Cancel



All done

validate validate copy runtime Debug Add trigger

OnlySelectedFiles > ForEach CSV > If_files_is_matches > True activities

Click

Copy data

CopyFactData

General Source Sink Mapping Settings User properties

Sink dataset *

reporting_sink

Open + New Learn more

Dataset properties

Name	Value
P_File_Name	@item().name

Copy behavior

Select...

Max concurrent connections

Block size (MB)

All done solution is completed now click on debug

Check migration readiness Run an assessment to identify which resources could be upgraded to Microsoft Fabric. Learn more about upgrading to Fabric

Start assessment (preview)

OnlySelectedFiles parameter_source_csv reporting_sink

Activities

Validate Debug Add trigger

Get Metadata

ForEach CSV

Parameters Variables Settings Output

Pipeline run ID 9d550652-65fe-48b2-89d4-091899bdcf15 Pipeline status Succeeded

View debug run consumption Monitor in Azure Metrics Export to CSV

Activity name	Activity status	Activity...	Run start	Duration	Integration runtime	User prop...	Activity run ID
If_files_is_matches	Succeeded	If Condition	12/31/2025, 9:30:54 PM	Less than 1s			64e40bfb-baa9-4f16-8110
CopyFactData	Succeeded	Copy data	12/31/2025, 9:30:26 PM	26s	AutoResolveIntegrationRuntime (East US)		edf985df-1faf-43eb-9844-c
If_files_is_matches	Succeeded	If Condition	12/31/2025, 9:30:25 PM	29s			084ba377-ea5f-4c5d-bd1f
CopyFactData	Succeeded	Copy data	12/31/2025, 9:29:45 PM	38s	AutoResolveIntegrationRuntime (East US)		ac939fec-1c5e-4a88-8c7f-
If_files_is_matches	Succeeded	If Condition	12/31/2025, 9:29:45 PM	40s			fd550095-2714-4483-98dt
ForEach CSV	Succeeded	ForEach	12/31/2025, 9:29:44 PM	1m 13s			9b16969d-2461-43cc-834
Get Metadata	Succeeded	Get Metadata	12/31/2025, 9:29:04 PM	39s	AutoResolveIntegrationRuntime (East US)		beff21b7-50e8-4cbb-9578

Cancel

All is done

Home > pwdatastorageaccount | Containers >

reporting

Container

Search

+ Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

reporting > csv_files

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 2 items

Name	Last modified	Access tier	Blob type	Size	Lease state
[...]					
Fact_Sales_1.csv	12/31/2025, 9:30:03 PM	Hot (inferred)	Block blob	319.93 KiB	Available
Fact_Sales_2.csv	12/31/2025, 9:30:48 PM	Hot (inferred)	Block blob	2.6 KiB	Available

Data Flow in Azure Data Factory:

✓ What is Data Flow in Azure Data Factory (ADF)?

Data Flow in Azure Data Factory is a **visual, code-free data transformation service** that allows you to **transform data at scale** using **Apache Spark**—without writing Spark code.

👉 Simply:

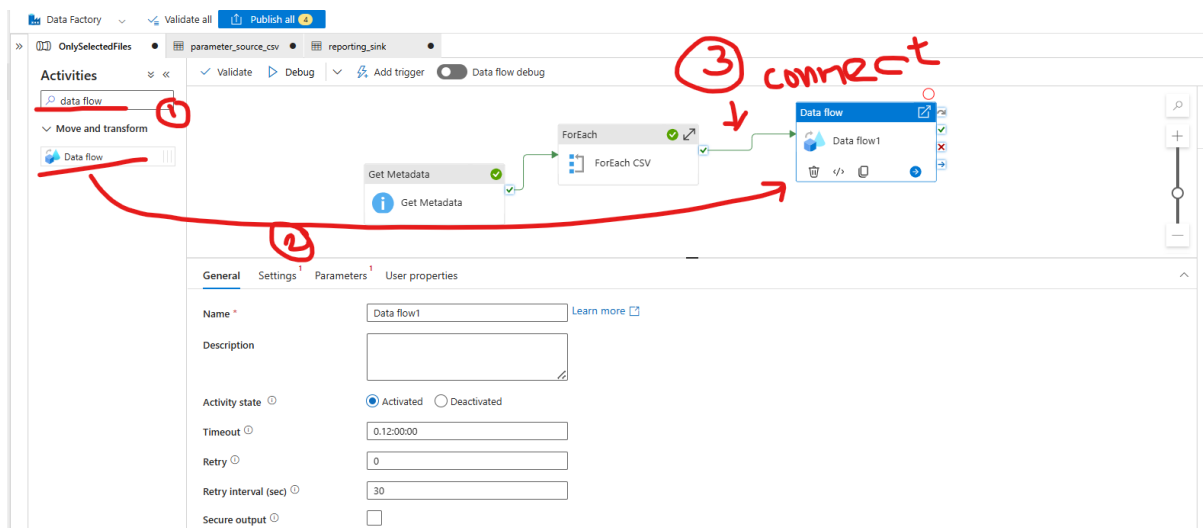
Data Flow = Transformation Layer in ADF

◆ Why is Data Flow used?

Use **Data Flow** when you need to **transform data**, not just copy it.

✓ Common Use Cases

- Filter rows
- Join multiple datasets
- Aggregate data
- Remove duplicates
- Derive new columns
- Handle schema drift
- Data cleansing & enrichment



After uploading or adding data in reporting container we want to Transformation like, Group By, any transformation. For this operation we need to used spark

✓ Validate ▶ Debug ▼ ⚙ Add trigger ⏻ Data flow debug

General Settings Parameters User properties

Name * Data Transformation [Learn more](#)

Description

Activity state ☒ Activated ☐ Deactivated

Timeout 0.12:00:00

Retry 0

Retry interval (sec) 30

Secure output ☐

Secure input ☐

General is Done.

General **Settings** Parameters User properties

Data flow * Select... [+ New](#)

Run on (Azure IR) * ☒ AutoResolveIntegrationRuntime

Compute size * Small

> Advanced

Logging level * ☒ Verbose ☐ Basic ☐ None

Microsoft Azure | Data Factory | ADF-Course-PW

Check migration readiness Run an assessment to identify which resources could be upgraded to Microsoft Fabric. Learn more about upgrading to Fabric

➤ Data Factory ➤ Validate all ➤ Publish all

➤ OnlySelectedFiles ➤ parameter_source_csv ➤ reporting_sink ➤ TransformCSV

✓ Validate ⏻ Data flow debug

Add Source

Parameters Settings

+ New

Properties

General Related (1)

Name * TransformCSV

Description

Cancel

Check migration readiness Run an assessment to identify which resources could be upgraded to Microsoft Fabric. [Learn more about upgrading to Fabric](#)

Data Factory Validate all Publish all 5

OnlySelectedFiles parameter_source_csv reporting_sink TransformCSV

✓ Validate Data flow debug

Add Source

Parameters Settings

+ New

Add Source ✓
Add Flowlet

Data Factory validate all Publish all 5

OnlySelectedFiles parameter_source_csv reporting_sink TransformCSV

✓ Validate Data flow debug

sourceCSV
Columns:
0 total

Add Source

Source settings Source options Projection Optimize Inspect Data preview

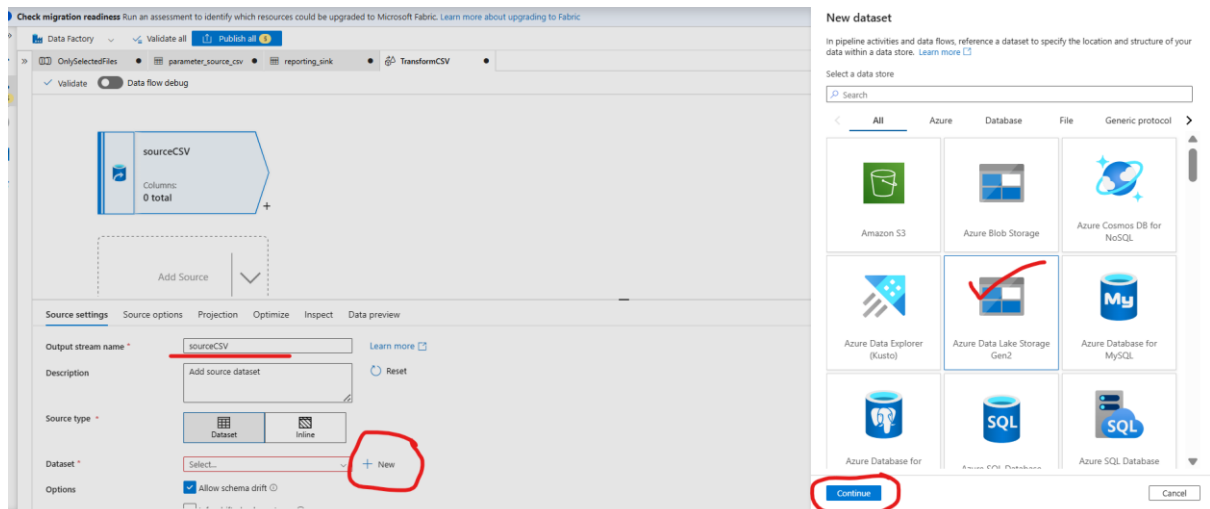
Output stream name * sourceCSV [Learn more](#)

Description Add source dataset [Reset](#)

Source type * Dataset Inline

Dataset * Select... + New

Options ☒ Allow schema drift



Select format of CSV

Set properties

Name

Linked service *

File path
 / /

First row as header ☒

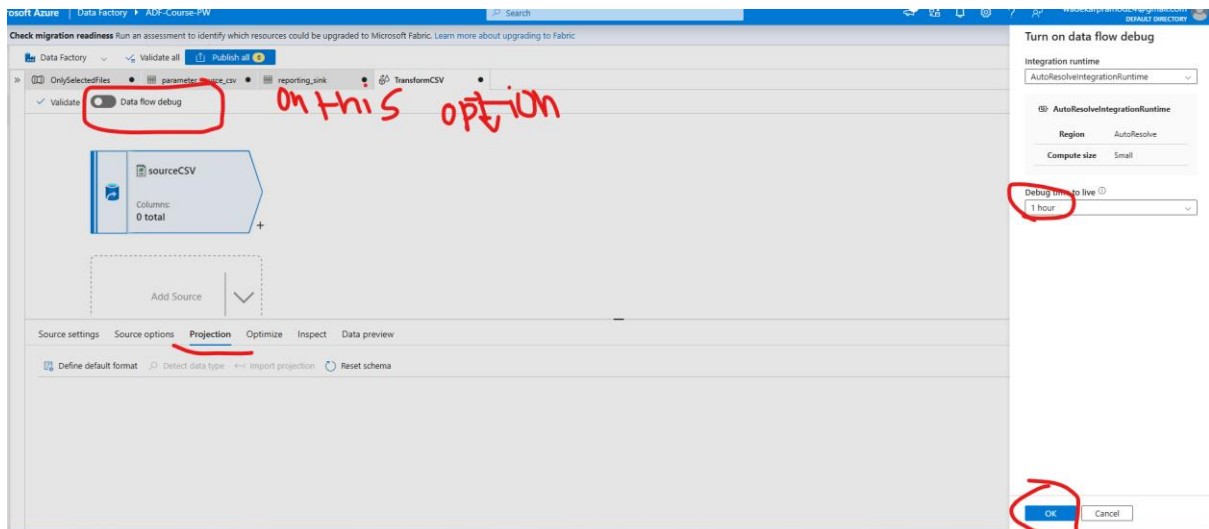
Import schema
☐ From connection/store ☐ From sample file ☒ None

> Advanced

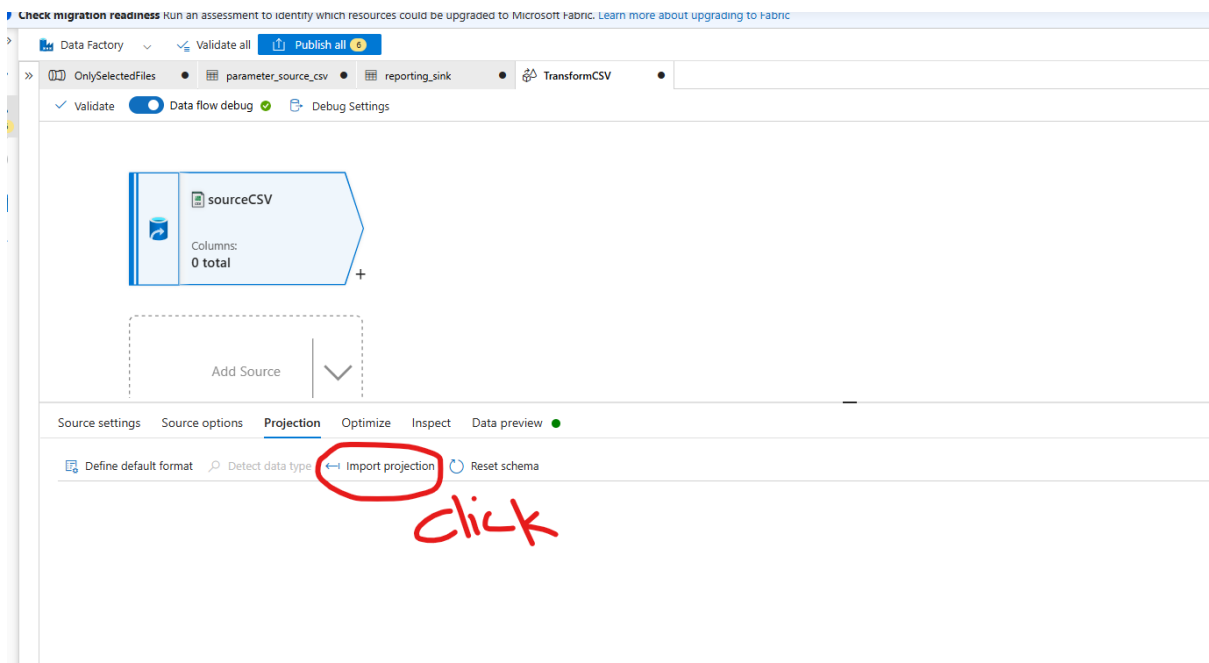
OK Back Cancel

Source setting Tab done.

Source option Tab not needed to change

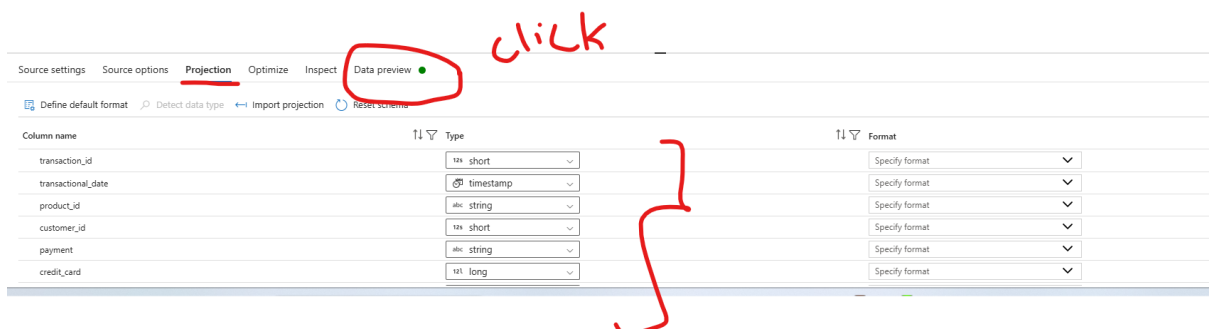


Wait some time to start cluster



Wait some time

Import projection is import schema



Columns: 10 total

Add Source

Source settings Source options Projection Optimize Inspect **Data preview**

Number of rows **INSERT** 100 **UPDATE** 0 **DELETE** 0 **UPSERT** 0 **LOOKUP** 0 **ERROR** 0 **TOTAL** 1000

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

	transaction_id	transactional_date	product_id	customer_id	payment	credit_card	loyalty_card	cost	quantity	price
1	1	2021-05-04 02:00:00.000	P0494	4	visa	4041593010498829	×	17.33	2	18.29
2	2	2021-05-04 03:04:00.000	P0221	5	visa	4041596151234556	×	0.59	1	1.49
3	3	2021-05-04 03:56:00.000	P0625	5	visa	4041594885335898	×	5.15	3	5.89
4	4	2021-05-04 05:20:00.000	P0431	8	mastercard	5108753677552345	×	10.67	2	11.59
5	5	2021-05-04 05:45:00.000	P0058	5	mastercard	5108752372298261	✓	11.38	2	12.39

if data not loaded click Refresh

Now we can do some tranformation:

Select Tranformation:

OnlySelectedFiles parameter_source_csv reporting_sink TransformCSV

✓ Validate Data flow debug Debug Settings

sourceCSV Columns: 10 total

Add Source

Source settings Source options Projection Opti

Number of rows **INSERT** 100

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

transaction_id transactional_date product_id customer_id payment credit_card

1 2021-05-04 02:00:00.000 P0494 4 visa 4041593010498829

2 2021-05-04 03:04:00.000 P0221 5 visa 4041596151234556

3 2021-05-04 03:56:00.000 P0625 5 visa 4041594885335898

4 2021-05-04 05:20:00.000 P0431 8 mastercard 5108753677552345

5 2021-05-04 05:45:00.000 P0058 5 mastercard 5108752372298261

click +

Clicking on + button showing all tranformation

OnlySelectedFiles parameter_source_csv reporting_sink TransformCSV

✓ Validate Data flow debug Debug Settings

sourceCSV Columns: 10 total

Add Source

Source settings Source options Projection Opti

Number of rows **INSERT** 100

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

transaction_id transactional_date product_id customer_id payment credit_card

1 2021-05-04 02:00:00.000 P0494 4 visa 4041593010498829

2 2021-05-04 03:04:00.000 P0221 5 visa 4041596151234556

3 2021-05-04 03:56:00.000 P0625 5 visa 4041594885335898

4 2021-05-04 05:20:00.000 P0431 8 mastercard 5108753677552345

5 2021-05-04 05:45:00.000 P0058 5 mastercard 5108752372298261

+

Multiple inputs/outputs

- Join
- Conditional Split
- Exists
- Union
- Lookup

Schema modifier

- Derived Column
- Select
- Aggregate
- Surrogate Key
- Pivot
- Unpivot
- Window
- Rank

Add Source

Select settings

Optimize

Inspect

Data preview

Output stream name *

selectcolumns

Learn more

Description

transactional_date, product_id, customer_id, payment, credit_card, loyalty_card, cost, quantity, price

Reset

Incoming stream *

sourceCSV

Options

Skip duplicate input columns

Skip duplicate output columns

Input columns *

Auto mapping

Reset

Add mapping

Delete

Select settings

Optimize

Inspect

Data preview

Input columns *

Auto mapping

Reset

Add mapping

Delete

sourceCSV's column

transaction_id

transactional_date

product_id

customer_id

payment

credit_card

loyalty_card

Name as

transaction_id

transactional_date

product_id

customer_id

payment

credit_card

loyalty_card

Select settings

Optimize

Inspect

Data preview

Refresh

number of rows

INSERT 100

UPDATE 0

DELETE 0

UPSERT 0

LOOKUP 0

ERROR 0

TOTAL 1000

transaction_id	transactional_date	product_id	customer_id	payment	price
1	2021-05-04 02:00:00.000	P0494	4	visa	18.29
2	2021-05-04 03:04:00.000	P0221	5	visa	1.49
3	2021-05-04 03:56:00.000	P0625	5	visa	5.89
4	2021-05-04 05:20:00.000	P0431	8	mastercard	11.59
5	2021-05-04 05:45:00.000	P0058	5	mastercard	12.39

Now select is done.

Now we can do filter operation like customer-id not equal to 12

✓ Validate ☒ Data flow debug ☒ Debug Settings

sourceCSV Import data from datatransform_flow_CSV

selectcolumns Columns: 6 total

click

Rank

External Call

Cast

Formatters

Flatten

Parse

Stringify

Row modifier

Filter

Sort

Alter Row

Assert

Flowlets

Flowlet

Destination

Sink

Select settings Optimize Inspect **Data preview**

Number of rows **INSERT** 100 **UPDATE** 0

Refresh Typecast Modify Map drifted Statistics Remove

	transaction_id	transactional_date		customer_id
+	1	2021-05-04 02:00:00.000	P0494	4
+	2	2021-05-04 03:04:00.000	P0221	5
+	3	2021-05-04 03:56:00.000	P0625	5
+	4	2021-05-04 05:20:00.000	P0431	8

Check migration readiness Run an assessment to identify which resources could be upgraded to Microsoft Fabric. Learn more about upgrading to Fabric

Data Factory Validate all Publish all

OnlySelectedFiles parameter_source_csv reporting_sink TransformCSV

✓ Validate ☒ Data flow debug ☒ Debug Settings

sourceCSV Import data from datatransform_flow_CSV

selectcolumns Renaming sourceCSV to selectcolumns with columns transaction_id, transactional_date, product_id

filterRow Columns: 6 total

Add Source

Filter settings Optimize Inspect **Data preview**

Output stream name filterRow

Description Add expression for the filter

Incoming stream selectcolumns

Filter on Enter filter...

Open expression builder

Dataflow expression builder

filterRow

Expression

customer_id != 12

2

Expression elements

All

Functions

Input schema

Parameters

Cached lookup

Data flow library functions

Expression values

Filter by keyword

Create new

123 transaction_id

transactional_date

abc product_id

123 customer_id

abc payment

123 price

Data preview Refresh

Save and finish Cancel Clear contents

3

1

OnlySelectedFiles • parameter_source_csv • reporting_sink • TransformCSV

✓ Validate Data flow debug Debug Settings

```

graph LR
    sourceCSV[sourceCSV  
Import data from  
datatransform_flow_CSV] --> selectcolumns[selectcolumns  
Renaming sourceCSV to  
selectcolumns with columns  
'transaction_id',  
transactional_date, product_id']
    selectcolumns --> filterRow[filterRow  
Columns:  
6 total]
  
```

Add Source

Filter settings Optimize Inspect Data preview

Output stream name * filterRow [Learn more](#)

Description Filtering rows using expressions on columns 'customer_id' [Reset](#)

Incoming stream * selectcolumns

Filter on * customer_id != 12

Data Factory Validate all Publish all

OnlySelectedFiles • parameter_source_csv • reporting_sink • TransformCSV

✓ Validate Data flow debug Debug Settings

Filter settings Optimize Inspect **Data preview**

Number of rows INSERT 100 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL 994

	transaction_id	transactional_date	product_id	customer_id	payment	price
+	1	2021-05-04 02:00:00.000	P0494	4	visa	18.29
+	2	2021-05-04 03:04:00.000	P0221	5	visa	1.49
+	3	2021-05-04 03:56:00.000	P0625	5	visa	5.89
+	4	2021-05-04 05:20:00.000	P0431	8	mastercard	11.59
+	5	2021-05-04 05:45:00.000	P0058	5	mastercard	12.39

Now we can do after filter, split the data:

Data Factory Validate all Publish all

OnlySelectedFiles • parameter_source_csv • reporting_sink • TransformCSV

✓ Validate Data flow debug Debug Settings

sourceCSV selectcolumns filterRow

Multiple inputs/outputs

- Join
- Conditional Split**
- Exists
- Union
- Lookup
- Schema modifier
- Derived Column

Data Factory | Validate all | Publish all

Only Selected Files | parameter_source_csv | reporting_link | TransformCSV

Validate | Data flow debug | Debug Settings

Conditional split settings | Optimize | Inspect | Data preview

Output stream name * | split1 | Learn more

Description | Add cases to distribute the data in multiple groups | Reset

Incoming stream * | filterRow

Split on | ☒ First matching condition | ☐ All matching conditions

Split condition

Stream names	Condition
Visa	payment == 'visa' Open expression builder
Enter stream name for all other rows...	Rows that do not meet any condition will use this output stream

Microsoft Azure | Data Factory | ADF-Course-PW

Dataflow expression builder

split1

Expression

payment == 'visa'

Expression elements

- All
- Functions
- Input schema
- Parameters
- Cached lookup
- Data flow library functions

Expression values

Filter by keyword

+ Create new

- transaction_id
- transactional_date
- product_id
- customer_id
- payment
- price
- abs(numeric_value)
- acos(numeric_value)
- add(first_expression , second_expression)
- addDays(date/timestamp , days to add)

Data preview | Refresh

Save and finish | Cancel | Clear contents

Data Factory | Validate all | Publish all | TransformCSV

OnlySelectedFiles | parameter_source_csv | reporting_link

Validate | Data flow debug | Debug Settings

```

graph LR
    sourceCSV[sourceCSV] --> selectcolumns[selectcolumns]
    selectcolumns --> filterRow[filterRow]
    filterRow --> Visa[Visa]
    filterRow --> Otherthanvisa[Otherthanvisa]
  
```

Conditional split settings | Optimize | Inspect | Data preview

Output stream name: split1 [Learn more](#) [Reset](#)

Description: Conditionally distributing the data in payment groups, based on columns ['1']

Incoming stream: filterRow

Split on: ☒ First matching condition ☐ All matching conditions

Split condition

Stream names	Condition
Visa	payment == 'visa'
Otherthanvisa	Rows that do not meet any condition will use this output stream

Conditional split settings | Optimize | Inspect | **Data preview**

Number of rows: INSERT 100 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL 331

Refresh | Statistics | Export to CSV

Output stream: Visa

transaction_id	transaction_date	transaction_id	customer_id	payment	price
1	2021-05-04 02:00:00.000	P0494	4	visa	18.29
2	2021-05-04 03:04:00.000	P0221	5	visa	1.49
3	2021-05-04 03:56:00.000	P0625	5	visa	5.89
7	2021-05-04 07:03:00.000	P0575	4	visa	3.99
15	2021-05-05 05:22:00.000	P0684	6	visa	7.49
17	2021-05-05 07:40:00.000	P0450	9	visa	4.79
21	2021-05-05 14:01:00.000	P0282	7	visa	11.59
22	2021-05-05 14:31:00.000	P0480	5	visa	13.49
24	2021-05-05 19:15:00.000	P0347	6	visa	6.29

Conditional split settings | Optimize | Inspect | Data preview

Output stream name: split1 [Learn more](#) [Reset](#)

Description: Conditionally distributing the data in payment groups, based on columns ['1']

Incoming stream: filterRow

Split on: ☒ First matching condition ☐ All matching conditions

Split condition

Stream names	Condition
Visa	payment == 'visa'
mastercard	payment == 'mastercard'
Otherthanvisa	Rows that do not meet any condition will use this output stream

Number of rows: **INSERT** 100 **UPDATE** 0 **DELETE** 0 **UPSERT** 0 **LOOKUP** 0 **ERROR** 0 **TOTAL** 33

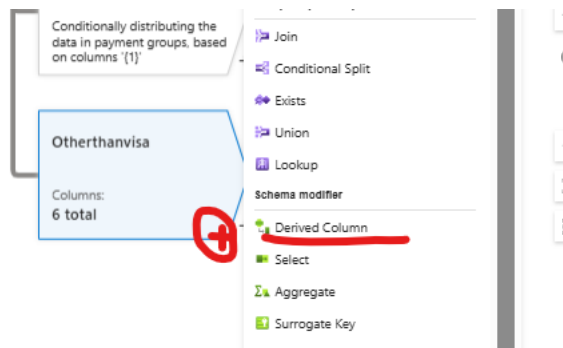
Refresh | Statistics | Export to CSV

Output stream: Visa

transaction_id	transactional_date	product_id	customer_id	payment	price
1	2021-05-04 03:56:00.000	P0625	4	visa	18.29
2	2021-05-04 07:03:00.000	P0575	5	visa	1.49
3	2021-05-04 03:56:00.000	P0625	5	visa	5.89
7	2021-05-04 07:03:00.000	P0575	4	visa	3.99
15	2021-05-05 05:22:00.000	P0684	6	visa	7.49

Done.

Now we can do column activity tranformation: this is used to tranform a column and or add a new column



Derived column's settings | Optimize | Inspect | Data preview

Output stream name: **derivedColumn** [Learn more](#)

Description: Creating/updating the columns 'transaction_id, transactional_date, product_id, customer_id, payment, price' [Reset](#)

Incoming stream: split1@Otherthanvisa

Columns:

Column	Expression
payment	Enter expression... Open expression builder

Dataflow expression builder Expression ref

derivedColumn

Derived Columns

+ Create new

ANY payment

Column name *

payment

Expression

`coalesce(payment, 'N/A')`

Expression elements

- All
- Functions
- Input schema
- Parameters
- Cached lookup
- Data flow library functions
- Locals

Expression values

Filter by keyword

+ Create new

- transaction_id
- transactional_date
- product_id
- customer_id
- payment
- price
- abs(123 numeric_value)
- acos(123 numeric_value)
- add(ANY first_expression, ANY second_expression)
- addDays(date/timestamp, ANY days to add)

Data preview Refresh

Save and finish Cancel Clear contents

Replace null value to N/A in Payment column

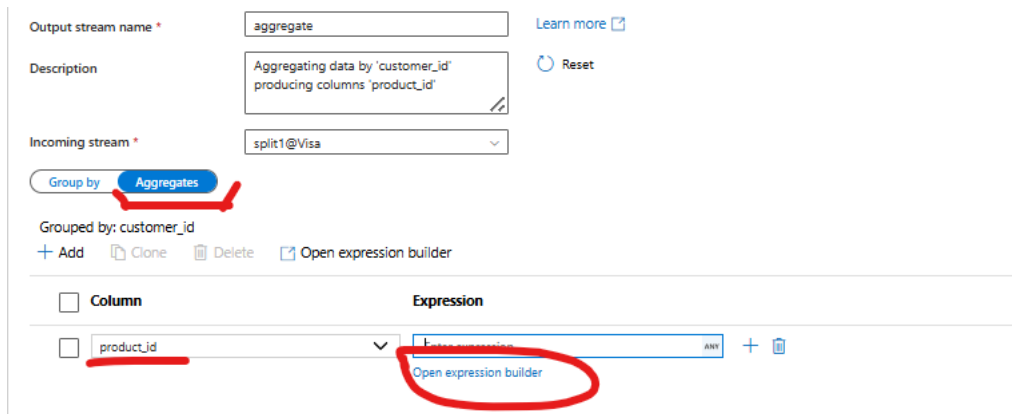
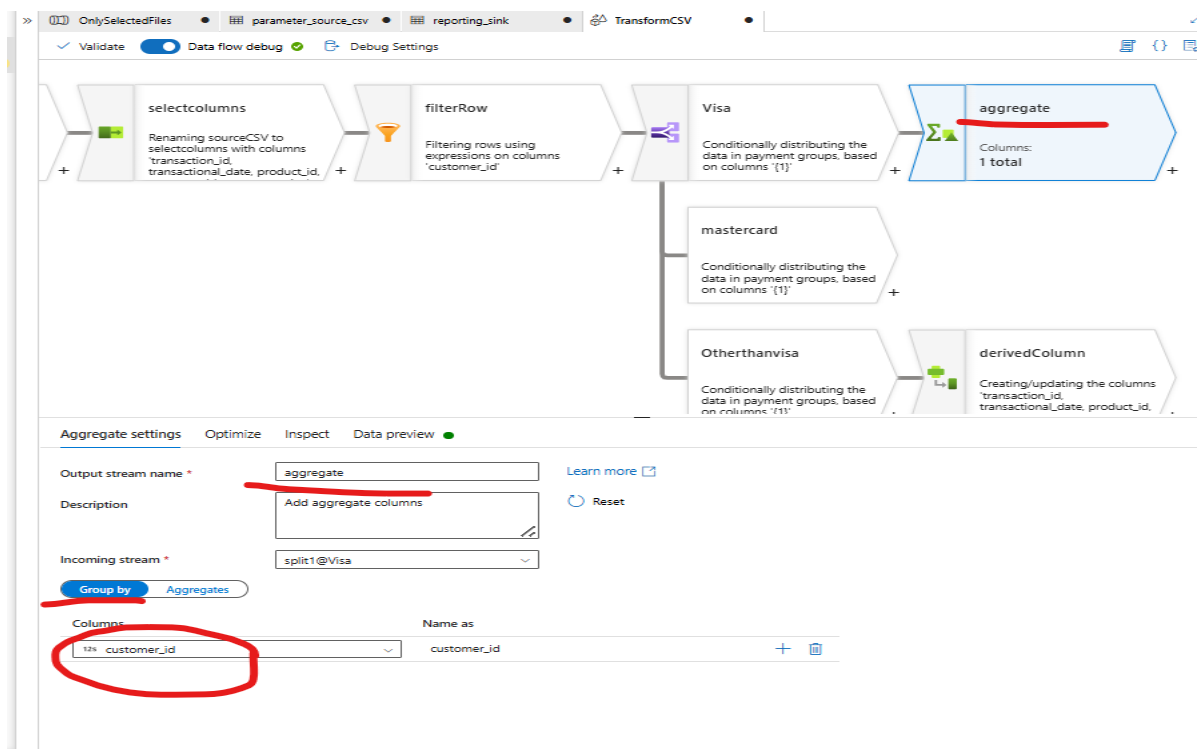
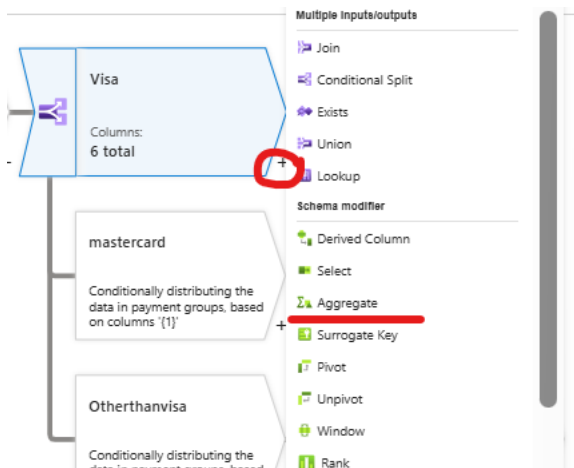
Derived column's settings Optimize Inspect **Data preview**

Number of rows + INSERT 100 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL 343

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

	transaction_id	transactional_date	product_id	customer_id	payment	price
+	6	2021-05-04 06:58:00.000	P0385	6	americanexpress	14.69
+	8	2021-05-04 07:45:00.000	P0187	5	americanexpress	4.89
+	13	2021-05-04 19:57:00.000	P0550	5	americanexpress	17.09
+	16	2021-05-05 05:58:00.000	P0321	7	americanexpress	18.29
+	18	2021-05-05 07:58:00.000	P0457	10	americanexpress	2.59
+	20	2021-05-05 10:11:00.000	P0152	7	americanexpress	17.99
+	30	2021-05-06 05:58:00.000	P0524	4	americanexpress	11.59
+	35	2021-05-07 02:32:00.000	P0413	7	N/A	8.19
+	42	2021-05-07 22:59:00.000	P0445	4	americanexpress	1.19
+	43	2021-05-08 00:23:00.000	P0097	6	americanexpress	11.29

Now we can perform group by transformation:



Dataflow expression builder Expressi

aggregate

Aggregate Columns

+ Create new

ANY product_id

Column name *

product_id

Expression

max(product_id)

Expression elements

All

Functions

Input schema

Parameters

Cached lookup

Data flow library functions

Locals

Expression values

Filter by keyword

+ Create new

123 transaction_id

transactional_date

product_id

123 customer_id

payment

price

123 abs(123 numeric_value)

123 acos(123 numeric_value)

ANY add(ANY first_expression , ANY second_expression)

addDays(date/timestamp , ANY days to add)

Save and finish Cancel Clear contents

Click on data preview

✓ Validate Data flow debug Debug Settings

selectcolumns Renaming sourceCSV to selectcolumns with columns 'transaction_id, transactional_date, product_id, ...'

filterRow Filtering rows using expressions on columns 'customer_id'

Visa Conditionally distributing the data in payment groups, based on columns '1'

mastercard Conditionally distributing the data in payment groups, based on columns '1'

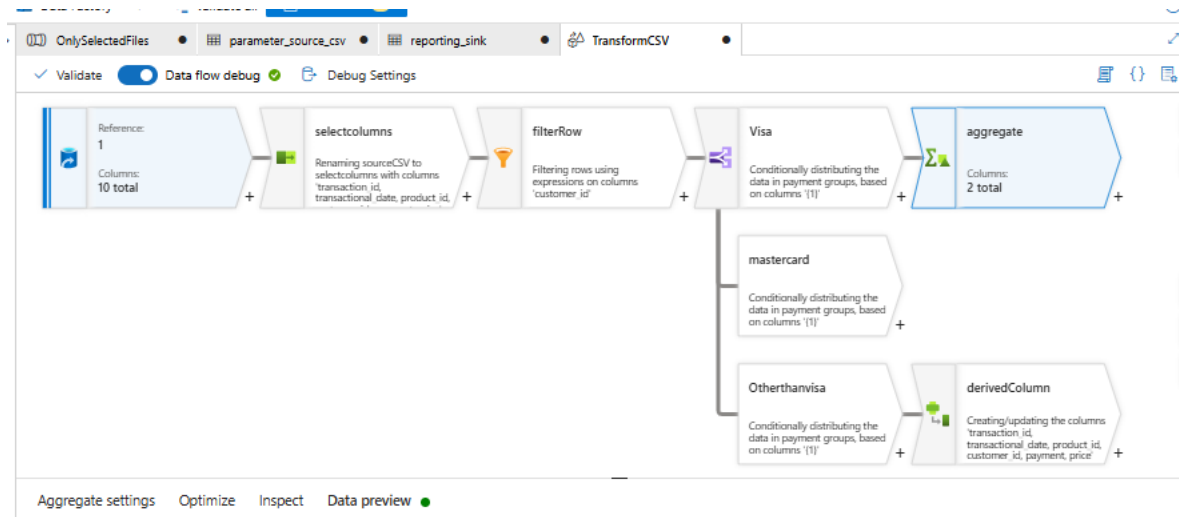
aggregate Columns: 2 total

Aggregate settings Optimize Inspect Data preview

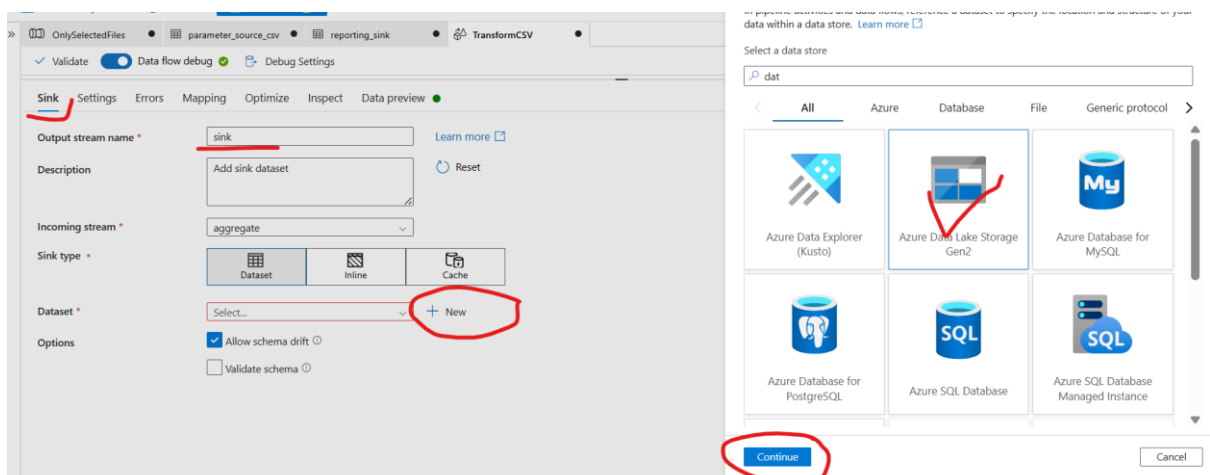
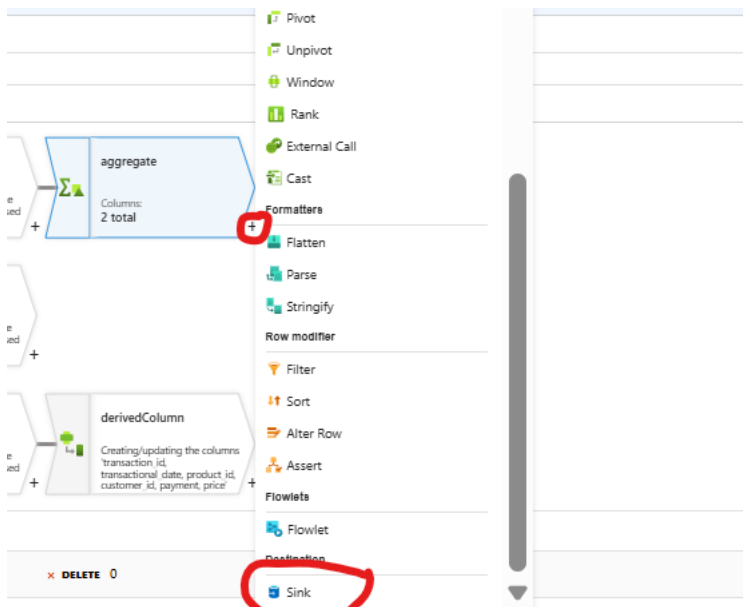
Number of rows INSERT 13 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL 1

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

customer_id	product_id
-1	P0022
0	P0314
1	P0179
2	P0640
3	P0746
4	P0736
5	P0708
6	P0714
7	P0735
8	P0745
9	P0659
10	P0727



All done. now write this data using sink



Select csv

Set properties

Name

Linked service *

File path
 / /

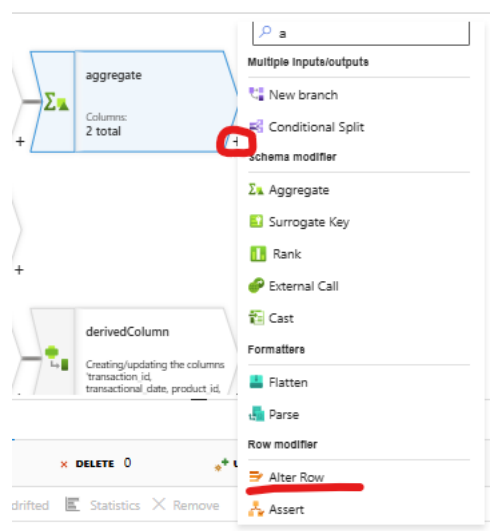
First row as header ☒

Import schema
☐ From connection/store ☐ From sample file ☒ None

> Advanced

After sink you can perform the some operations like, insert, update, alter etc

We can Perform alter operation:



Conditionally distributing the data in payment groups, based on columns: [T]

Columns: 6 total

Alter row settings | Optimize | Inspect | Data preview

Output stream name * [Learn more](#)

Description [Reset](#)

Incoming stream *

Alter row conditions * [+](#) [-](#) [Insert if](#) [+](#) [-](#)

click

Dataflow expression builder

alterRow

Expression

Expression elements

- All
- Functions
- Input schema
- Parameters
- Cached lookup
- Data flow library functions

Expression values

[+ Create new](#)

- customer_id
- product_id
- abs(123 numeric_value)
- acos(123 numeric_value)
- add(ANY first_expression, ANY second_expression)
- addDays(date/timestamp, ANY days to add)
- addMonths(date/timestamp, ANY months to add)
- and(ANY first_condition, ANY second_condition)
- approxDistinctCount(ANY expression, 123 numeric_value)
- array(ANY items)

Data preview [Refresh](#)

[Save and finish](#) [Cancel](#) [Clear contents](#)

Or you can add 1==1 we need to true condition eg 2=2, a=a etc click on validate button

Validate [Data flow debug](#) [Debug Settings](#) [TransformCSV](#)

filterRow [+](#) Visa [+](#) aggregate [+](#) alterRow [+](#) sink

mastercard [+](#) Otherthanvisa [+](#) derivedColumn

Alter row settings | Optimize | Inspect | Data preview

Output stream name * [Learn more](#)

Description [Reset](#)

Incoming stream *

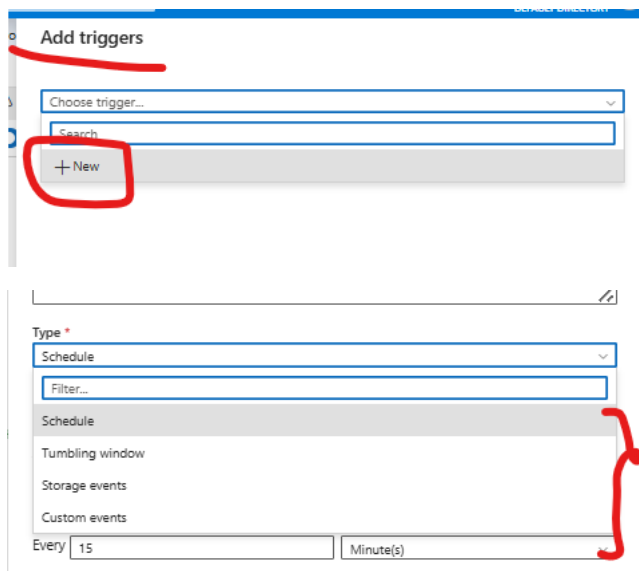
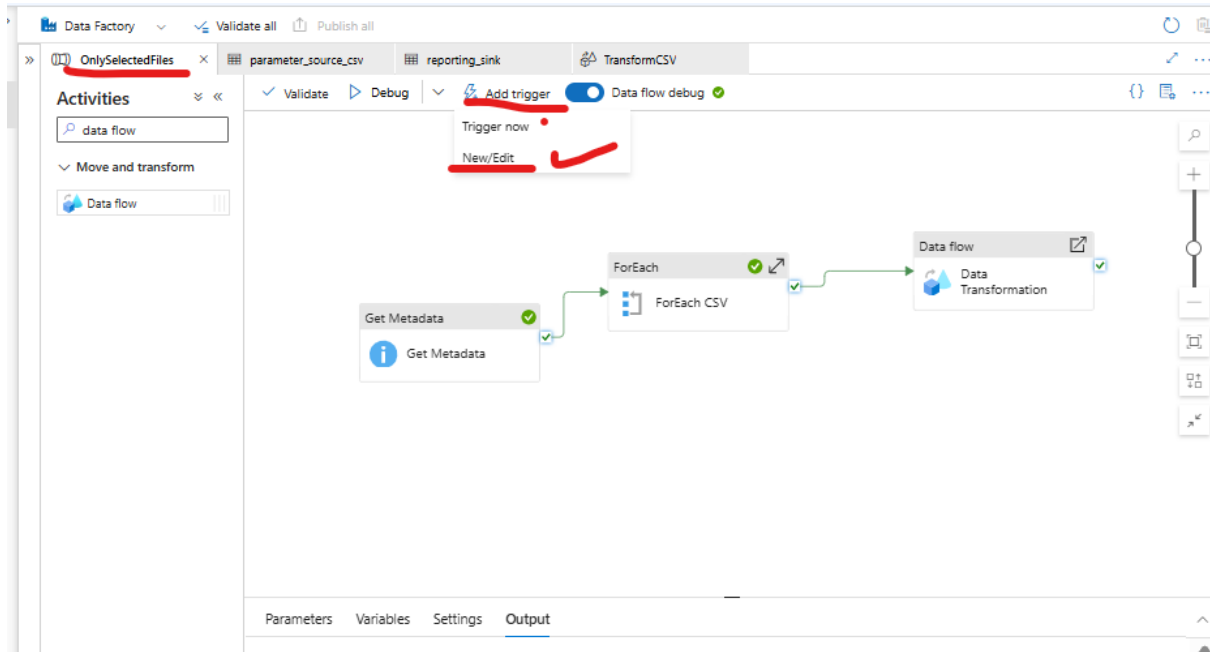
Alter row conditions * [+](#) [-](#) [Insert if](#) [+](#) [-](#)

Finally all done if you need perform any opteron you can do that

Now Triggering this pipeline after main pipeline:

Note: first publish your all data before doing this click on publish all button

Schedule Trigger in Azure Data Factory:



New trigger

Name *
selectedFilesTrigger

Description

Type *
Schedule

Start date * ⓘ
1/1/2026, 1:48:35 AM

Time zone * ⓘ
Chennai, Kolkata, Mumbai, New Delhi (UTC+5:30)

Recurrence * ⓘ
Every 15 Minute(s)

☒ Specify an end date

End On * ⓘ
1/1/2026, 1:50:35 AM

Annotations
+ New

Start trigger ⓘ
☒ Start trigger on creation

OK Cancel

Check or publish all first the trigger done.

Now go to manage tab -> triggers

Data Factory | validate all | publish all

Triggers

To execute a pipeline set the trigger. Triggers represent a unit of processing that determines when a pipeline execution needs to be kicked off.

+ New Refresh

Filter by name

Showing 1 - 1 of 1 items

Name ↑↓	Type ↑↓	Status ↑↓	Related ↑↓	Annotations ↑
selectedFilesTrigger	Schedule	Started	1	

General

- Connector upgrade adv...
- Factory settings

Connections

- Linked services
- Integration runtimes
- Microsoft Purview
- ADF in Microsoft Fabric

Source control

- Git configuration
- ARM template

Author

- Triggers
- Global parameters
- Data flow libraries

If want to check automatic trigger when time is come or as of time :

monitor tab

Pipeline name	Run start	Run end	Duration	Triggered by	Status	Run	Parameters	Annotations	Run ID
OnlySelectedFiles	1/1/2026, 7:37:00 PM	1/1/2026, 6:53:48 PM	17s	SelectedFileTrigger	In progress	Original			b265c806-22a1-47a1-9609-5164c9939
OnlySelectedFiles	1/1/2026, 6:49:00 PM	1/1/2026, 6:53:48 PM	4m 38s	trigger1	Succeeded	Original			b7961293-a439-4d2c-a70a-3425a7307f
OnlySelectedFiles	1/1/2026, 1:48:00 AM	1/1/2026, 1:53:00 AM	5m 0s	selectedFilesTrigger	Succeeded	Original			4ab84f22-6f41-44e1-8d8c-e9e10b4f17f
OnlySelectedFiles	1/1/2026, 1:48:00 AM	1/1/2026, 1:52:37 AM	4m 38s	selectedFilesTrigger	Succeeded	Original			ea409e2b-6b14-4d67-851d-590339b0d6

monitor tab

click

Pipeline name	Run start	Run end	Duration	Triggered by	Status	Run	Parameters	Annotations	Run ID
OnlySelectedFiles	1/1/2026, 7:37:00 PM	1/1/2026, 6:53:48 PM	17s	SelectedFileTrigger	In progress	Original			b265c806-22a1-47a1-9609-5164c9939
OnlySelectedFiles	1/1/2026, 6:49:00 PM	1/1/2026, 6:53:48 PM	4m 38s	trigger1	Succeeded	Original			b7961293-a439-4d2c-a70a-3425a7307f
OnlySelectedFiles	1/1/2026, 1:48:00 AM	1/1/2026, 1:53:00 AM	5m 0s	selectedFilesTrigger	Succeeded	Original			4ab84f22-6f41-44e1-8d8c-e9e10b4f17f
OnlySelectedFiles	1/1/2026, 1:48:00 AM	1/1/2026, 1:52:37 AM	4m 38s	selectedFilesTrigger	Succeeded	Original			ea409e2b-6b14-4d67-851d-590339b0d6

All pipeline runs > OnlySelectedFiles - Activity runs

Activity runs

Pipeline run ID: b265c806-22a1-47a1-9609-5164c993947

Activity name	Activity status	Activity...	Run start	Duration	Integration runtime	User proper...	Activity run ID	Log
Get Metadata	Succeeded	Get Metadata	1/1/2026, 7:37:03 PM	22s	AutoResolveIntegrationRuntime (East US)		98e9610d-eb8b-442d-83a1-b247da5564c9	
ForEach CSV	Succeeded	ForEach	1/1/2026, 7:37:26 PM	51s			969eb469-cfff-46c5-9031-4734615756d8	
If_files_is_matches	Succeeded	If Condition	1/1/2026, 7:37:26 PM	27s			3326dffb-6968-4721-b987-c58772bf668a	
CopyFactData	Succeeded	Copy data	1/1/2026, 7:37:27 PM	26s	AutoResolveIntegrationRuntime (East US)		789b2b03-06a3-4edf-8fcb-69a5e187c228	
If_files_is_matches	Succeeded	If Condition	1/1/2026, 7:37:54 PM	19s			335b2950-2ca3-4eb0-b681-6a165ce69084	
CopyFactData	Succeeded	Copy data	1/1/2026, 7:37:54 PM	18s	AutoResolveIntegrationRuntime (East US)		20c4f9b3-d13e-48d8-b3f5-1fb0077202da	
If_files_is_matches	Succeeded	If Condition	1/1/2026, 7:38:13 PM	Less than 1s			04149b4b-4b81-432f-8578-ac4fc989c8de	
Data Transformation	Succeeded	Data flow	1/1/2026, 7:38:17 PM	3m 59s	AutoResolveIntegrationRuntime (East US)		e53d6429-1cce-49a7-8302-f2ffe3fdd5d4	

reporting > Transform_data_flow_Output

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 6 items

Name	Last modified	Access tier	Blob type	Size	Lease state
[.]					...
_SUCCESS	1/1/2026, 7:41:57 PM	Hot (Inferred)	Block blob	0	Available
part-00000-02f58720-b00c-4a32-bac8-6eb45ae4648b-c000.csv	1/1/2026, 1:52:44 AM	Hot (Inferred)	Block blob	130 B	Available
part-00000-0e7a1268-dd6d-4469-80ad-14cfe0266ede-c000.csv	1/1/2026, 6:56:41 PM	Hot (Inferred)	Block blob	130 B	Available
part-00000-5399771d-5024-4c15-bfa2-756300a9de9e-c000.csv	1/1/2026, 6:53:29 PM	Hot (Inferred)	Block blob	130 B	Available
part-00000-9520151c-9b44-41d9-921c-4f9c8b76f49c-c000.csv	1/1/2026, 7:41:57 PM	Hot (Inferred)	Block blob	130 B	Available
part-00000-ea35222f-6a57-4b86-bb06-00c5ec8381c-c000.csv	1/1/2026, 1:52:31 AM	Hot (Inferred)	Block blob	130 B	Available

Set Variable Activate:

✓ What is Set Variable Activity in Azure Data Factory (ADF)?

Set Variable Activity in ADF is used to **assign or update the value of a pipeline variable during pipeline execution.**

👉 Simply:

Set Variable Activity = Change variable value at runtime

◆ Why is Set Variable Activity used?

It helps in **dynamic pipeline control** like:

- Storing intermediate values
 - Passing values between activities
 - Controlling conditions & loops
 - Building reusable pipelines
-

◆ What are Variables in ADF?

Variables are **pipeline-level temporary storage.**

Supported Variable Types:

- String
- Boolean
- Array

📌 Variables exist **only during pipeline execution.**

◆ How Set Variable Activity works

1. Create a variable in **Pipeline → Variables tab**
2. Use **Set Variable activity**
3. Assign a value using:
 - Static value
 - Expression
 - Output of another activity

When we can use complex solution then need to used this activity

Factory resources

Filter resources by name

Pipelines

- copy_Data_lake_to_Datalake_pipeline
- Copy_CSV_git_to_Distination_datalake
- OnlySelectedFiles
- variablePipeline

Change Data Capture (preview) 0

Datasets 9

Data flows 1

Power Query 0

Activities

Search activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Properties

General Related

Name *

variablePipeline

Description

Annotations

+ New

Pipelines

- copy_Data_lake_to_Datalake_pipeline
- Copy_CSV_git_to_Distination_d...
- OnlySelectedFiles
- variablePipeline

Change Data Capture (preview) 0

Datasets 9

Data flows 1

Power Query 0

Activities

Search activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics

Get Metadata

Get Metadata

ForEach

ForEach CSV

Activities

Data flow

Data Transformation

copy and paste variable pipeline

Pipelines

- copy_Data_lake_to_Datalake_pipeline
- Copy_CSV_git_to_Distination_datalake
- OnlySelectedFiles
- variablePipeline

Change Data Capture (preview) 0

Datasets 9

Data flows 1

Power Query 0

Activities

Search activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General

Get Metadata

Get Metadata

Activities

Search activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Get Metadata

Get Metadata

click

Parameters Variables Settings Output

+ New

3

Properties

General Related

Name *
variablePipeline

Description

Annotations
+ New

Data Factory

Validate all Publish all

OnlySelectedFiles variablePipeline CSV_Meta_Data

Activities

Search activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Get Metadata

Get Metadata

Parameters Variables Settings Output

+ New Delete

Name	Type	Default value
Var_files	Array	Value

Data Factory

Validate all Publish all

OnlySelectedFiles variablePipeline CSV_Meta_Data

Activities

Search activities

set

General

(X) Set variable

Get Metadata

Get Metadata

Set variable

(X) Set variable1

General Settings 1 User properties

Name *
Set variable1

Description

Activity state
Activated Deactivated

Secure output

Secure input

The image shows two screenshots of the Azure DevOps Pipelines interface. The top screenshot displays a pipeline graph with a 'Get Metadata' task connected to a 'Set variable' task. The 'Set variable' task is configured with the name 'Stores_Files_Set_variable'. The bottom screenshot shows the 'Settings' tab for the 'Set variable' task. The 'Variable type' is set to 'Pipeline variable'. The 'Name' field contains 'Var_files'. The 'Value' field is empty, and the 'Add dynamic content [Alt+Shift+D]' link is highlighted with a red circle. The 'General' tab is also visible, showing the 'Name' field as 'Stores_Files_Set_variable' and the 'Activity state' as 'Activated'.

General Settings¹ User properties

Name * Stores_Files_Set_variable [Learn more](#)

Description

Activity state ^① ☒ Activated ☐ Deactivated

Secure output ^① ☐

Secure input ^① ☐

General Settings¹ User properties

Variable type ^① ☒ Pipeline variable ☐ Pipeline return value

Name * Var_files [+](#) New

Value *

[Add dynamic content \[Alt+Shift+D\]](#)

Search related to pipeline variable Or pipeline return value

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

```
@activity('Get Metadata').output.childItems
```

Clear contents

Activity outputs Parameters System variables Functions Variables

Search

- Get Metadata
Get Metadata activity output
- Get Metadata childItems
List of subfolders and files in the given folder
- Get Metadata exists
Whether a file, folder, or table exists
- Get Metadata itemName
Name of the file or folder
- Get Metadata itemType
Type of the file or folder. Returned value is File or Folder
- Get Metadata lastModified
Last modified datetime of the file or folder

OK Cancel

Click on debug

Data Factory Validate all Publish all

Activities General (X) Set variable

Get Metadata Set variable (X) Stores_Files_Set_variable

Parameters Variables Settings Output

Pipeline run ID: 088b0197-0e39-48d5-8cbe-9ef601172441 Pipeline status: Succeeded

Showing 1 - 2 of 2 items

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime	User properties	Activity run ID	Log
Stores_Files_Set_variable	Succeeded	Set variable	1/1/2026, 8:28:54 PM	Less than 1s			082b2598-ed9d-44b2-8135-ad88eebe533c	
Get Metadata	Succeeded	Get Metadata	1/1/2026, 8:28:40 PM	14s	AutoResolveIntegrationRuntime (East US)		a664863a-345e-4874-aab1-5d923262a1a3	

Stores_Files_Set_variable Get Metadata

Output

Copy to clipboard

```
{
  "childItems": [
    {
      "name": "Fact_Sales_1.csv",
      "type": "File"
    },
    {
      "name": "Fact_Sales_2.csv",
      "type": "File"
    },
    {
      "name": "git_csv_file.csv",
      "type": "File"
    }
  ],
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (East US)",
  "executionDuration": 1,
  "durationInQueue": {
    "integrationRuntimeQueue": 10
  },
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": {
      "meterType": "AzureIR",
      "duration": 0.016666666666666666,
      "unit": "Hours"
    }
  }
}
```

get meta data pipeline

Output

Copy to clipboard

```
{
  "name": "Var_files",
  "value": [
    {
      "name": "Fact_Sales_1.csv",
      "type": "File"
    },
    {
      "name": "Fact_Sales_2.csv",
      "type": "File"
    },
    {
      "name": "git_csv_file.csv",
      "type": "File"
    }
  ]
}
```

stores_files_set-var pipeline

See the difference in set variable file output is we show file name now we can used this any where

Storage Event Trigger in ADF:

✓ What is Storage Event Trigger in Azure Data Factory (ADF)?

A **Storage Event Trigger** in ADF is used to **automatically start a pipeline when a file or folder event occurs in Azure Storage**.

👉 Simply:

Storage Event Trigger = Event-based pipeline execution

◆ Why is Storage Event Trigger used?

Instead of running pipelines on a schedule, it allows ADF to react **immediately** when:

- A file is **created**
- A file is **deleted**

✂ This is useful for **real-time or near real-time data ingestion**.

◆ Supported Storage Types

- **Azure Blob Storage**
- **Azure Data Lake Storage Gen2**

◆ How Storage Event Trigger works (Flow)

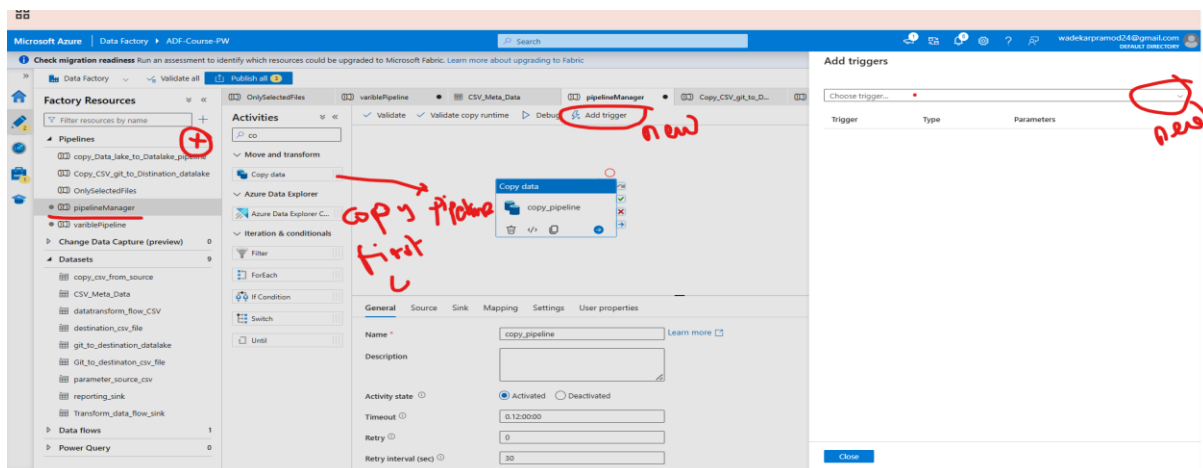
1. File lands in **ADLS Gen2 / Blob**
2. **Event Grid** captures the event
3. Event notifies **ADF Trigger**
4. ADF **automatically starts the pipeline**

✂ You don't need to poll or schedule anything.

The requirement is the when manager add some important file in container the we need get in out destination container before the work. This requi need to do by automate using this activity.

Then we need like a copy activity and this trigger every night. Or this is automatically start also and stop also just after manager upload file.

If manager add file then again tigger pipeline (start) automatically and after that stop also automatically. This is the power of storage event trigger.



New trigger

Name *

Managertrigger

Description

Type *

Storage events

Filter...

Schedule

Tumbling window

Storage events

Custom events

Container name * ⓘ

/containername/

✖ Container name is required

Blob path begins with ⓘ

Blob path ends with ⓘ

Event * ⓘ

Continue

Cancel

New trigger

Name *
Managertrigger

Description

Type *
Storage events

Account selection method *
☒ From Azure subscription ☐ Enter manually

Azure subscription *
Azure subscription 1 (cccc7002-f456-4a63-aedb-2fd621f67a10)

Storage account name *
pwdatastorageaccount

Container name *
source

Blob path begins with
CSV_Files/Fact_Sales_1.csv

Blob path ends with

Event *
☒ Blob created ☐ Blob deleted

Ignore empty blobs *
☒ Yes ☐ No

Annotations
+ New

Start trigger
☒ Start trigger on creation

Continue Cancel

Blob created : means when manger is upload or add file in source container then pipeline is trigger.

Data preview

Make sure you have specific filters. Configuring filters that are too broad can match a large number of files created/deleted and may significantly impact your cost.

Event Trigger Filters

Container name: source
Starts with: CSV_Files/Fact_Sales_1.csv
Ends with:

1 blobs matched in "source"

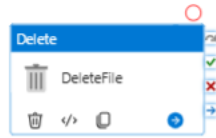
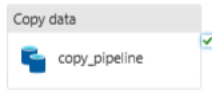
	Blob name
1	CSV_Files/Fact_Sales_1.csv

1 - 1 of 1 items

Continue Back Cancel

Now we can perform delete activity because when one file is available in source folder then its run again and again, then we want to when file is copied successfully then need to delete in source folder.

The screenshot displays the Azure Data Factory (ADF) interface. On the left, the 'Activities' pane shows a search bar with 'del' and a list of activities including 'Delete', which is highlighted with a red circle. A red arrow points from the 'Delete' activity in the list to the 'Delete' activity in the main canvas. The main canvas shows a 'Copy data' activity named 'copy_pipeline' connected to a 'Delete' activity named 'Delete1'. Below the canvas, the 'Delete1' activity is configured in the 'General' tab. The 'Name' field is 'Delete1', and the 'Activity state' is set to 'Activated' (indicated by a red circle and a red underline). The 'Timeout' is set to '0.12:00:00', 'Retry' is '0', and 'Retry interval (sec)' is '30'. The 'Logging settings' and 'Error properties' tabs are also visible.



General Source Logging settings User properties

Dataset *  [Open](#) [New](#) [Preview data](#) [Learn more](#)

File path type ☒ File path in dataset ☐ Wildcard file path ☐ List of files

Filter by last modified

Recursively ☒

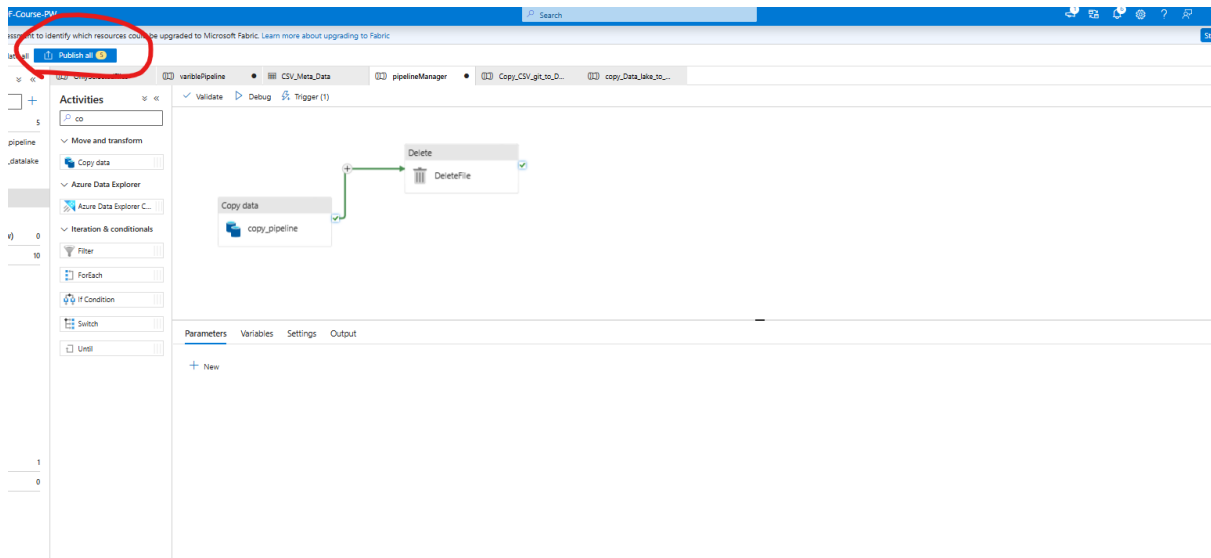
Max concurrent connections



General Source **Logging settings** User properties

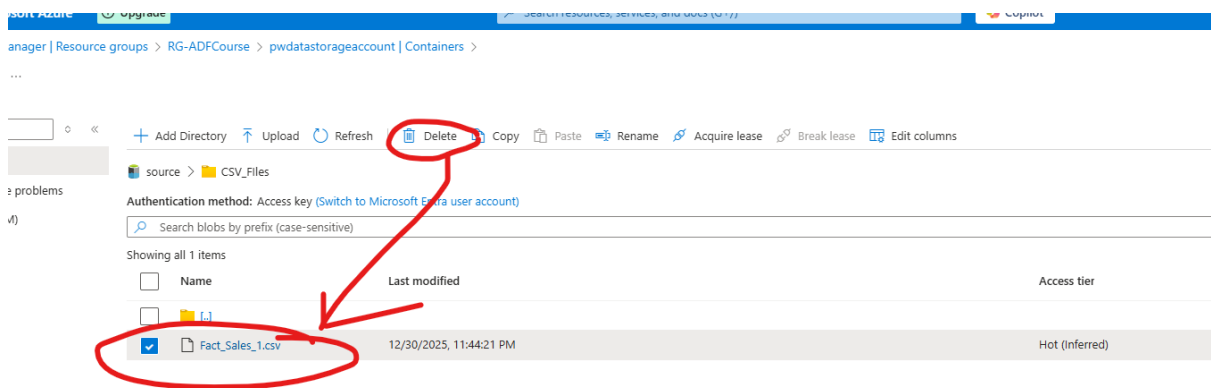
Enable logging ☐

unchecked

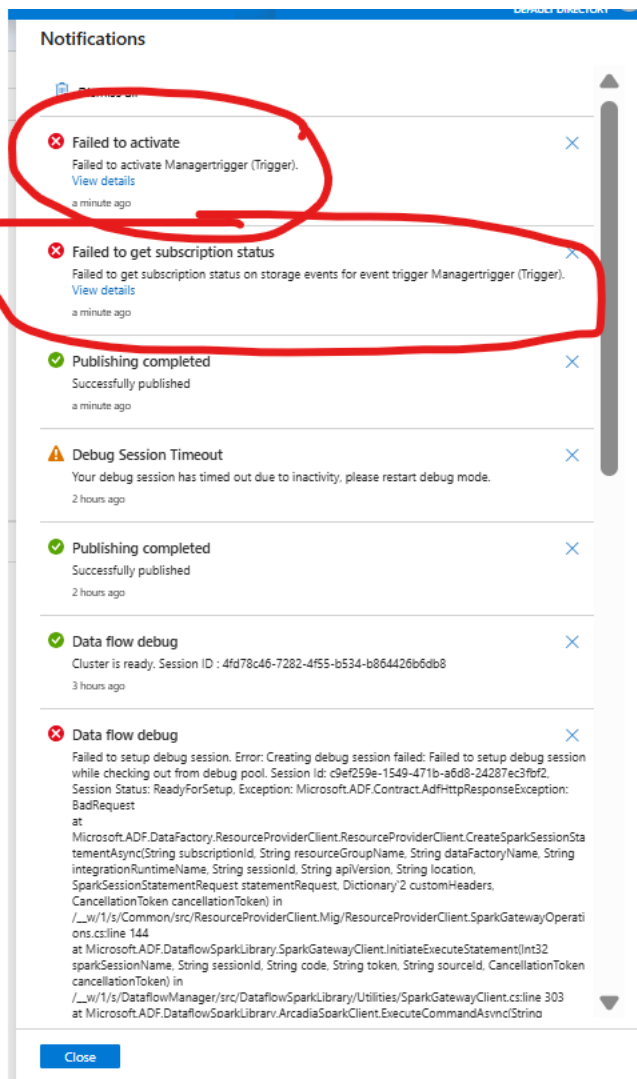


Click on publish all.

Before click on publish all do the following steps for testing



Now click on publish all button



This is showing for new big trigger for need gesture

Now solve this error:

Go to home tab and click on search subscription

Subscription name ↑↓	Subscription ID ↑↓	My role ↑↓	Current cost	Secure S
Azure subscription 1	cccc7002-f456-4a63-aedb-2fd621f67a10	Owner		100%

click



Azure subscription 1

Subscription



What security features does this Subscription provide?

+2

Search

Cancel subscription

Rename

Change directory

Feedback

Resource visualizer

Events

Resource groups

Resources

Cost Management

Billing

Settings

Programmatic deployment

Billing properties

Preview features

Usage + quotas

Policies

My permissions

Resource providers

Deployments

Deployment stacks

Resource locks

Monitoring

Your remaining \$192.05 of free credit expires in 28 days. Upgrade to keep going with

Essentials

Subscription ID

cccc7002-f456-4a63-aedb-2fd621f67a10

Subscription name

[Azure subscription 1](#)

Directory

Default Directory (wadekarpramod24gmail.o...

My role

Owner

Status

Active

Plan

Azure Plan

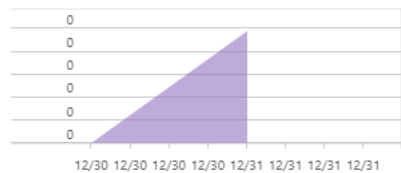
Parent management group

82fc89f7-0821-43f1-879b-4f140b589e39

Secure Score

[100%](#)

Spending rate and forecast

Current cost
₹0.00[View details](#)

Costs by resource

[View details](#)

Top products by number of resources

Azure Defender

Azure subscription 1 | Resource providers

Subscription

Search

Register

Unregister

Refresh

Feedback

Resource visualizer

Events

Resource groups

Resources

Cost Management

Billing

Settings

Programmatic deployment

Billing properties

Preview features

Usage + quotas

Policies

My permissions

Resource providers

Deployments

eve

Status : All

Registration Policy : All

Provider ↑

Status

Registration Policy

[Microsoft.DevelopmentWindows365](#)

NotRegistered

RegistrationRequired ⓘ

[Microsoft.EventGrid](#)

NotRegistered

RegistrationRequired ⓘ

[Microsoft.EventHub](#)

Register

Unregister

RegistrationRequired ⓘ

Provider	Status	Registration Policy
☐ Microsoft.DevelopmentWindows365	...	NotRegistered RegistrationRequired ⓘ
☐ Microsoft.EventGrid	...	registered RegistrationRequired ⓘ
☐ Microsoft.EventHub	...	NotRegistered RegistrationRequired ⓘ

Now we can trigger the last pipeline

New trigger

Name *
managerFiletrigger

Description

Type *
Storage events

Account selection method *
☒ From Azure subscription ☐ Enter manually

Azure subscription
Azure subscription 1 (cccc7002-f456-4a63-aedb-2fd621f67a10)

Storage account name *
pwdatastorageaccount

Container name *
source

Blob path begins with
CSV_Files/Fact_Sales_1.csv

Blob path ends with

Event *
☒ Blob created ☐ Blob deleted

Ignore empty blobs *
☒ Yes ☐ No

Annotations
+ New

Start trigger
☒ Start trigger on creation

Continue Cancel

Next again continue and publish all

Publish all

You are about to publish all pending changes to the live environment. [Learn more](#)

Pending changes (2)

NAME	CHANGE	EXISTING
Managertrigger	(Edited)	Managertrigger
managerFiletrigger	(New)	-

Publish **Cancel**

After publish done go to monitor tab and check

Check migration readiness Run an assessment to identify which resources could be upgraded to Microsoft Fabric. [Learn more about upgrading to Fabric](#) **Start assessment (preview)**

Trigger runs

All Schedule Tumbling window Storage events Custom events Refresh Edit columns

Chennai, Kolkata, Mu... : Last 24 hours Trigger name : All Status : All Runs : Latest runs Export to CSV

Showing 1 - 11 items

Trigger n...	Trigger type	Trigger ti...	Status	Pipelines	Run	Message	Properties	Run ID
SelectedfileTi...	Schedule trig...	1/1/2026, 9:22:00	Succeeded	1	Original			08584343241...
SelectedfileTi...	Schedule trig...	1/1/2026, 9:07:00	Succeeded	1	Original			08584343250...
SelectedfileTi...	Schedule trig...	1/1/2026, 8:52:00	Succeeded	1	Original			08584343259...
SelectedfileTi...	Schedule trig...	1/1/2026, 8:36:50	Succeeded	1	Original			08584343268...
SelectedfileTi...	Schedule trig...	1/1/2026, 8:22:00	Succeeded	1	Original			08584343277...
SelectedfileTi...	Schedule trig...	1/1/2026, 8:07:00	Succeeded	1	Original			08584343286...
SelectedfileTi...	Schedule trig...	1/1/2026, 7:52:00	Succeeded	1	Original			08584343295...
SelectedfileTi...	Schedule trig...	1/1/2026, 7:36:50	Succeeded	1	Original			08584343304...
trigger1	Schedule trig...	1/1/2026, 6:49:00	Succeeded	1	Original			08584343333...
selectedFilesT...	Schedule trig...	1/1/2026, 1:48:00	Succeeded	1	Original			085843433946...
selectedFilesT...	Schedule trig...	1/1/2026, 1:47:50	Succeeded	1	Original			085843433946...

not showing

Trigger not showing because file is not available in source container.

+ Add Directory ↑ Upload ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename ⚙ Acquire lease ⚙ Break lease 🛠 Edit columns					
source > CSV_Files					
Authentication method: Access key (Switch to Microsoft Entra user account)					
Search blobs by prefix (case-sensitive)					
Showing all 1 items					
☐	Name		Last modified	Access tier	Blob type
☐	[-]				
☐	Fact_Sales_1.csv		1/1/2026, 11:08:56 PM	Hot (Inferred)	Block blob

Now upload file in source container and again go to monitor tab-> trigger runs

All Schedule Tumbling window Storage events Custom events ↻ Refresh 🛠 Edit columns									
Chennai, Kolkata, Mu... : Last 24 hours Trigger name : All Status : All Runs : Latest runs									
Showing 1 - 12 items									
Trigger name	Trigger type	Trigger time	Status	Pipelines	Run	Message	Properties	Run ID	
managerFileTrigger	Storage events ...	1/1/2026, 11:06:44	✔ Succeeded	1	Original			c5de90d8-3944...	
SelectedFileTrigger	Schedule trigger	1/1/2026, 9:22:00 F	✔ Succeeded	1	Original			085843432416...	
SelectedFileTrigger	Schedule trigger	1/1/2026, 9:07:00 F	✔ Succeeded	1	Original			085843432506...	
SelectedFileTrigger	Schedule trigger	1/1/2026, 8:52:00 F	✔ Succeeded	1	Original			085843432596...	
SelectedFileTrigger	Schedule trigger	1/1/2026, 8:36:59 F	✔ Succeeded	1	Original			085843432686...	
SelectedFileTrigger	Schedule trigger	1/1/2026, 8:22:00 F	✔ Succeeded	1	Original			085843432776...	

This is trigger automatically when file is uploaded.

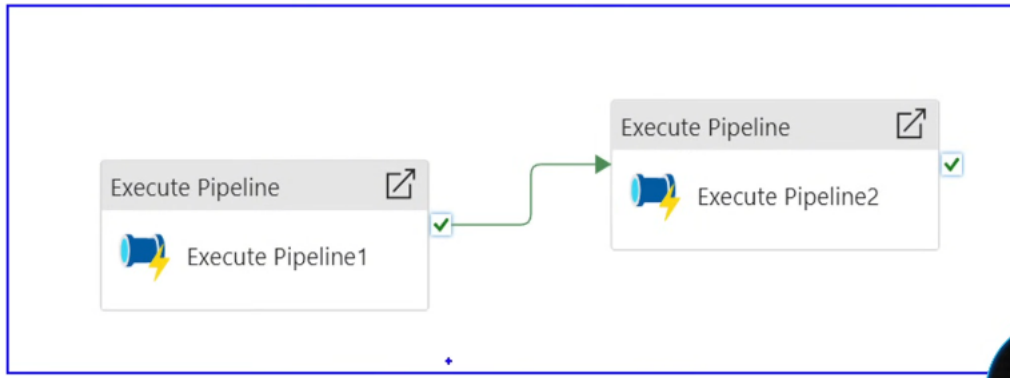
After copy in destination folder then delete automatically as per our pipeline

+ Add Directory ↑ Upload ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename ⚙ Acquire lease ⚙ Break lease 🛠 Edit columns					
source > CSV_Files					
Authentication method: Access key (Switch to Microsoft Entra user account)					
Search blobs by prefix (case-sensitive)					
Showing all 0 items					
☐	Name		Last modified	Access tier	
☐	[-]				

Rerun Cancel Refresh Update pipeline List Gantt									
Copy data Delete									
copy_pipeline DeleteFile									
Activity runs									
Pipeline run ID fffb7b27-bc7f-4a61-a46f-b0ee8170f643									
All status Monitor in Azure Metrics Export to CSV									
Showing 1 - 2 items									
Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID		
DeleteFile	✔ Succeeded	Delete	1/1/2026, 11:09:17 PM	15s	AutoResolveIntegrationRuntime (East US)		a6b32eff-1dcf-4f19-ba21-75e17d2aa59d		
copy_pipeline	✔ Succeeded	Copy data	1/1/2026, 11:09:02 PM	14s	AutoResolveIntegrationRuntime (East US)		c2c640c4-1aab-41f6-8cb9-ba4ebeb9225b		

Execute Pipeline Activity in Azure Data Factory:

EXECUTE PIPELINE ACTIVITY



Requirement: if manage add data on 2am -> it will copy the data to the destination then after it is adding on destination this activity pull the data from the github that is fact_sales.csv then both the files will go to the get meta data activity then they will go to for each activity then go to if else condition then go to copy activity then our data flow is running. Do this automatically and trigger

Now detach all the trigger pipelines

2 new edit

1

4

3

5

Close

Onlyselectdefile pipeline run after manager pipeline

- Now remove all file folder in containers noy delete only souce contained file and folder

Now we can perform End to End Pipeline:

The screenshot shows the Microsoft Fabric Data Factory interface. On the left, the 'Factory Resources' pane lists various pipelines, with 'ProdPipeline' highlighted and circled in red. In the center, the 'Activities' pane shows the 'Execute Pipeline' activity, also circled in red. The main canvas displays a diagram of the 'Execute Pipeline' activity, which is connected to a 'ProdPipeline' resource. The 'Properties' pane on the right shows the configuration for the 'ProdPipeline', with the 'Name' field circled in red.

This section provides a detailed view of the 'Execute Pipeline' activity configuration. The 'General' tab is selected, showing the 'Name' field set to 'ExecuteManagerPipeline' and the 'Activity state' set to 'Activated'. The 'Settings' tab is also visible, showing the 'Invoked pipeline' set to 'pipelineManager' and the 'Wait on completion' checkbox checked. The 'User properties' tab is also visible, showing the 'Name' field set to 'ExecuteManagerPipeline' and the 'Activity state' set to 'Activated'. The 'Invoked pipeline' field is circled in red, and the 'Wait on completion' checkbox is also circled in red.

publish all

pipelineManager copy_Data_lake_to... copy_csv_from_source Copy_CSV_gif_to_D... OnlySelectedFiles variablePipeline ProdPipeline

Activities

exe

General

Execute Pipeline

Execute SSIS package

Machine Learning

Machine Learning Exe...

Execute Pipeline

ExecuteManagerP... pipelineManager

Execute Pipeline

ExecuteOnSelected FilePipeline

Properties

General Related

Name *

ProdPipeline

Description

Annotations

+ New

Cancel

General Settings User properties

Name *

ExecuteOnSelectedFilePipeline

Description

Activity state

Activated Deactivated

Secure input

pipelineManager copy_Data_lake_to... copy_csv_from_source Copy_CSV_gif_to_D... OnlySelectedFiles variablePi

Activities

exe

General

Execute Pipeline

Execute SSIS package

Machine Learning

Machine Learning Exe...

Execute Pipeline

ExecuteManagerP... pipelineManager

Execute Pipeline

ExecuteOnSelecte... OnlySelectedFiles

General Settings User properties

Settings

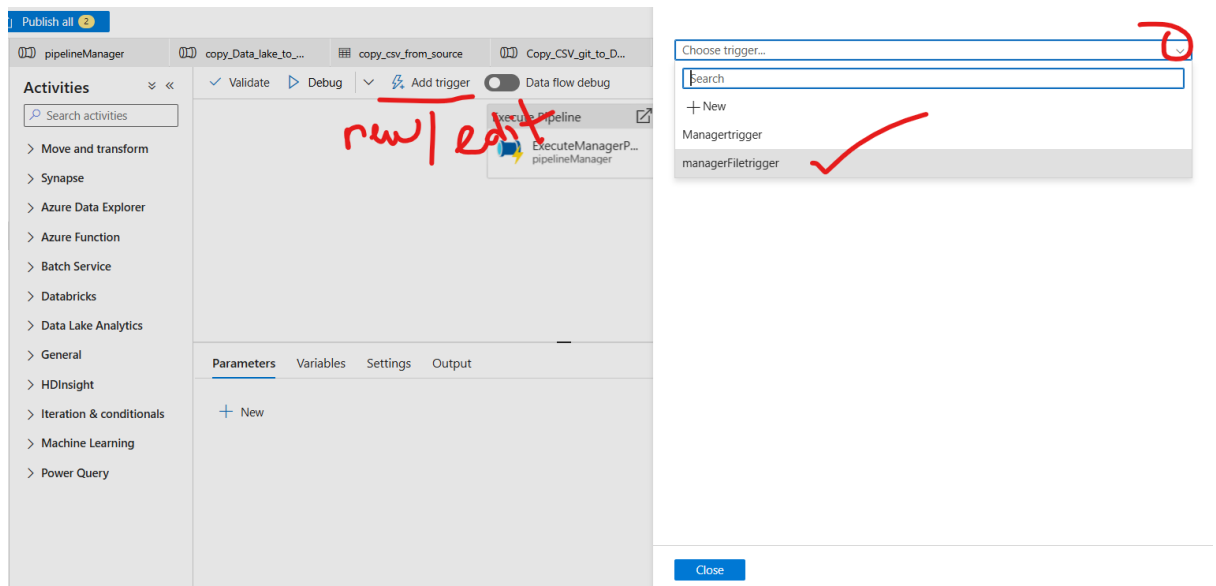
Invoked pipeline *

OnlySelectedFiles

Wait on completion

Open + New

Now add new trigger



Edit trigger

Name *
managerFiletrigger

Description

Type *
BlobEventsTrigger

Account selection method * ①
☒ From Azure subscription ☐ Enter manually

Azure subscription ①
Azure subscription 1 (cccc7002-f456-4a63-aedb-2fd621f67a10)

Storage account name * ①
pwdatastorageaccount

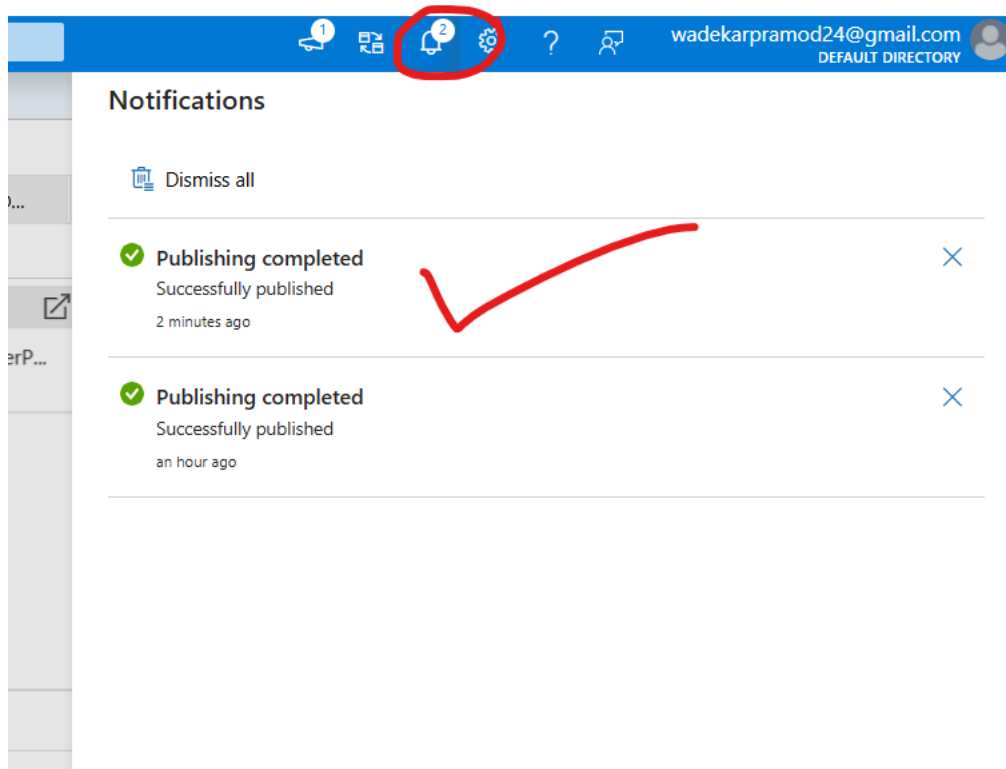
Container name * ①
source

Blob path begins with ①
CSV_Files/Fact_Sales_1.csv

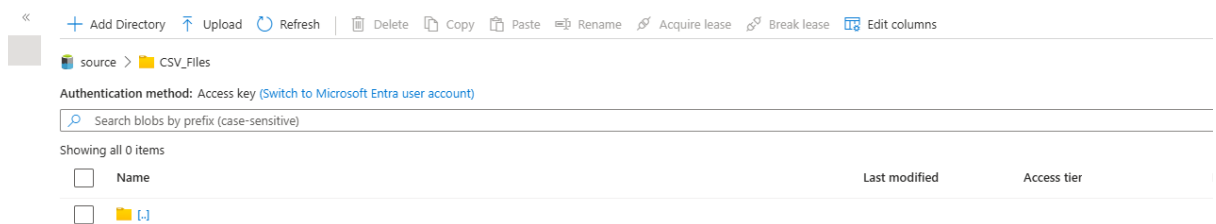
Blob path ends with ①

Continue Cancel

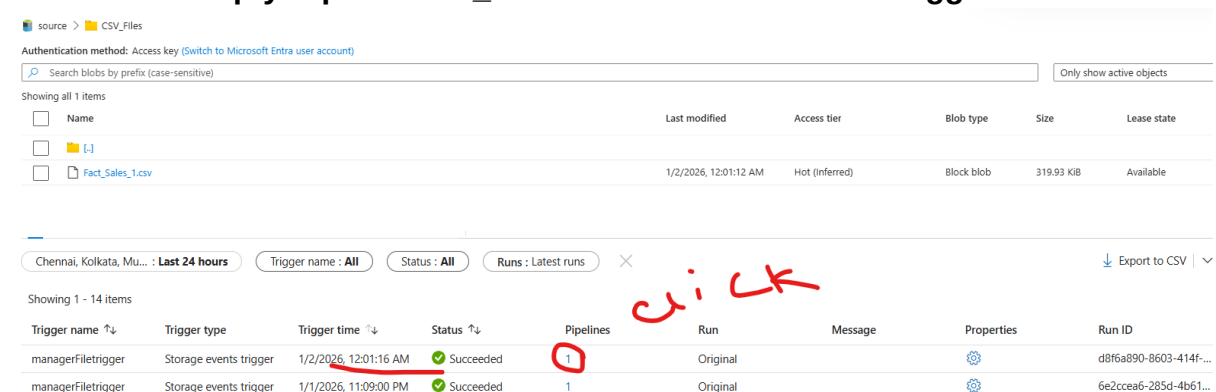
After that again continue ->ok -> publish all



Now pipeline is done now we can test all this production pipeline



Now this is empty. Upload fact_sales.csv here nad check trigger section



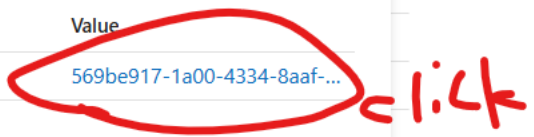
	Status ↑↓	Pipelines	Run
2:01:16 AM	✓ Succeeded	1	Original

1:05
1:06
:22
:07
:52
:36
:22
:07

Pipeline runs



Name	Value
ProdPipeline	569be917-1a00-4334-8aaf-...



Rerun
 Cancel
 Refresh
 Update pipeline
 List Gantt



Activity runs

Pipeline run ID 569be917-1a00-4334-8aaf-8e0855b8150c

All status

[Monitor in Azure Metrics](#)
[Export to CSV](#)

Showing 1 - 2 items

Activity name ↑↓	Activity sta... ↑↓	Activity type	Run start ↑↓	Duration	Integration runtime	User properties	Activity run ID
ExecuteOnSelectedFilePipeline	🔄 In progress	Execute Pipelin	1/2/2026, 12:01:47 AM	1m 32s			99a8b727-1b95-4bda-be4d-c61066715d7c
ExecuteManagerPipeline	✓ Succeeded	Execute Pipelin	1/2/2026, 12:01:21 AM	26s			a78ad6b8-d1be-435d-a6ee-a4d586411797



Rerun
 Cancel
 Refresh
 Update pipeline
 List Gantt



Activity runs

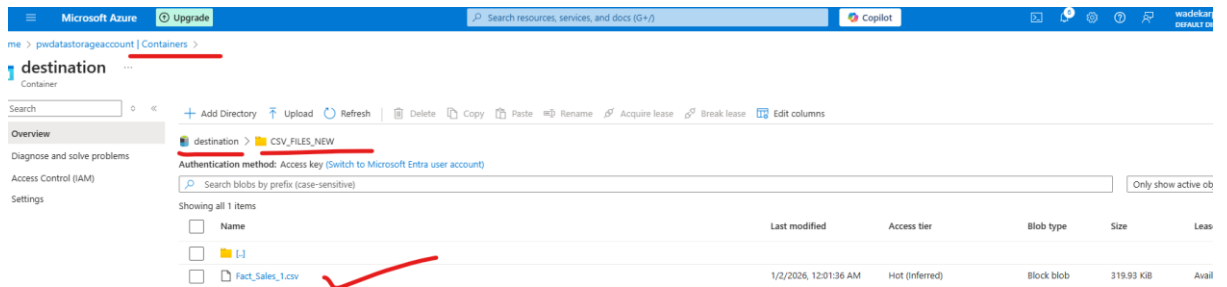
Pipeline run ID 569be917-1a00-4334-8aaf-8e0855b8150c

All status

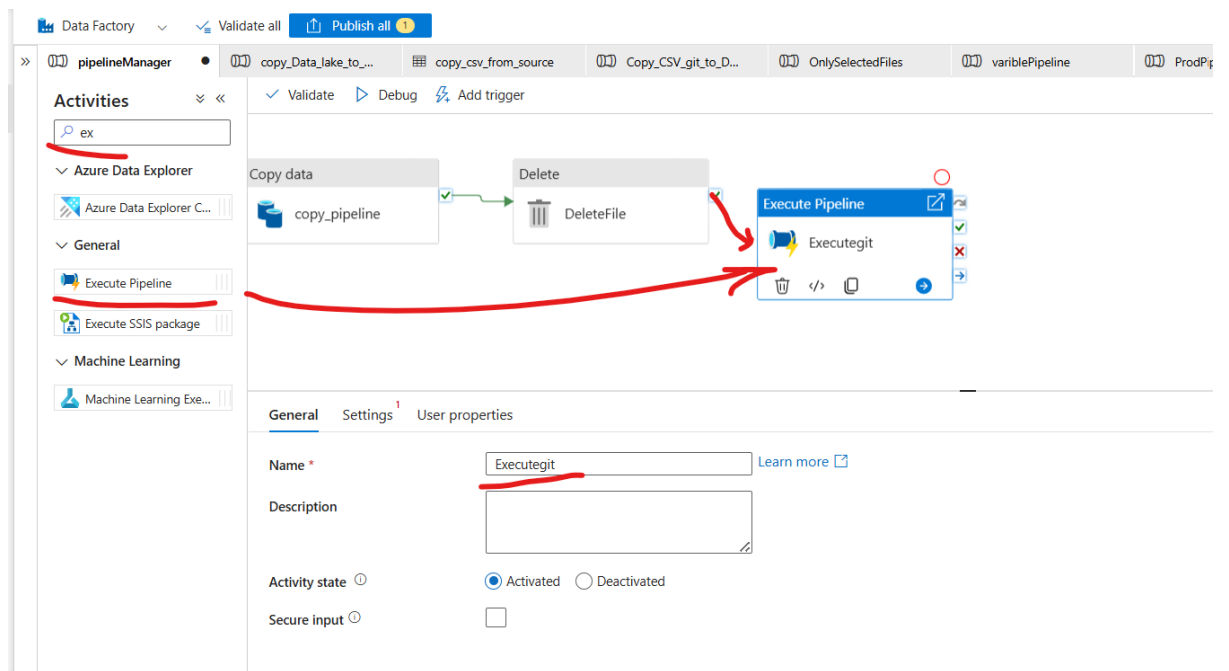
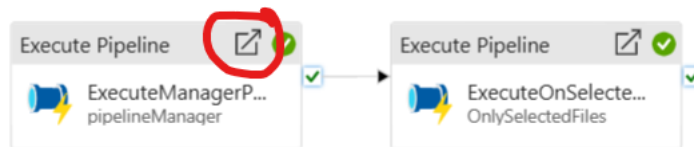
[Monitor in Azure Metrics](#)
[Export to CSV](#)

Showing 1 - 2 items

Activity name ↑↓	Activity sta... ↑↓	Activit... ↑↓	Run start ↑↓	Duration ↑↓	Integration runtime ↑↓	User prop... ↑↓	Activity run ID ↑↓
ExecuteOnSelectedFilePipeline	✓ Succeeded	Execute Pipelin	1/2/2026, 12:01:47 AM	4m 44s			99a8b727-1b95-4bda-be4d-c61066715d7c
ExecuteManagerPipeline	✓ Succeeded	Execute Pipelin	1/2/2026, 12:01:21 AM	26s			a78ad6b8-d1be-435d-a6ee-a4d586411797



But the github have two files but here add one because we are not add git pipeline, now we can do



assessment to identify which resources could be upgraded to Microsoft Fabric. [Learn more about upgrading to Fabric](#) Start assessment (preview)

Rerun Cancel Refresh Update pipeline List Gantt

Execute Pipeline ExecuteManagerP... pipelineManager Execute Pipeline ExecuteOnSelecte... OnlySelectedFiles

Activity runs

Pipeline run ID 4cbe9d6-a9e3-4d40-abb7-0cab310626b3

All status Monitor in Azure Metrics Export to CSV

Showing 1 - 2 items

Activity name	Activity sta...	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID
ExecuteOnSelectedFilePipeline	In progress	Execute Pipeline	1/2/2026, 12:19:32 AM	23s			d6bbb48d-c25b-4a6e-9b27-e17994c31234
ExecuteManagerPipeline	Succeeded	Execute Pipeline	1/2/2026, 12:18:33 AM	59s			e4314da5-adac-4f36-aa86-e2cd52959908

Rerun Cancel Refresh Update pipeline List Gantt

Execute Pipeline ExecuteManagerP... pipelineManager Execute Pipeline ExecuteOnSelecte... OnlySelectedFiles

Activity runs

Pipeline run ID 4cbe9d6-a9e3-4d40-abb7-0cab310626b3

All status Monitor in Azure Metrics Export to CSV

Showing 1 - 2 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
ExecuteOnSelectedFilePipeline	Succeeded	Execute Pipeline	1/2/2026, 12:19:32 AM	4m 54s			d6bbb48d-c25b-4a6e-9b27-e17994c31234
ExecuteManagerPipeline	Succeeded	Execute Pipeline	1/2/2026, 12:18:33 AM	59s			e4314da5-adac-4f36-aa86-e2cd52959908

Microsoft Azure Upgrade Search resources, services, and docs (Ctrl) Copilot wadekurpramos24@ym... DEFAULT DIRECTORY PROVISIONER

Home > pwdatastorageaccount1 Containers > destination

Search + Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

destination CSV_FILES_NEW

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 2 items Only show active objects

Name	Last modified	Access tier	Blob type	Size	Lease state
[.]					
Fact_Sales_1.csv	1/2/2026, 12:19:01 AM	Hot (Inferred)	Block blob	319.93 KiB	Available
Fact_Sales_2.csv	1/2/2026, 12:19:26 AM	Hot (Inferred)	Block blob	2.8 KiB	Available

+ Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

reporting csv_files

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 2 items Only show active objects

Name	Last modified	Access tier	Blob type	Size	Lease state
[.]					
Fact_Sales_1.csv	1/2/2026, 12:20:01 AM	Hot (Inferred)	Block blob	319.93 KiB	Available
Fact_Sales_2.csv	1/2/2026, 12:20:15 AM	Hot (Inferred)	Block blob	2.8 KiB	Available

reporting > Transform_data_flow_Output

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 3 items

Name	Last modified	Access tier	Blob type	Size	Lease state
[-]					
[-] SUCCESS	1/2/2026, 12:24:15 AM	Hot (Inferred)	Block blob	0	Available
part-00000-19f799a2-0a21-4711-9fa1-e8a9e0c1bf5e-c000.csv	1/2/2026, 12:24:14 AM	Hot (Inferred)	Block blob	130 B	Available
part-00000-7a1d26c2-5974-44ec-89e8-285334104fba-c000.csv	1/2/2026, 12:06:21 AM	Hot (Inferred)	Block blob	130 B	Available

source > CSV_Files

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 0 items

Name	Last modified	Access tier	Blob type	Size	Lease state
[-]					

After copy file is deleted.

✓ What is a Tumbling Window Trigger in Azure Data Factory (ADF)?

A **Tumbling Window** is a **time-based trigger** in ADF that runs pipelines in **fixed, non-overlapping** time intervals.

👉 Simply:

Tumbling Window = Scheduled trigger with strict time windows

◆ Why is Tumbling Window used?

It is mainly used for:

- **Batch processing**
- **Time-partitioned data**
- **Incremental loads**
- **Reliable, ordered execution**

Each window runs **exactly once** — no duplicates, no overlap.

◆ How Tumbling Window works

- Time is divided into fixed windows (e.g. every 1 hour)
- Each window has:
 - Start time
 - End time
- Pipeline processes data **only for that window**
- Next window starts **after previous completes**

📌 Windows never overlap.

◆ Example (Simple)

Trigger: Every 1 hour

Window	Time
--------	------

Window 1	10:00 – 11:00
----------	---------------

Window 2	11:00 – 12:00
----------	---------------

Window 3	12:00 – 13:00
----------	---------------

Each run processes data for that specific hour.

◆ Real-world Use Case

📌 **Incremental sales load**

- Load sales data every hour
- Use `windowStart` & `windowEnd`
- Ensure **no duplicate data**

..... All Done